TH650A/TS2000

Industrial Robot

MAINTENANCE MANUAL

Notice

1. Make sure that this instruction manual is delivered to the final user of SHIBAURA MACHINE's industrial robot.
2. Before operating the industrial robot, read through and completely understand this manual.
3. After reading through this manual, keep it nearby for future reference.

Rev.5
March 19, 2020

NUMAZU, JAPAN

Shibaura Machine

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WARRANTY

This product is delivered to each customer only after it is inspected very carefully to make sure that it satisfies the SHIBAURA MACHINE's standard. Should it cause an inconvenience, we will guarantee as described below.

1. Warranty period
   SHIBAURA MACHINE agrees to repair or replace as necessary all defective material or workmanship up to the period shown below, whichever comes first.
   1) Twenty-four (24) months from the date of dispatch from our plant.
   2) Eighteen (18) months from the date of machine installation at customer's job site.
   3) 4,000 running hours from the date of initial machine operation.

2. Contents of warranty
   1) Only this product is subject to SHIBAURA MACHINE's Guarantee. Such Guarantee covers the specifications and functions as defined in the product specifications manual, catalog, instruction manual, etc. In no event does the Guarantee cover any secondary or associated damage caused by malfunction of this product.
   2) SHIBAURA MACHINE repairs the product free of charge only when it malfunctioned after handling or use according to the instruction manual attached to the product within the specified warranty period.

3. Exemption from responsibility
   SHIBAURA MACHINE's Guarantee shall not cover the following cases.
   1) Incorrect use not described in the instruction manual, and trouble or damage caused by negligent use.
   2) Inconvenience caused by aged deterioration or long-term usage (natural fading of coating or painting, deterioration of consumable parts (*1), etc.).
   3) Inconvenience caused by sensuous phenomena (noise generation, etc. which will not affect the function).
   4) Remodeling or disassembly which SHIBAURA MACHINE will not permit.
   5) Trouble and damage caused by insufficient maintenance/inspection or improper repair.

6) Trouble and damage caused by disaster, fire or other external factor.

7) Internal data such as program and point which were created by the customer.

8) When the product purchased in Japan was shipped overseas.

(*1) The consumable parts signify the replacement parts for maintenance as listed in Section 5 of this manual.

4. Precautions

1) Unless this product was used pursuant to its specifications, SHIBAURA MACHINE will not guarantee the basic performance of the product.

2) If the customer did not observe the warnings and cautions described in this manual, SHIBAURA MACHINE will not assume the responsibility for any consequential accident resulting in injury or death, damage or trouble.

3) Please note that the warnings, cautions and other descriptions stipulated in this manual are only those which can be assumed by SHIBAURA MACHINE as of now.

INTRODUCTION

This manual describes the maintenance and inspection of the TH650A industrial robot and TS2000 controller.
The maintenance and inspection are essential to maintain the robot performance for long years to prevent trouble and improve the safety.
Before starting actual operation, it is strongly recommended to read through this manual and draw up a maintenance schedule.

CAUTIONS ON SAFETY

This manual contains the important information on the robot and controller to prevent injury to the operators and persons nearby, to prevent damage to assets and to assure correct use.
Make sure that the following details (indications and symbols) are well understood before reading this manual. Always observe the information that is noted.

[Explanation of indications]

Indication	Meaning of indication
 **DANGER**	This means that "incorrect handling will imminently lead to fatalities or serious injuries."
 **WARNING**	This means that "incorrect handling may lead to fatalities or serious injuries."
 **CAUTION**	This means that "incorrect handling may lead to personal injuries *1) or physical damage *2)."

*1) Injuries refer to injuries, burns and electric shocks, etc., which do not require hospitalization or long-term medical treatment.

*2) Physical damage refers to damages due to destruction of assets or resources.

[Explanation of symbols]

Symbol	Meaning of symbol
	This means that the action is prohibited (must not be done). Details of the actions actually prohibited are indicated with pictures or words in or near the symbol.
	This means that the action is mandatory (must be done). Details of the actual actions that must be done are indicated with pictures or words in or near the symbol.
	This means danger and caution. The details of the actual caution are indicated with pictures or words in or near the symbol.

[Maintenance and inspection]

Be sure to observe the following items to use this product safely.

 **DANGER**	
 Prohibited	• DO NOT incinerate, disassemble or charge the batteries. Otherwise, they may rupture.
 Mandatory	• Be sure to turn off the main power switch of the controller before starting inspection or maintenance. • Batteries should be disposed of according to the user's in-house regulations.

! CAUTION	
 Disassembly prohibited	• The user must NEVER replace or modify parts other than those described in the instruction manual. Otherwise, the performance may deteriorate or faults or accidents will be caused.
 Mandatory	• Always use the SHIBAURA MACHINE's designated spare parts when replacing the parts. • Maintenance and inspection should be performed regularly. Otherwise, the system may malfunction or accidents will be caused.

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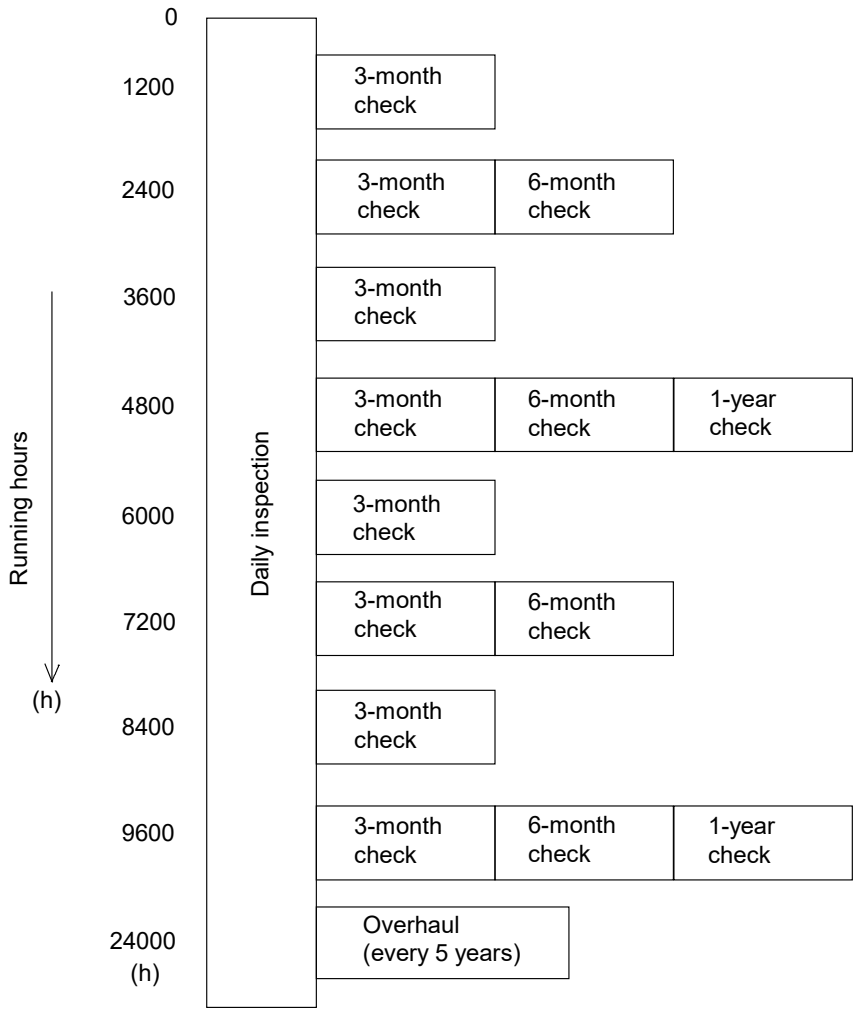
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1. Maintenance

1.1 Maintenance Schedule

Maintenance comes in the two (2) types; daily inspection, and regular inspection and maintenance. For the regular inspection and maintenance, inspection items are added every 1,200 running hours.



Inspection schedule

- When the system is operated for 16 hours per day:
  16 hours × 25 days × 3 months = 1,200 hours
- When the system is operated for 24 hours per day:
  24 hours × 25 days × 3 months = 1,800 hours

1.2 Items for Maintenance and Inspection

This section describes the items for maintenance and inspection.
For the executing procedures, see the relevant paragraph listed in the table below.

1.2.1 Inspection at Power OFF (at Non-Operation)

D: Daily inspection Q: Quarterly inspection
S: Semi-annual inspection A: Annual inspection

Description	Check position	D	Q	S	A	Refer to
Make sure that all bolts are completely tightened and secured. Otherwise, tighten the bolts completely.	Tool set bolts	O	O	O	O	Para. 2.2.1
	Robot installation bolts	O	O	O	O	
	Bolts for each joint reduction gear				O	
	Motor set bolts				O	
Make sure that all connectors are tightened completely	Base connector	O	O	O	O	Para. 2.2.2
	External interconnecting wiring for axis 5	O	O	O	O	
Make sure that no cut or scratch is found. Clean the contaminant, if any.	Entire robot	O	O	O	O	
	Arm 2 interior			O	O	
Make sure that enough volume of grease is filled. Otherwise, fill the grease.	Ball screw shaft	O	O	O	O	Para. 2.7.3
	Reduction gear interior				O	Para. 2.8.3
Make sure that each timing belt is not worn out or cracked. Check each timing belt for tension.	Arm 2 interior			O	O	Para. 2.6.5
Make sure that the cables and air tubing are not worn out.	Cables and air tubing of each unit			O	O	Para. 2.2.3
Make sure by moving the axis by hand that each axis can be operated properly.	Entire robot			O	O	Para. 2.2.4

1.2.2 Inspection at Power ON (at Operation)

D: Daily inspection Q: Quarterly inspection
S: Semi-annual inspection A: Annual inspection

Description	Check position	D	Q	S	A	Refer to
Push each arm by hand in the servo ON condition to make sure that the arm is secured.	Each joint				O	
Make sure that no abnormal vibration or noise is caused.	Entire robot	O	O	O	O	
Check for the repeatability.	Entire robot		O	O	O	
Replace the batteries for position detection with new ones.	Battery box				O	Para. 2.9.2

1.2.3 Overhaul

Description	Maintenance parts	Every 5 years	Refer to
Check for the consumable parts and replace as necessary.	Reduction gears, ball screws, motors, belts	O	Para. 2.4 ~ 2.7
Replace the battery with a new one.	Backup battery in the controller	O	Para. 3.3.3
Replace the switching power supply unit with a new one.	In the controller	O	Para. 3.3.4

1.3 Maintenance Tools

We recommend to use the following tools for maintaining the robot and controller.

- Screwdrivers (Phillips head screwdriver and slotted screwdriver)
- Hexagonal wrench key set: M3 ~ M16
- Torque wrench set: M4 ~ M10
- Nippers

Recommended provisions other than the tools:

- Grease gun
- Waste cloth
- Vice (clamp)
- Liquid gasket (recommendation: 1211 made by ThreeBond)
- Seal tape
- Loctite adhesives (242; mid adhesive force)
- Belt tension meter
- TSPC program creation/teaching support software (option)

1.4 Maintenance Contract and Repair

1.4.1 Maintenance Contract

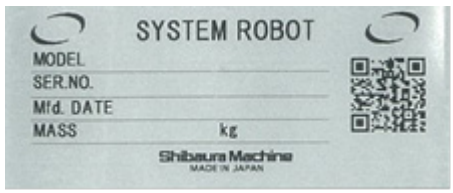
For the inspection items for maintenance and inspection to be done semi-annually and after, it is rather difficult for the user to execute, except for greasing and lubrication. When this happens, contact the SHIBAURA MACHINE Service Department. We recommend that the user conclude an after-sale service contract with us.

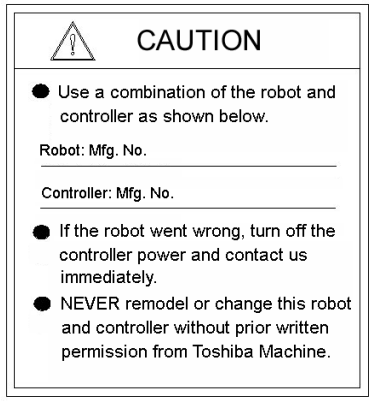
1.4.2 Repair

If a fault has occurred or if repair is necessary, turn off the controller power and contact the SHIBAURA MACHINE Service Department. At this time, advise us of the details of the fault and the following information stated on the robot and controller.

- Robot model
- Robot manufacture number
- Controller manufacture number
- Robot manufacture date
- Running hours

For the manufacture number and manufacture date, see the following labels attached to the main robot base.





1.4.3 Modification

The robot and controller MUST NOT be changed or disassembled without prior consent from SHIBAURA MACHINE.

CAUTION

- The user must NEVER replace or modify parts other than those described in the instruction manual. Otherwise, the performance may deteriorate or faults or accidents will be caused.

2. Maintenance of the Main Robot

2.1 Cautions on Maintenance and Inspection

When performing inspection and maintenance of the robot, strictly observe the following precautions to protect yourself and coworkers.

! DANGER

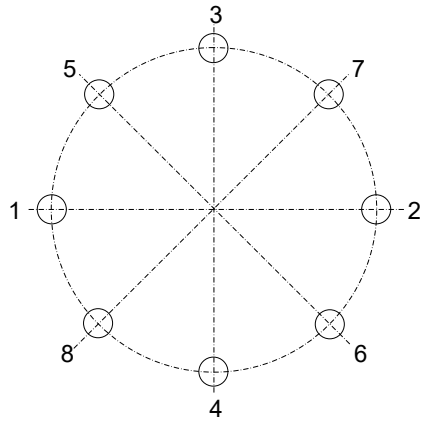
- Be sure to turn off the main power switch of the controller before approaching the robot for maintenance and inspection.
- When moving the robot by hand while the power is turned on, be sure to assure the safe work and effect an emergency stop beforehand.
- In the above situation, if the work is to be done while the axis 3 motor brake OFF switch is pressed, be sure to perform the work by two (2) persons.
  One person should carry out the work while the other is monitoring the work outside the hazardous zone. The latter person should watch the work and be ready to turn off the controller switch if any abnormality occurs. Once the controller power is turned off, the motor brake actuates even if the axis 3 motor brake OFF switch is kept pressed.
- If the axis 3 motor brake OFF switch is pressed while the robot is carrying a heavy workpiece, the axis 3 may drop suddenly. To avoid this, the customer should take all necessary measures by themselves.

2.1.1 Clamping Hexagon Socket Head Cap Screws

This robot uses hexagon socket head cap screws at places requiring mechanical strength. At the time of factory-assembly, lock adhesives are applied and each screw is tightened with the following clamping torque.
When these screws are tightened again according to the inspection and maintenance as stipulated in this manual, use a torque wrench, etc. to assure the appropriate clamping torque.
The places which do not fall under the table below are detailed in the paragraph of the replacement procedures.

Screw	M4	M5	M6	M8	M10
Clamping torque	4.7 N·m	8.8 N·m	17 N·m	27 N·m	54 N·m

For the screws arranged on a circle for mounting the reduction gear, etc., tighten them in the diagonal order, as shown below.



DO NOT tighten one (1) screw at a time. Tighten each screw in multiple steps, using a hexagonal wrench key, and secure with appropriate clamping torque by means of a torque wrench.

2.2 Details of Inspection

2.2.1 Check of Each Bolt (or Screw) for Clamping

(1) Tool Set Bolts

Make sure, using the hexagonal wrench key, that the hexagon socket head cap screws of the tool set flange, which are clamped to the tool shaft are tightened completely. If loosened, tighten them completely.

Make sure, using the hexagonal wrench key, that the hexagon socket head cap screws securing the tool flange and tool are tightened completely. Otherwise, tighten them completely.

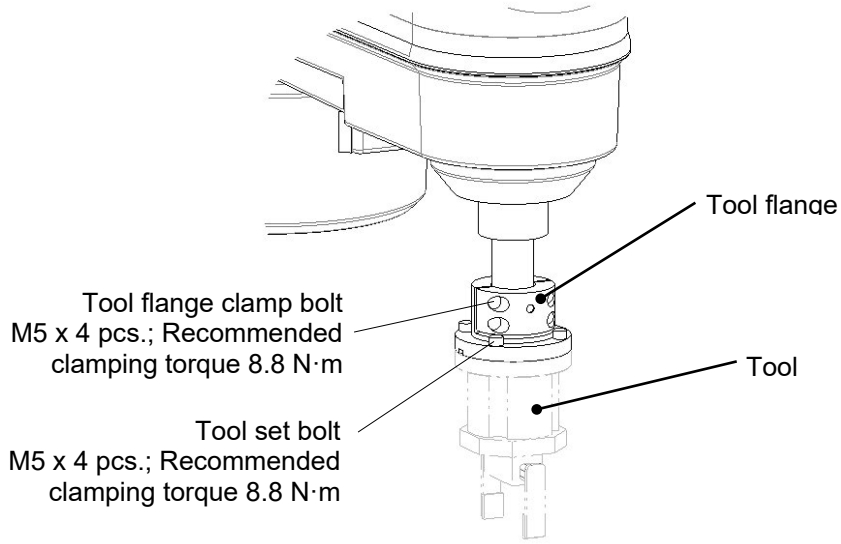


Fig. 2.1 Tool set bolts

(2) Robot Installation Bolts

Make sure, using the hexagonal wrench key, that the installation bolts of the main robot base are tightened completely. If loosened, tighten them completely.

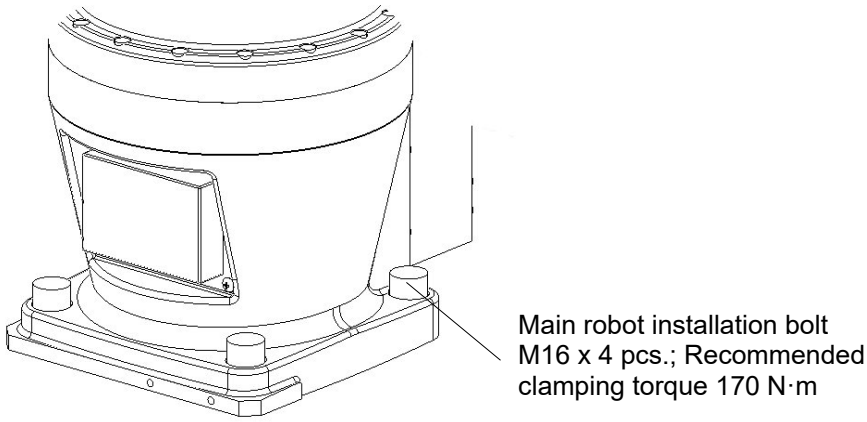


Fig. 2.2 Robot base set bolts

(3) Bolts for each joint

Make sure, using the hexagonal wrench key, that the bolts for the axis 1 and 2 joints (i.e., bolts for clamping each reduction gear and arm) are tightened completely. If loosened, tighten them completely.

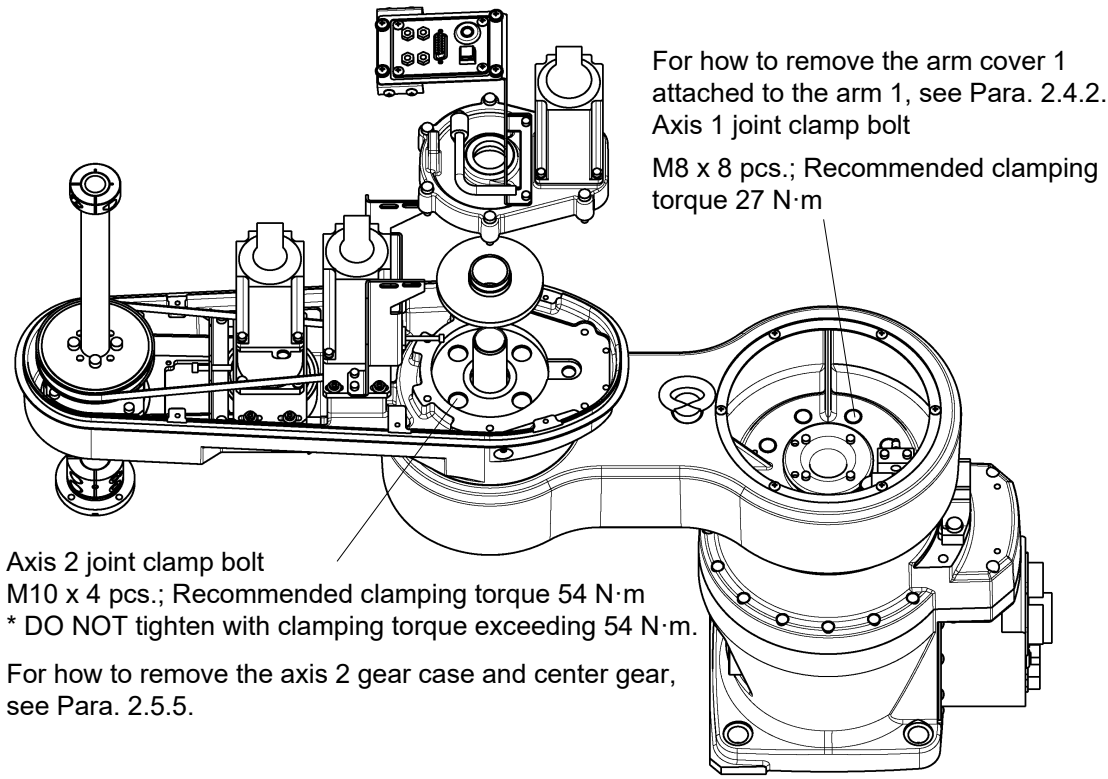


Fig. 2.3 Axes 1 and 2 set bolts

(4) Motor Set Bolts

Make sure, using the hexagonal wrench key, that the bolts for securing each axis drive motor are tightened completely. If loosened, tighten them completely. Also make sure that the motor and motor base set bolts are tightened completely. For the places where the following bolts are used, see Para. 2.5.

Axis 1 motor set bolt:	M5 × 20, 4 pcs.	Recommended clamping torque: 8.8 N·m
Axis 1 motor base set bolt:	M5 × 25, 4 pcs.	Recommended clamping torque: 8.8 N·m
Axis 2 motor set bolt:	M4 × 16, 4 pcs.	Recommended clamping torque: 4.7 N·m
Axis 2 gear case set bolt:	M6 × 35, 6 pcs.	Recommended clamping torque: 17 N·m
Axis 3 motor set bolt:	M4 × 16, 4 pcs.	Recommended clamping torque: 4.7 N·m
Axis 3 motor base set bolt:	M5 × 16, 4 pcs.	Recommended clamping torque: 8.8 N·m
Axis 4 motor set bolt:	M4 × 16, 4 pcs.	Recommended clamping torque: 4.7 N·m
Axis 4 reduction gear set bolt:	M5 × 16, 4 pcs.	Recommended clamping torque: 8.8 N·m
Axis 4 motor base set bolt:	M5 × 16, 4 pcs.	Recommended clamping torque: 8.8 N·m

For the axis 4 reduction gear set bolts and axis 4 motor base set bolts, see Para. 2.8.8 and Para. 2.6.4, respectively.

2.2.2 Check of Each Connector for Clamping

Make sure that the cable connectors connected to the main robot base are tightened completely. If loosened, tighten them completely.

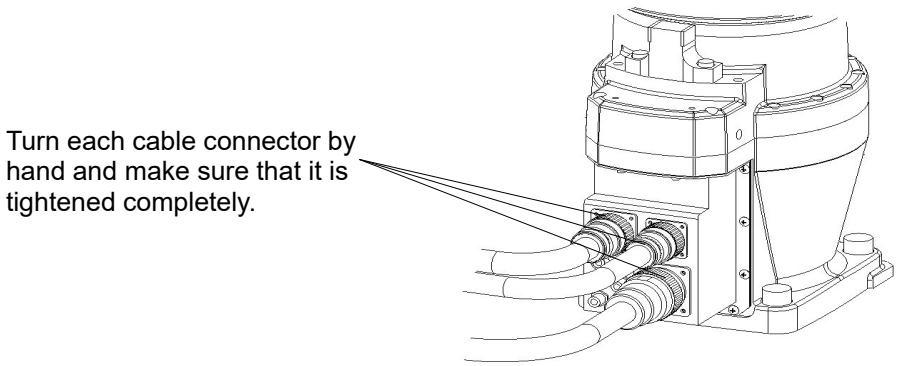


Fig. 2.4 Base connectors

2.2.3 Check of Each Cable and Air Tube for Abrasion

Disassemble the arm 2 cover and arm 1 covers (two (2) different covers are provided) and make sure that each cable and air tube are not worn out, broken or cracked. Especially, take careful precautions to inspect the vicinity of the cable outlets of the axis 1 joint and axis 2 joint.
For how to disconnect the covers, see Para. 2.4.

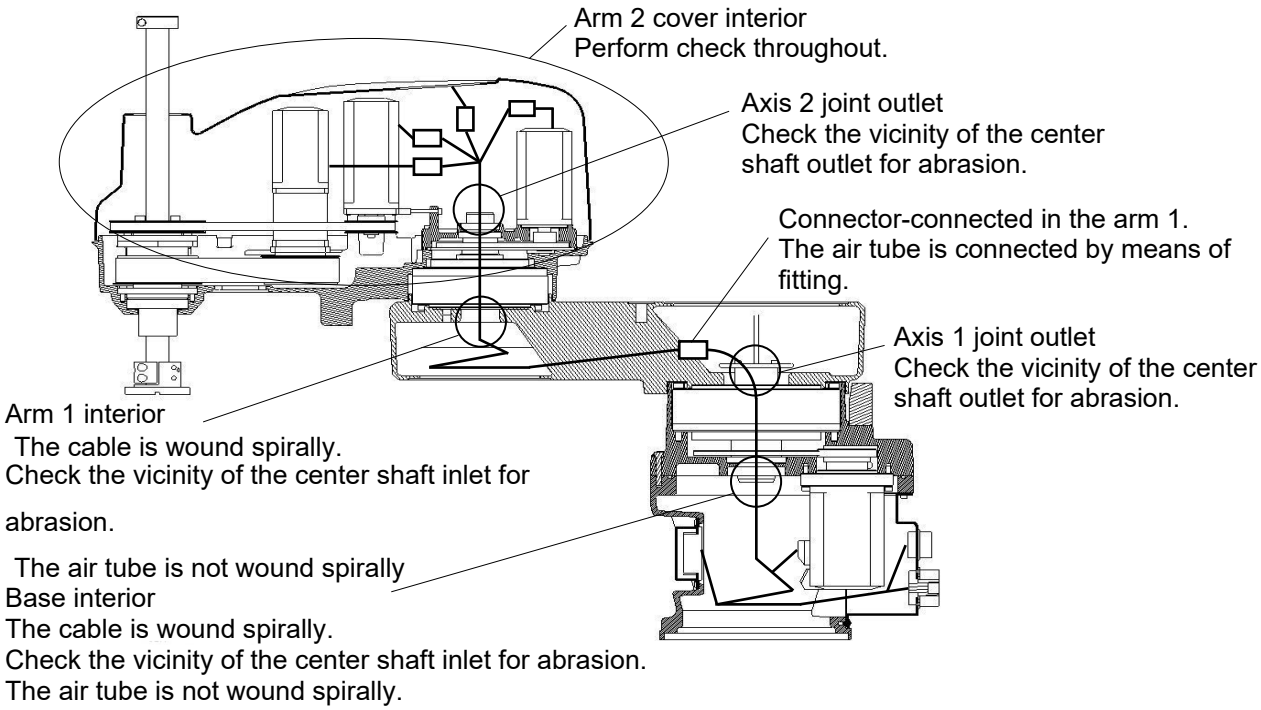


Fig. 2.5 Cable and air tube check points

2.2.4 Check of Each Axis for Operation

Turn on the main power switch of the controller, then keep the EMERGENCY pushbutton switch in the depressed condition. Move each axis by hand and make sure that it can move smoothly.
For the axes 3 and 4, when the brake OFF pushbutton switch is pressed, the brake is released. Take careful precautions at this time as the tool shaft may drop according to the weight of the hand and tool.

2.3 Layout of Robot Components and Drive Mechanism

The layout of the robot mechanical components is shown in Fig. 2.6.

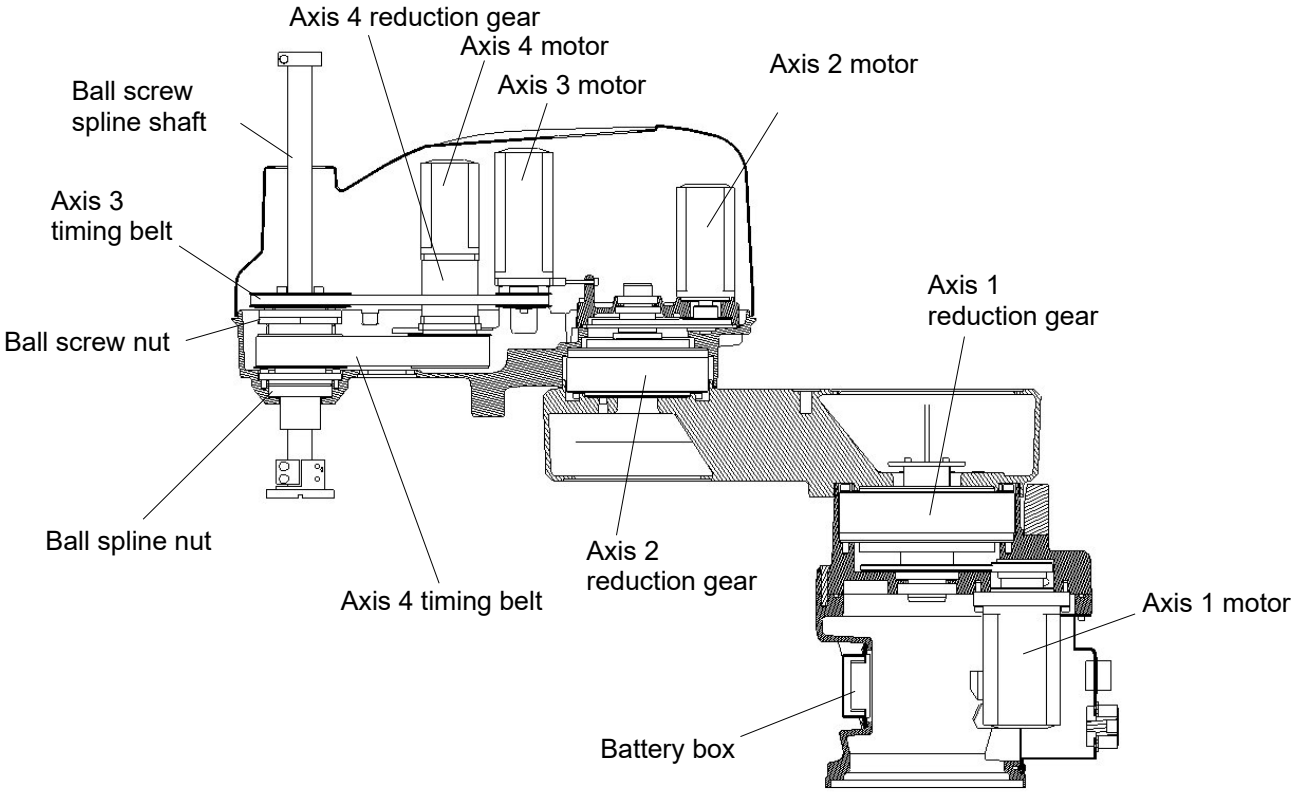


Fig. 2.6 Layout of robot mechanical components

The drive mechanism of each axis is described below.

1) Axis 1
   The axis 1 AC servo motor is mounted on the base.
   The motor revolution is input to the reduction gear via the input gear and reduction center gear. The output of the reduction gear directly controls the arm 1 motion.

2) Axis 2
   The axis 2 AC servo motor is mounted on the arm 2.
   The motor revolution is input to the reduction gear via the input gear and reduction center gear. The output of the reduction gear directly controls the arm 2 motion.

3) Axis 3
The axis 3 AC servo motor is mounted on the arm 2.
The motor revolution is input to the ball screw nut through the timing belt to move the ball screw spline up and down.

4) Axis 4
The axis 4 AC servo motor is mounted on the arm 2.
The motor revolution is input to the ball spline nut through the reduction gear and timing belt to revolve the ball screw spline shaft.

2.4 Dismounting and Mounting Each Cover

This paragraph describes the dismounting and mounting of the covers, which are common to the maintenance and replacement of each unit.

! DANGER

- When opening the cover, take careful precautions not to allow entry of moisture or contaminant into the robot. If the power is turned on while moisture or contaminant is left, you may get an electric shock or the robot may malfunction, which is very dangerous.

! CAUTION

- When mounting the arm 2 cover, take careful precautions not to catch any cable in it. If the cable is bent and pushed by force, it will be broken. Each cable is secured to the plate, etc. with TY-RAP. After the cover is disconnected, make sure of the cable layout and return the cables to natural wiring state.
- When the arm cover 2 is disconnected, you can see that the cables are wound spirally and connected to the arm 2 through the hollow shaft of the arm 2. After the cover is disconnected, make sure of the cable layout and return the cables as originally arranged, then mount the arm cover 2.
- The cables coming from the cover which also serves as the connector panel cover are wound spirally in the base, then connected to the arm 1 through the hollow shaft of the axis 1. After the cover is disconnected, make sure of the cable layout and return the cables as originally arranged, then mount the cover serving as the connector panel cover also.

2.4.1 Arm 2 Cover

The arm 2 cover is secured to the cover set bracket with four (4) cross truss head screws (M4 × 10). As the arm 2 cover is inserted into the arm 2, it can be lifted once the screws are removed. The inserted unit may not be disengaged easily, however. When this happens, it is recommended that the axis 3 be lowered beforehand until the ball screw stopper enters the cover so that you can hold the ball screw hole ("A" part in Fig. 2.7) when lifting up the cover. The packing attached to the cover can be disconnected together with the cover.

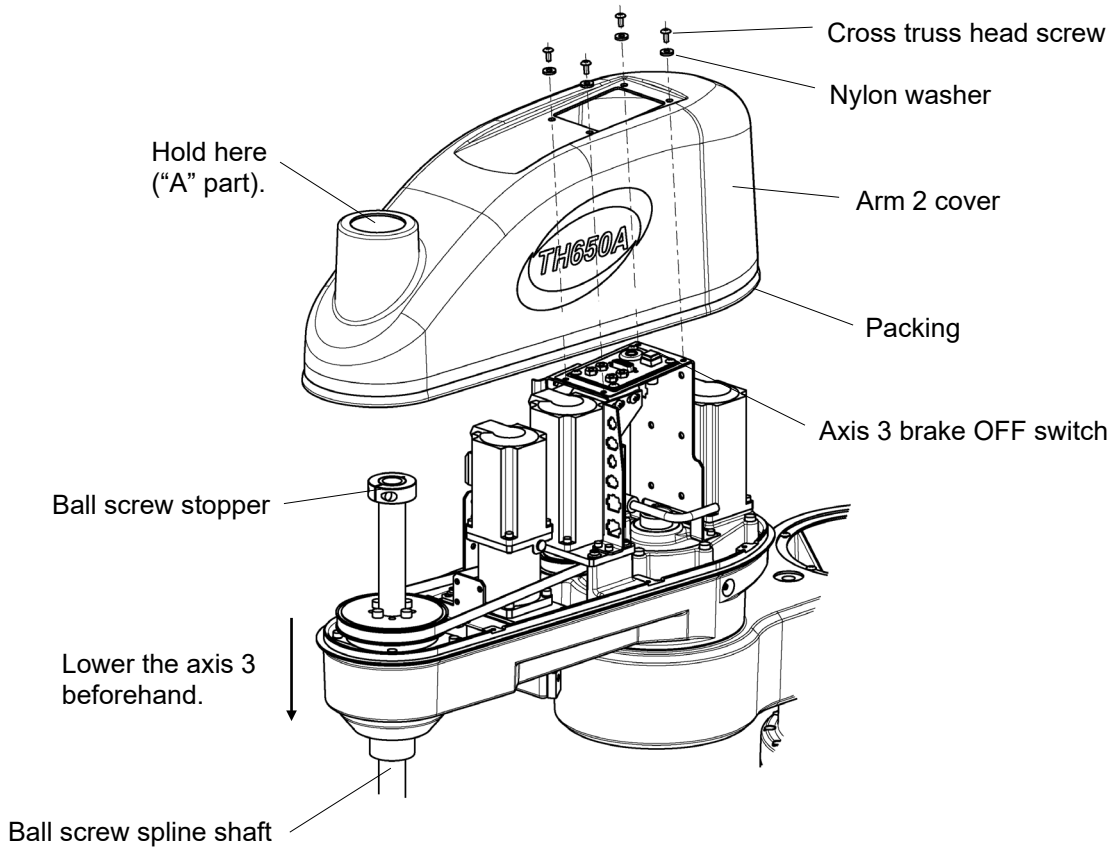


Fig. 2.7 Arm 2 cover

Nylon washers are used to protect the cover. When mounting the cover, be sure to set them. Also, take utmost care not to turn up the packing attached under the cover. After the cover is mounted, manually move up and down the ball screw shaft spline while pressing the brake OFF switch, and make sure that the ball screw hole will not interfere with the ball screw stopper.

2.4.2 Arm 1 Covers

The arm 1 covers are provided on top of the axis 1 (arm cover 1) and under the axis 2 (arm cover 2). Each cover is secured to the arm 1 with six (6) cross truss head screws (M4 × 6).

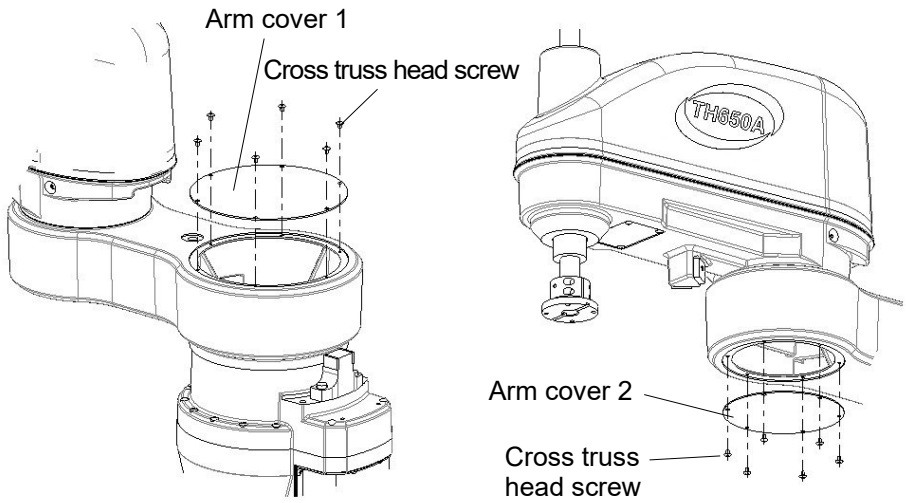


Fig. 2.8 Arm 1 covers

2.4.3 Base Covers

In all, two (2) base covers are provided; the cover also serving as the connector panel cover and cover also used as the battery box cover.
The former cover is secured to the base with nine (9) cross truss head screws (M4 × 6) and four (4) hexagon socket head cap screws (M4 × 6).
The latter cover is secured to the base with four (4) cross truss head screws (M4 × 6). Though each cover can be disconnected when the clamp bolts are removed, it is connected with the cables inside. DO NOT pull the cover by force.
The cover also serving as the connector panel cover is provided with packing. Take careful precautions when handling this cover.

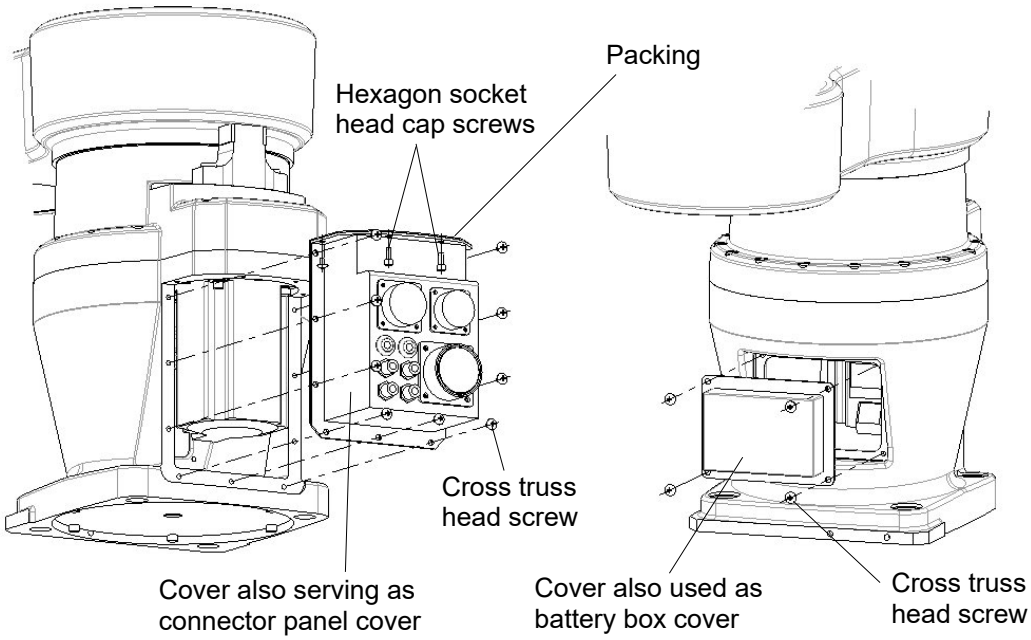


Fig. 2.9 Base covers

2.5 Replacing Motor

The motor is to be replaced by our service engineer. If it is replaced by the customer, we will not guarantee any consequential trouble or accident.

! DANGER

- The motor should be replaced with a new one only after the controller power is turned off with the power supply plug removed. If the work is done while the power is turned on, you may get an electric shock or the robot may malfunction, which is very dangerous.

! CAUTION

- When replacing the motor, take careful precautions not to exert a heavy impact on the motor shaft. Otherwise, the motor and encoder may be damaged.
- NEVER disassemble the motor and encoder. Otherwise, they cannot be used any further due to positional shift, etc.
- Once the motor has been changed, the mechanical home point (or origin) will shift and precise control will not be possible. To avoid this, home return operation is necessary after motor replacement.
  For the home return procedures, see Section 4.

2.5.1 Type of Motor

The motors employed in this robot are shown below. When you place an order for a new motor, make sure of the robot model (TH650A), axis name, type and our drawing number according to the following table.
As the input gears are used for the axes 1 and 2, a key is provided between each motor shaft and input gear. At motor replacement, key alignment is also necessary. It is recommended, therefore, that the input gear be replaced at the same time with motor replacement.
For the axis 3 also, a pulley on the motor side is driven by means of the key.
For our drawing numbers on the input gear and pulley and type of the key, also see the table below.

Description	Axis name	Type	Our drawing No.
AC servo motor	Axis 1	MSMD092S2U	S746326
	Axis 2	MSMD062S2U	S746327
	Axis 3	MSMD042S2V	S746336
	Axis 4	MSMD042S1C	S816701
Input gear	Axis 1		S816839
	Axis 2		S816840
Pulley	Axis 3		M198789
Key	Axis 1	Square key 6×6×20	
Key	Axes 2, 3	Square key 5×5×16	

2.5.2 Motor Locations

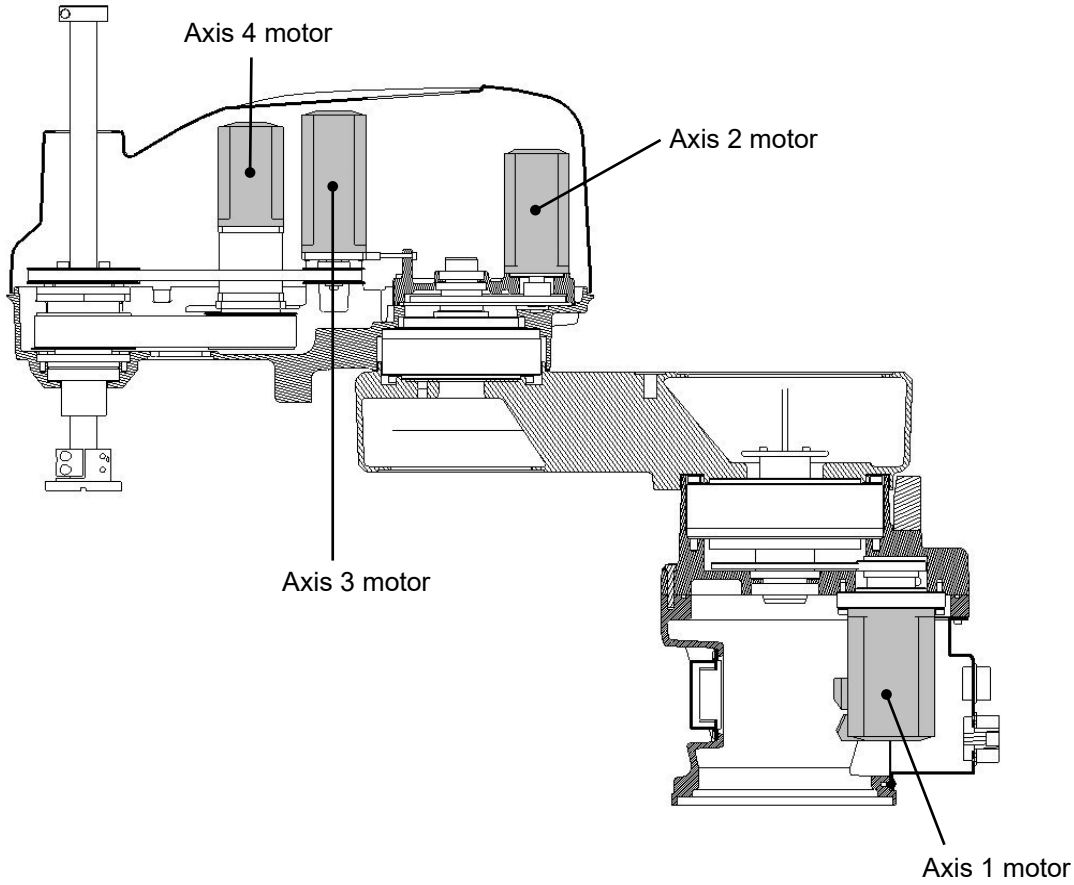


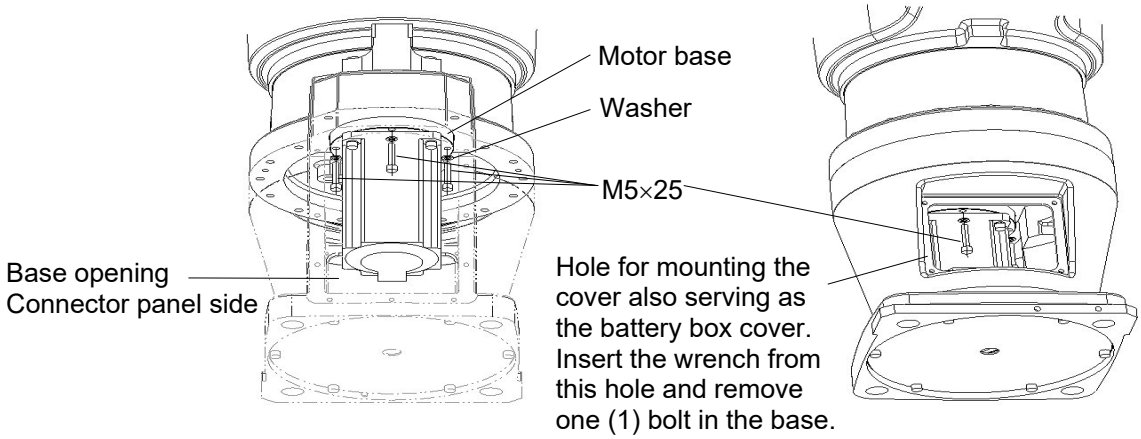
Fig. 2.10 Motor locations

2.5.3 Dismounting Axis 1 Motor

1) Disconnect the two (2) covers attached to the base. (See Para. 2.4.3.)

2) Disconnect connectors J1AS and J1AP (power drive cables) and connectors J1BS and J1BP (encoder cables) of the axis 1 motor.

3) Remove the four (4) bolts (M5 × 25) securing the motor base, then draw the motor downward. At this time, insert the hexagonal wrench key from the front side hole where the cover also serving as the battery box cover is attached, then remove the one (1) bolt in the base.

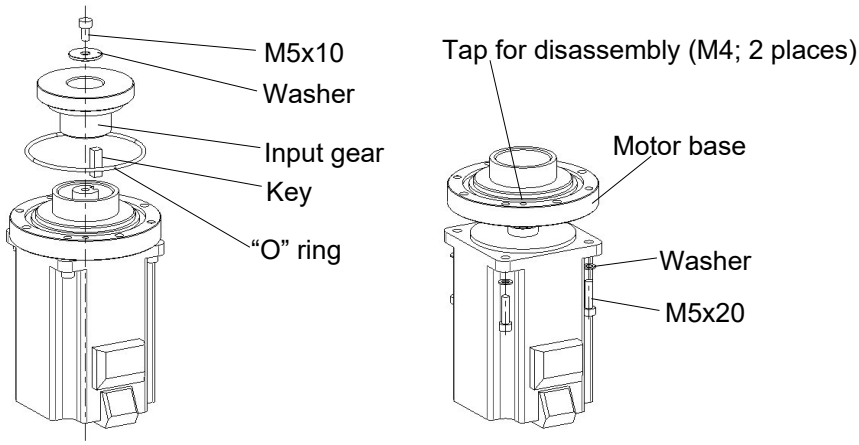


4) Remove the disconnected motor base from the base opening.
At this time, use a waste cloth, etc. because the input gear is grease-lubricated. Take careful precautions not to stain the cables with grease.

5) Secure the input gear, using a vice, etc. and remove the bolt (M5 × 10) at the shaft end. When securing the input gear, use a shock-resistant material such as waste cloth in such a manner that the tooth top or tooth surface is not cut or scratched. After removing the bolt, draw the washer and input gear upward. Be sure to remove the key and “O” ring.

! CAUTION

- When dismounting the input gear, DO NOT knock it with excessive force. Otherwise, the encoder may be damaged.



6) Remove the four (4) bolts (M5 × 20) securing the motor.

7) Disengage the motor from the motor base. As the motor set surface is applied with liquid gasket, disengage them, using the taps for disassembly (M4; two (2) places).

2.5.4 Mounting Axis 1 Motor

1) Perform key alignment, using the new motor and input gear. At this time, no clearance should exist between the motor shaft and input gear.

! CAUTION

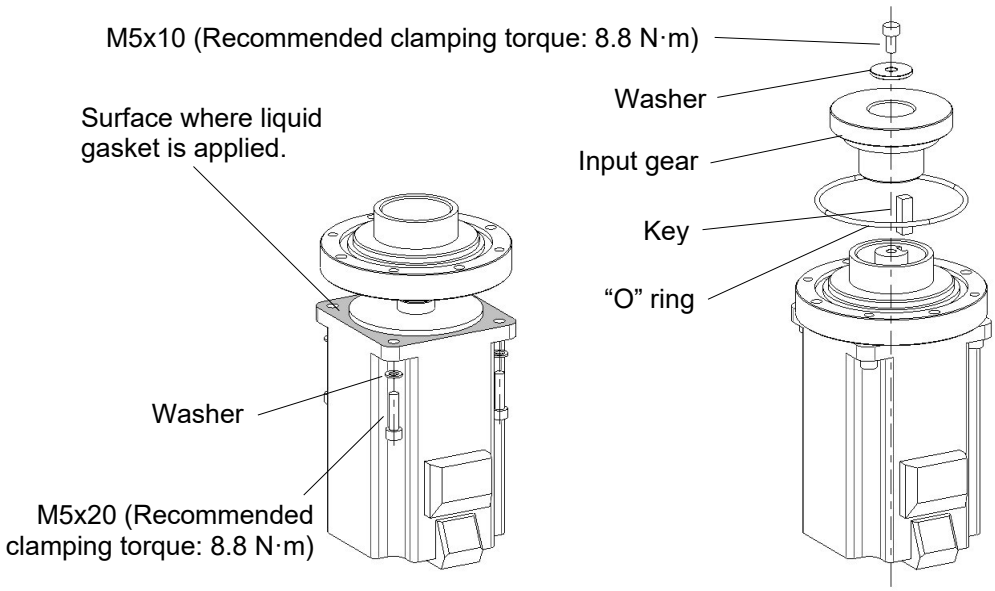
- Perform key alignment very carefully. If there is a clearance between the motor shaft and input gear, an abnormal noise may be heard at the time of operation. Also, positioning accuracy will drop and the life of the parts will shorten.

2) Mount the motor on the motor base with four (4) bolts (M5 × 20). At this time, apply the liquid basket to the motor set surface.

3) Mount the input gear and key, then secure them with the bolt (M5 × 10) and washer. Secure the input gear, using a vice, etc. and tighten the bolt (M5 × 10) at the shaft end. When securing the input gear, use a shock-resistant material such as waste cloth in such a manner that the tooth top or tooth surface is not cut or scratched.

! CAUTION

- When mounting the input gear, DO NOT knock it with excessive force. Otherwise, the encoder may be damaged.



4) Set the "O" ring in the "O" ring groove of the motor base to secure it to the base.

5) To secure the motor base, follow Steps 3) and 4) of Para. 2.5.3 in the reversed order.

6 Connect the axis 1 motor connectors (for the power drive cable and encoder cable).

7) Return the cables in the base as originally arranged, then attach the cover to the base.

8) After the axis 1 motor is replaced with a new one and before the power is turned on, move the arm 1 by hand to make sure that no abnormal noise is heard.

9) Turn the power on and perform home setting for the axis 1. Now, the motor replacement completes.
(Check for the coordinates of the axes 2, 3 and 4, and perform home setting when necessary.)

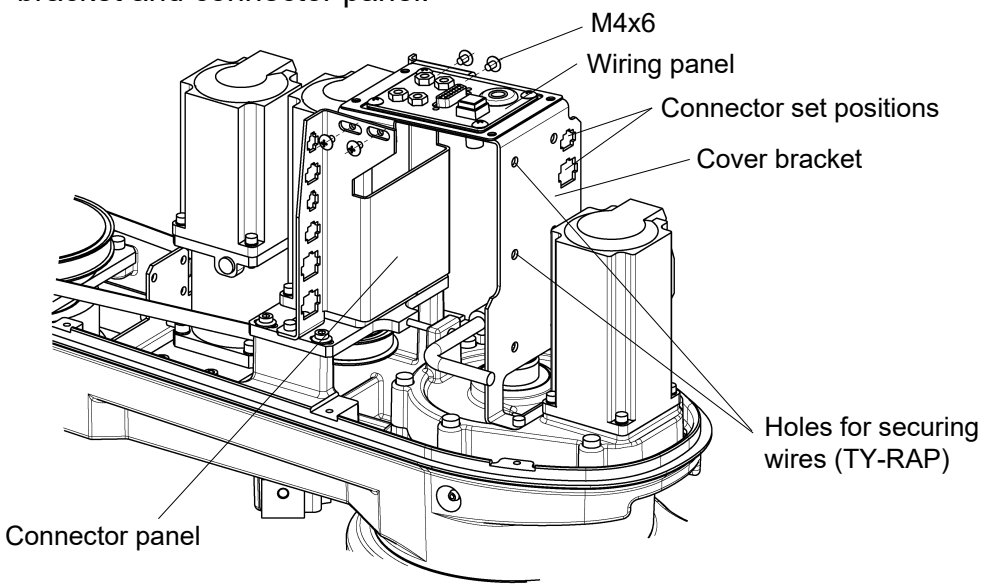
! CAUTION

- Be sure to mount the “O” ring and apply the liquid gasket. Otherwise, grease will leak from the motor base set surface.

2.5.5 Dismounting Axis 2 Motor

1) Disconnect the arm 2 cover. (See Para. 2.4.1.)

2) Cut the TY-RAP for the cables secured to the cover bracket, using nippers, etc., and remove the four (4) cross truss head screws (M4 × 6) securing the cover bracket and connector panel.



3) Disconnect connectors J2AS and J2AP (power drive cables) and connectors J2BS and J2BP (encoder cables) of the axis 2 motor.

4) Disconnect all connectors of the cables coming from the wiring panel, and air tubes. (JOEP, JOFP and JOES, JOFS (CN0), brake OFF switch cable, four (4) air tubes)

5) Loosen and remove the two (2) bolts (M4 × 8) securing the cover bracket.

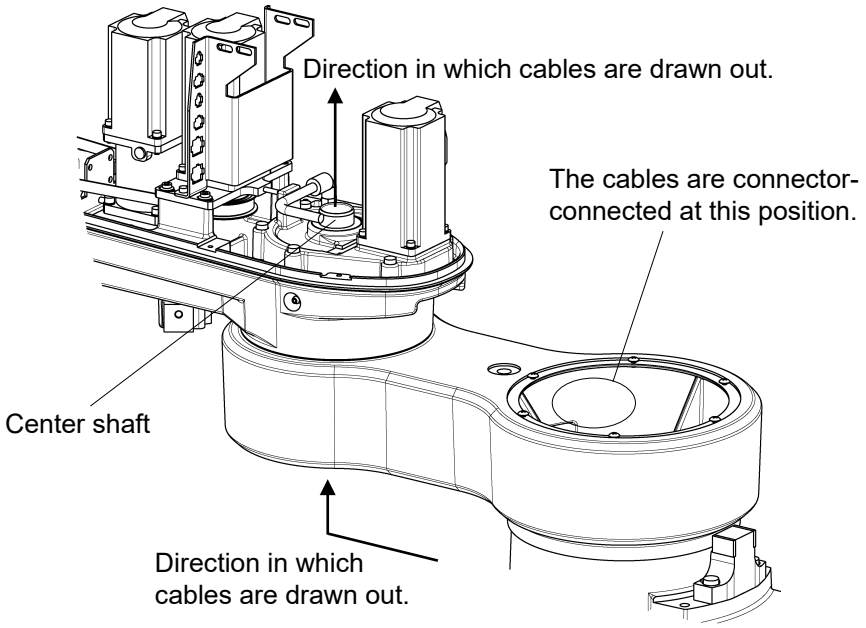
6) Disconnect the both arm 1 covers. (See Para. 2.4.2.)

7) As the cables are connector-connected on the upper side of the axis 1, disconnect all connectors. Also, disconnect the air tubes at the quick-operated joints.

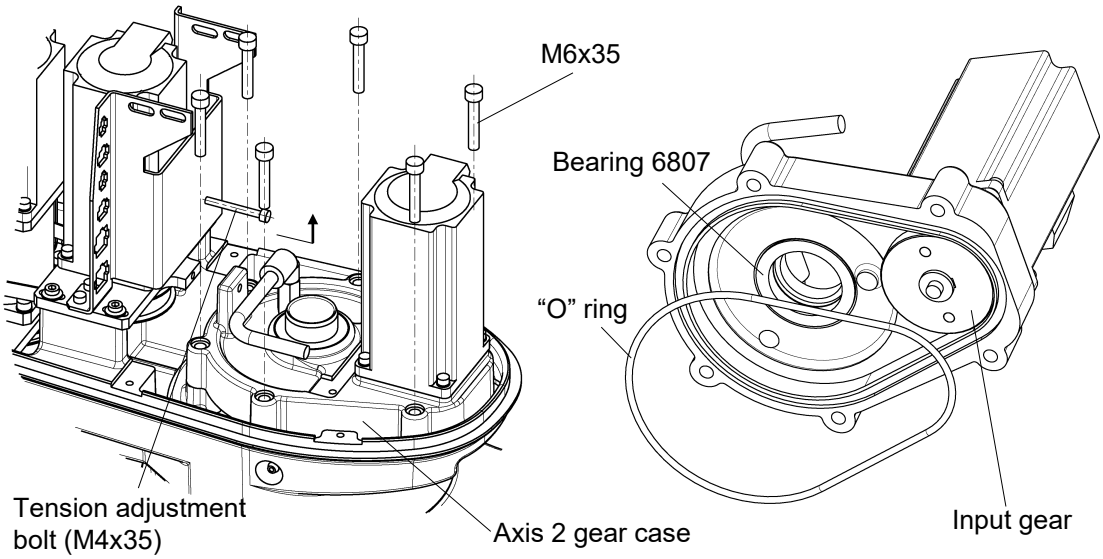
8) Draw out the cables and air tubes thus disconnected up to the upper side of the arm 2.
As the inner diameter of the center shaft is small, observe the following order, taking careful precautions not to damage the cables and connectors when disconnecting the cables.

Air tubes → Power drive cables (small connectors) → Encoder cables (large connectors)

Especially, when disconnecting the encoder cable connectors, pull them out one by one so as to arrange them vertically. DO NOT pull them out by force.



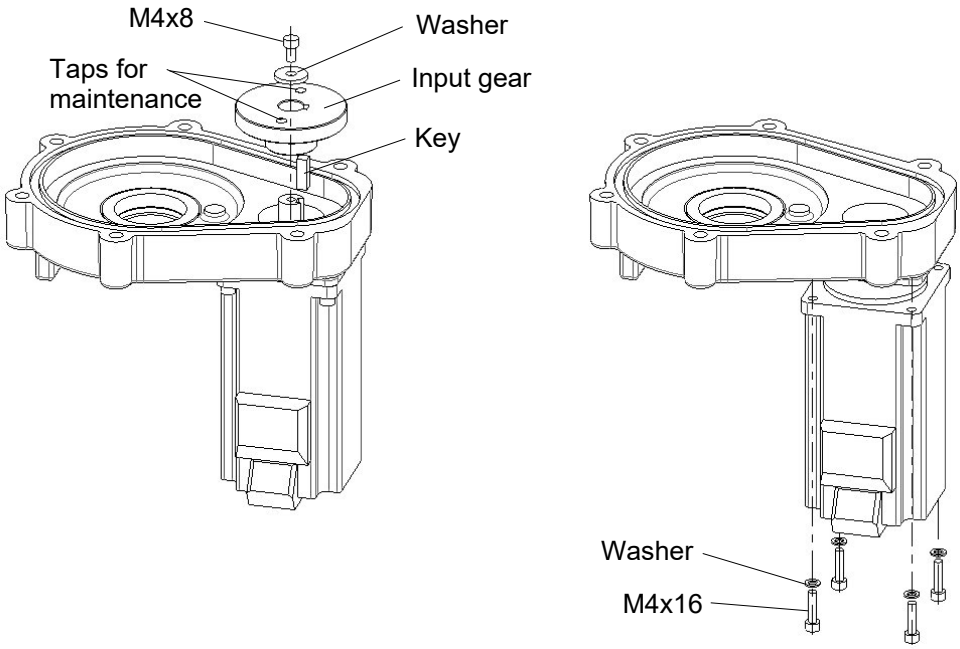
9) Remove the six (6) bolts (M6 × 35) securing the axis 2 gear case. Remove the axis 3 tension adjustment bolt (M4 × 35) also. When dismantling the gear case, provide a waste cloth, etc. as the grease is filled in the gear case. When this happens, be sure to remove the “O” ring. There is no problem when the bearing remains set.



10) The taps (M5) for maintenance are provided on the input gear. Secure the input gear, using them and remove the bolt (M4 × 8) at the shaft end.

11) Draw the input gear and key upward. When they cannot be drawn out easily, screw the bolts into the maintenance taps, then draw out the input gear.

12) Remove the four (4) bolts (M4 × 16) securing the motor.



13) Disengage the motor from the gear case. As the motor set surface is applied with liquid gasket, the motor may be hard to disconnect. Disconnect the motor while taking careful precautions not to exert an excessive impact on it.

2.5.6 Mounting Axis 2 Motor

1) Perform key alignment, using the new motor and input gear. At this time, no clearance should exist between the motor shaft and input gear.

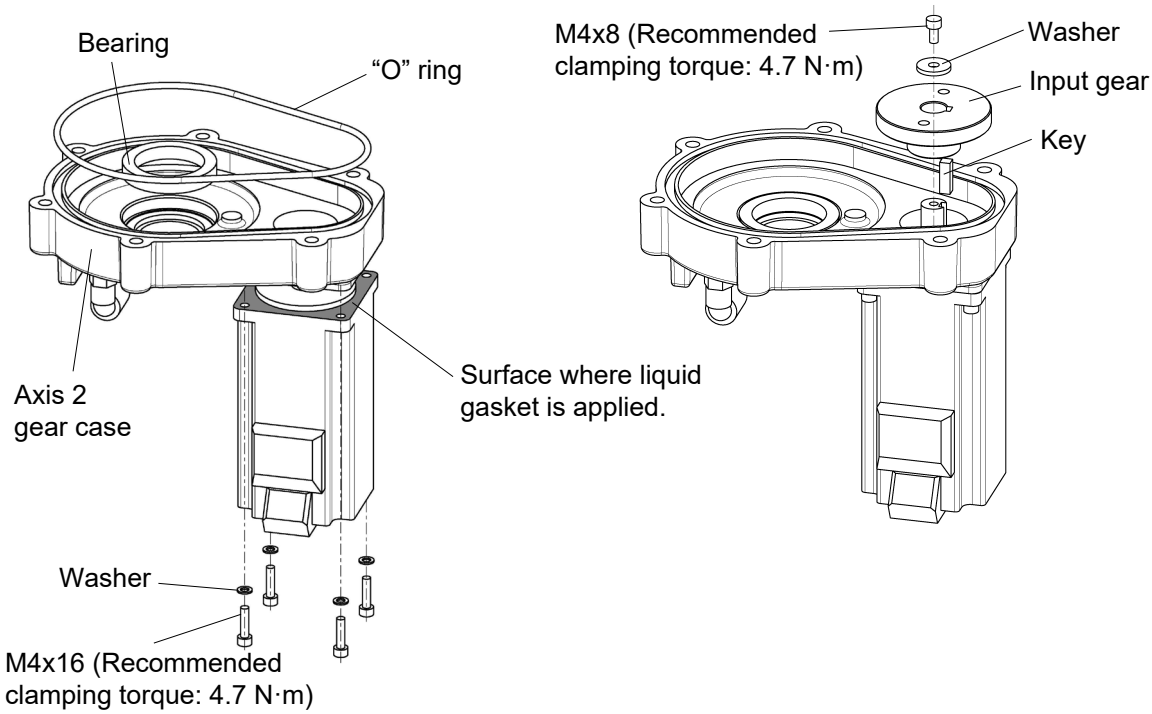
! CAUTION

- Perform key alignment very carefully. If there is a clearance between the motor shaft and input gear, an abnormal noise may be heard at the time of operation. Also, positioning accuracy will drop and the life of the parts will shorten.

2) Mount the motor on the gear case with four (4) bolts (M4 × 16). At this time, apply the liquid basket to the motor set surface.

3) Mount the input gear and key, then secure them with the bolt (M4 × 8) and washer. When this happens, use the taps for maintenance (M5) which are machined on the input gear.

4) Set the “O” ring and bearing in the gear case.



! CAUTION

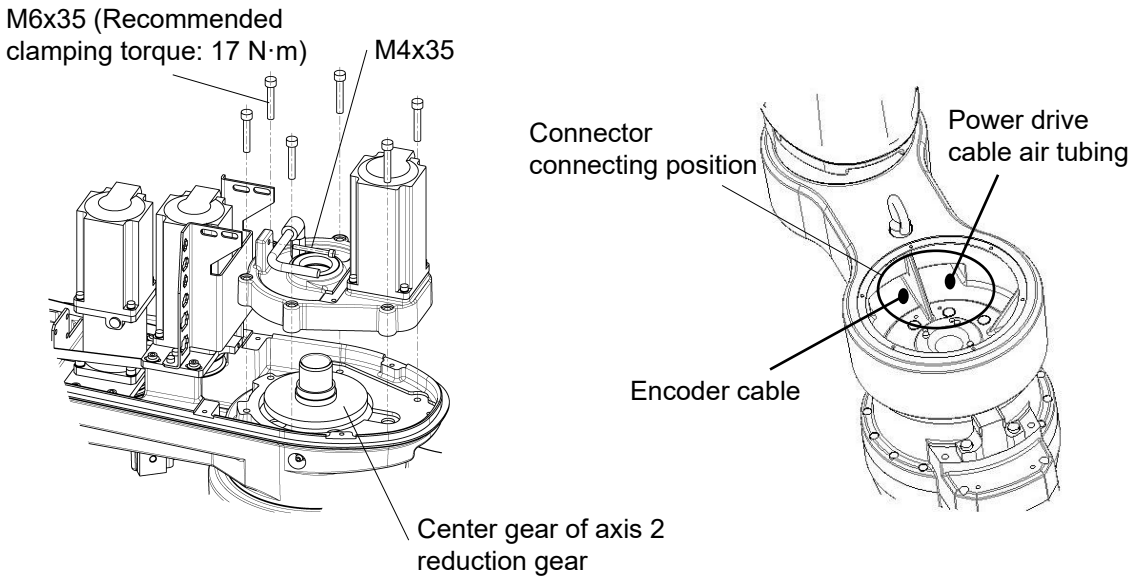
- Be sure to attach the “O” ring and apply the liquid gasket. Otherwise, grease will leak from the gear case set surface and motor set surface.

5) Attach the gear case to the arm 2 with six (6) bolts (M6 × 35). Make sure that the center gear of the axis 2 reduction gear is mounted. Mount the axis 3 tension adjustment bolt (M4 × 35) then. (Adjustment of the axis 3 tension is unnecessary.)

6) Pass the disconnected cables and air tubes up to the connector connecting position on the upper side of the arm 1.
As the inner diameter of the center shaft is small, observe the following order when passing the cables.

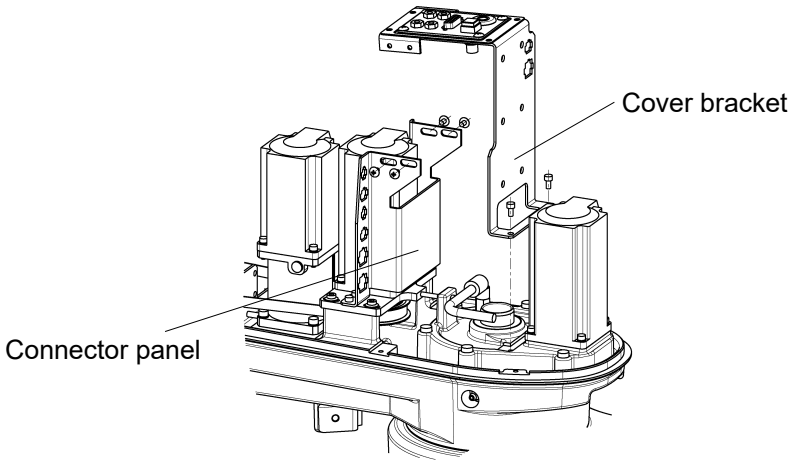
Encoder cables (large connectors) → Power drive cables (small connectors) → Air tubes

Especially, when passing the encoder cable connectors, pass them one by one so as to arrange them vertically. DO NOT pull them out by force.
In the arm 1, the encoder cables and power drive cables are separated on the right and left sides. See the figure below.



7) Connect the cables and air tubes thus passed as originally arranged.

8) Mount the cover bracket, and secure it together with the connector panel.



9) Connect the axis 2 motor connectors (for power drive cable and encoder cable).

10) Return the cables in the arm 2 as originally arranged, then attach all covers.

11) After the axis 2 motor is replaced with a new one and before the power is turned on, move the arm 2 by hand to make sure that no abnormal noise is heard.

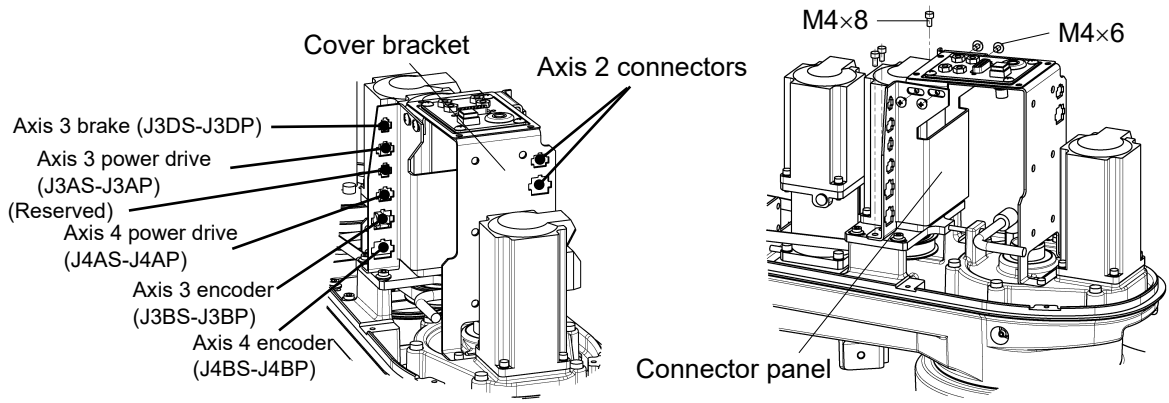
12) Turn the power on and perform home setting for the axes 2, 3 and 4. Now, the motor replacement completes.

2.5.7 Dismounting Axis 3 Motor

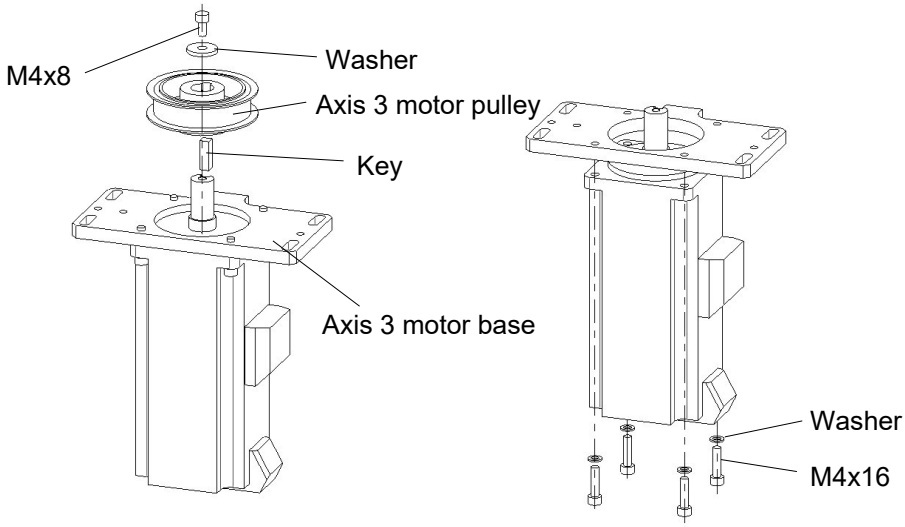
! CAUTION

- The axis 3 motor is provided with a brake. At replacement of the axis 3 motor, this brake becomes inoperative. Before starting the work, therefore, move down the ball screw spline to the lower limit. Otherwise, the shaft will drop due to the dead weight of the shaft or workpiece, and your hand or finger may be caught.

1) Remove the arm 2 cover. (See Para. 2.4.1.)
2) Disconnect all connectors for the axes 3 and 4, which are connected to the connector panel. For the connector layout, see the figure below.
3) Remove the four (4) cross truss head screws (M4 × 6) securing the cover bracket and connector panel, and the three (3) bolts (M4 × 8) securing the connector panel to dismount the connector panel.



4) Remove the axis 3 tension adjustment bolt (M4 × 35) and loosen the flange head bolts (M5 × 16) securing the axis 3 motor base to cancel the axis 3 timing belt tension. Then draw out the motor together with the motor base upward.
5) Secure the axis 3 motor pulley, using a vice, etc. and remove the bolt (M4 × 8) at the shaft end. When securing the pulley, take careful precautions not to distort the pulley flange and not to damage the pulley.
6) After removing the bolt, draw out the washer, pulley and key.
7) Remove the four (4) bolts (M4 × 16) securing the motor, and separate the axis 3 motor from the motor base.



2.5.8 Mounting Axis 3 Motor

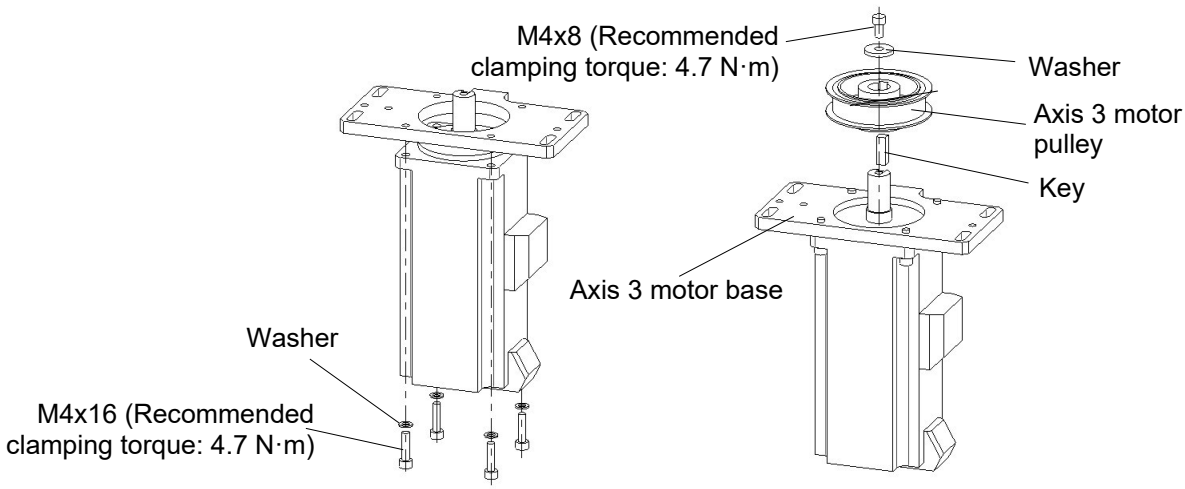
1) Perform key alignment, using the new motor and input gear. At this time, no clearance should exist between the motor shaft and pulley.

! CAUTION

- Perform key alignment very carefully. If there is a clearance between the motor shaft and pulley, positioning accuracy will drop and the life of the parts will shorten.

2) Mount the motor on the axis 3 motor base with four (4) bolts (M4 × 16).

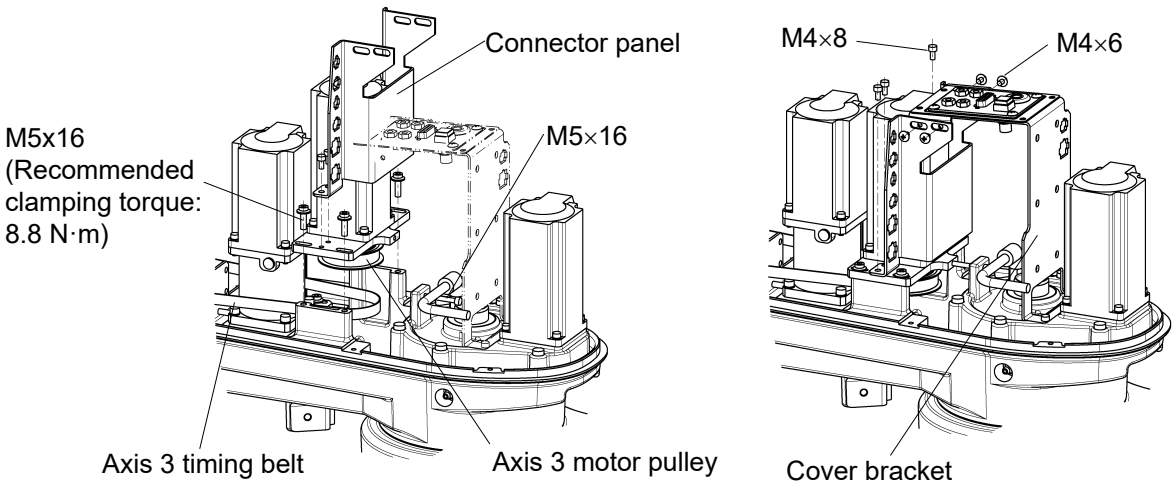
3) Mount the axis 3 motor pulley and key, then secure them with the washer and bolt (M4 × 8).



4) Temporarily clamp the motor base to the arm 2 with four (4) flange head bolts (M5 × 16). Pass the axis 3 timing belt to the axis 3 motor pulley, then exert a tension, using the axis 3 tension adjustment bolt (M4 × 35).
(For the belt replacement procedures and tension adjustment value, see Para. 2.6.3.) Then completely tighten and secure the bolts which were set temporarily.

5) Mount the connector panel, connect the connectors, then secure the connector panel and cover bracket.

6) Mount the arm 2 cover and perform home setting for the axes 3 and 4. Now, the motor replacement completes.



7) Carry out a test operation of the axis 3 and make sure that each part operates properly.

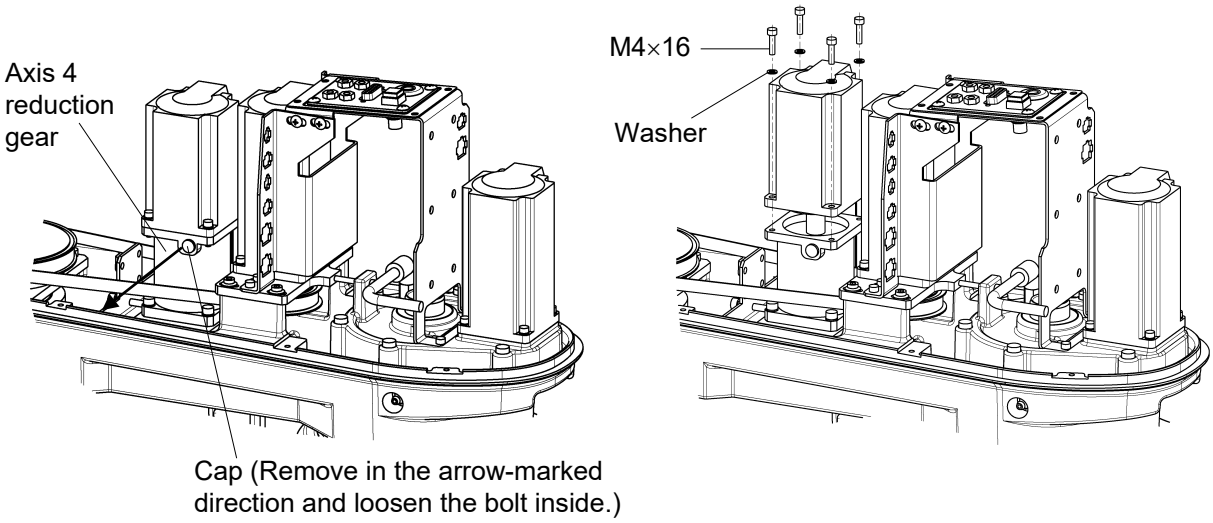
2.5.9 Dismounting Axis 4 Motor

1) Remove the arm 2 cover. (See Para. 2.4.1.)

2) Disconnect connectors J4AS and J4AP (power drive cables) and connectors J4BS and J4BP (encoder cables) of the axis 4 motor. For the connector layout, see Para. 2.5.7.

3) Remove the cap attached to the lateral side of the axis 4 reduction gear and loosen the coupling bolt (M4 × 12) securing the axis 4 motor shaft and input shaft of the axis 4 reduction gear.
If the phases disagree, turn the ball screw shaft by hand to align the phases.

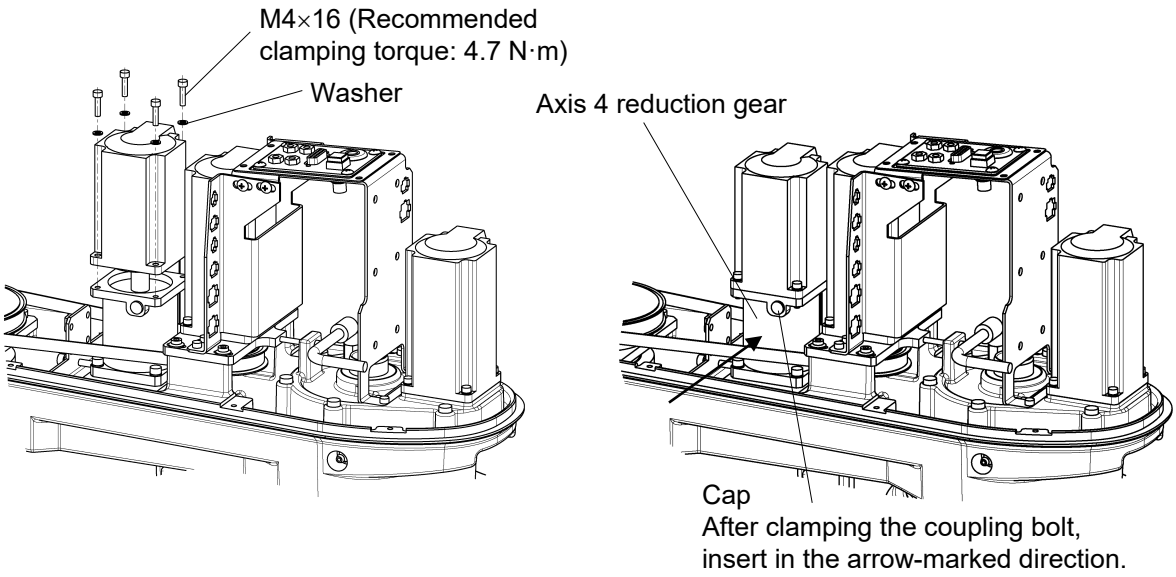
4) Remove the four (4) bolts (M4 × 16) securing the motor, then draw out the motor upward.



2.5.10 Mounting Axis 4 Motor

1) Mount the new motor on the axis 4 reduction gear with four (4) bolts (M4 × 16).

2) Tighten the coupling of the axis 4 reduction gear by means of the attached bolt (M4 × 12) to mount the cap. (Recommended clamping torque: 4.7 N·m)



3) Connect the axis 4 motor connectors.

4) Mount the arm 2 cover and perform home setting for the axes 3 and 4. Now, the motor replacement completes.
After the axis 4 motor has been replaced, be sure to perform home setting for the axis 3.

5) Carry out a test operation of the axis 4 and make sure that each part operates properly.

2.6 Adjusting and Replacing Timing Belt

The timing belt is to be replaced by our service engineer. If it is replaced by the customer, we will not guarantee any consequential trouble or accident.

! DANGER

- The timing belt should be replaced with a new one only after the controller power is turned off with the power supply plug removed. If the work is done while the power is turned on, you may get an electric shock or the robot may malfunction, which is very dangerous.

! CAUTION

- The axis 3 motor is provided with a brake. At replacement of the axis 3 timing belt, this brake becomes inoperative. Before starting the work, therefore, move down the ball screw spline to the lower limit. Otherwise, the shaft will drop due to the dead weight of the shaft or workpiece, and your hand or finger may be caught.
- Because the timing belt is disconnected, the mechanical home point will shift and proper control cannot be done. To avoid this, home return operation is necessary after replacement of the timing belt. For the home return procedures, see Section 4.

2.6.1 Type of Timing Belt

The timing belts used in this robot are shown below.
When you place an order for the new timing belt, specify the robot model (TH650A), axis name, type and our drawing number.

Description	Axis name	Width	Type	Our drawing No.
Timing belt	Axis 3	10 mm	705–UP–3M–10–HC	S816773
	Axis 4	30 mm	575–UP–5M–30–HC	S816774

2.6.2 Timing Belt Locations

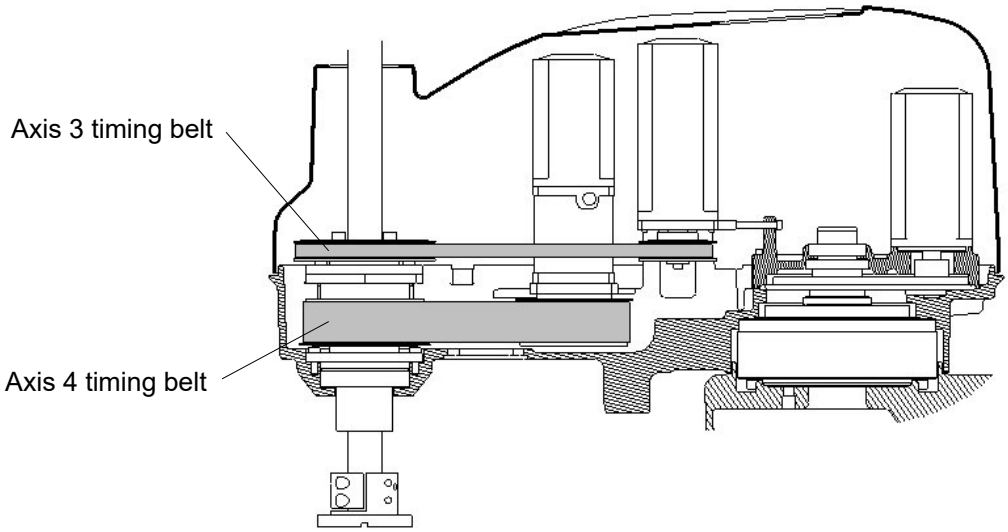


Fig. 2.11 Timing belt locations

2.6.3 Replacing Axis 3 Timing Belt

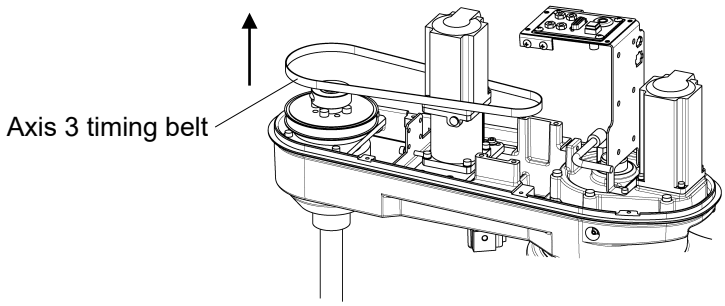
1) Remove the arm 2 cover. (See Para. 2.4.1.)

2) Disconnect the axis 3 motor base which remains attached with the motor and pulley. For the disconnecting procedures, see Para. 2.5.7.

! CAUTION

- The axis 3 motor is provided with a brake. At replacement of the axis 3 timing belt, this brake becomes inoperative. Before starting the work, therefore, move down the ball screw spline to the lower limit. Otherwise, the shaft will drop due to the dead weight of the shaft or workpiece, and your hand or finger may be caught.

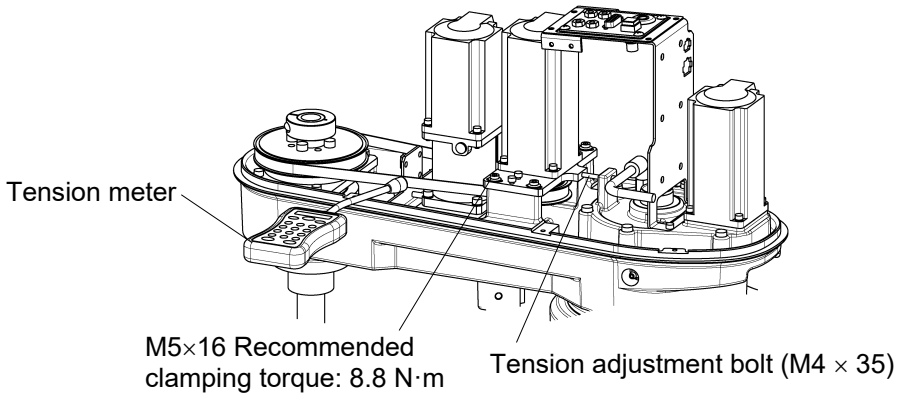
3) Disconnect the axis 3 timing belt.



4) Mount the new timing belt.

5) Temporarily clamp the motor set which was dismounted in Step 2) above to the arm 2 with four (4) flange head bolts (M5 × 16). Pass the axis 3 timing belt to the axis 3 motor pulley, then exert a tension, using the axis 3 tension adjustment bolt (M4 × 35).
While pulling the axis 3 motor base by means of the axis 3 tension adjustment bolt, measure the tension on the tension meter. At a position where the tension has reached 49 to 57 N, tighten the flange head bolts.
The values used for the tension meter are tabled below.

Value of tension [N]	Unit mass [g/m]	Belt width [mm]	Span [mm]
49 ~ 57	2.2	10	232



6) Connect the connectors and perform home point setting for the axis 3.

7) Carry out a test operation of the axis 3 and make sure that each part operates properly.

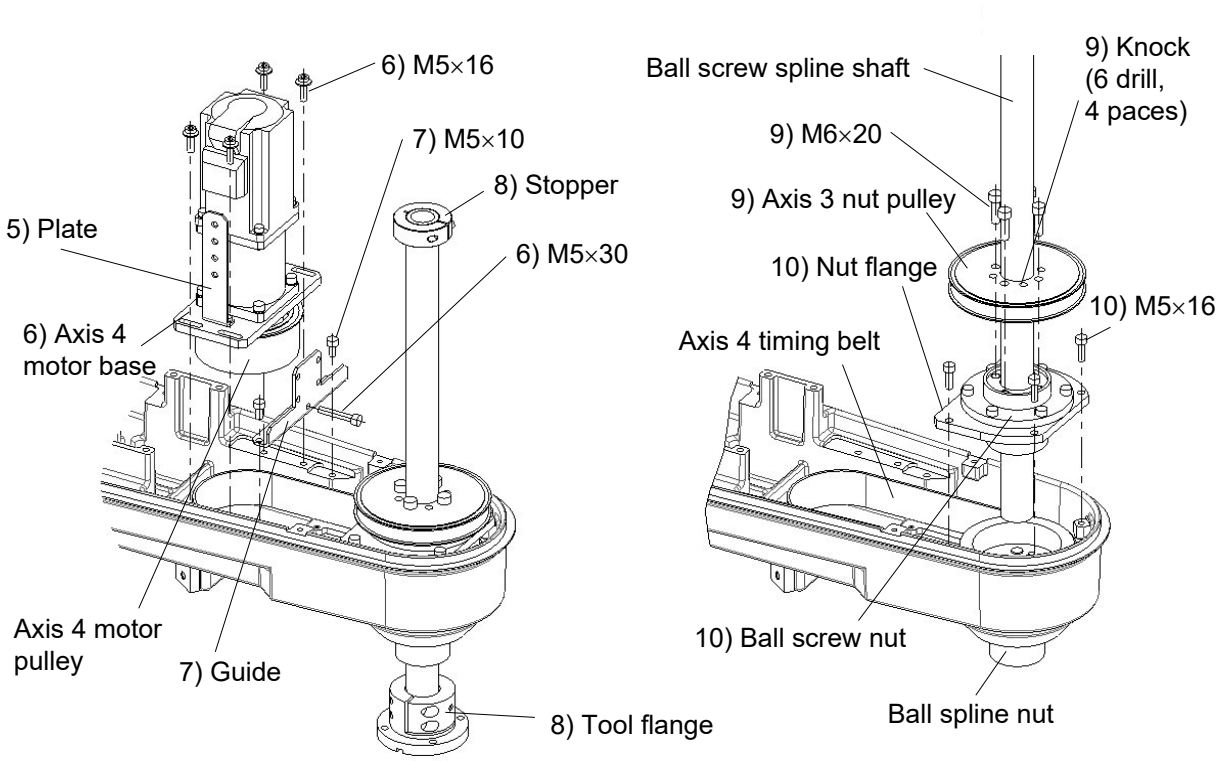
8) Mount the arm 2 cover. Now replacement of the axis 3 timing belt completes.

2.6.4 Replacing Axis 4 Timing Belt

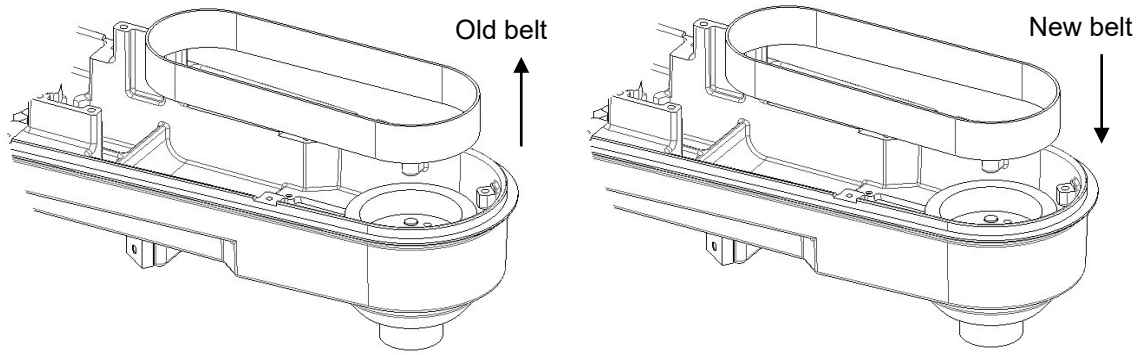
! CAUTION

- When the axis 4 timing belt is replaced with a new one, the axis 3 should be disassembled also due to the structure. Therefore, strictly observe the cautions on replacement of the axis 3 timing belt and motor also.
- When the ball screw nut integrated with the ball screw spline shaft is disconnected, take utmost care not to cause the ball screw spline shaft to come off. Otherwise, the ball in the ball screw nut will drop and the ball screw nut integrated with the ball screw spline shaft cannot function any further.

1) Remove the arm 2 cover. (See Para. 2.4.1.)
2) Disconnect all connectors which are set to the connector plate.
3) Disconnect the axis 3 motor base which remains attached with the motor and pulley. For the disconnecting procedures, see Para. 2.5.7.
4) Dismount the axis 3 timing belt.
5) Cut the TY-RAP for the cables secured to the axis 4 motor base plate.
6) Loosen the axis 4 tension adjustment bolt (M5 × 30), remove the flange head bolts (M5 × 16) securing the axis 4 motor base, then pull out the motor base upward.
7) Remove the two (2) bolts (M5 × 10) securing the guide to disconnect the guide.
8) Remove the tool flange and stopper. (The hand, etc. should be disconnected beforehand.)
9) Disconnect the axis 3 nut pulley secured to the ball screw nut. When removing the clamp bolts (M6 × 20), the pulley will rotate. Secure the pulley, using the holes (6mm-dia. drill; four (4) places) machined on the pulley.
10) Remove the four (4) bolts (M5 × 16) securing the nut flange, then carefully draw out the ball screw nut integrated with the shaft upward. If the ball screw nut integrated with the shaft is drawn out diagonally with an unusually large force, the ball in the ball screw nut may drop.



11) Draw out the axis 4 timing belt upward, then set a new belt.



12) Secure the nut flange, ball screw nut and ball screw spline shaft which were disconnected in Step 10) above to the arm 2, using the four (4) bolts (M5 × 16). At first, tighten the bolts temporarily. Move the shaft up and down by hand and make sure that it moves smoothly, then tighten the bolts completely.

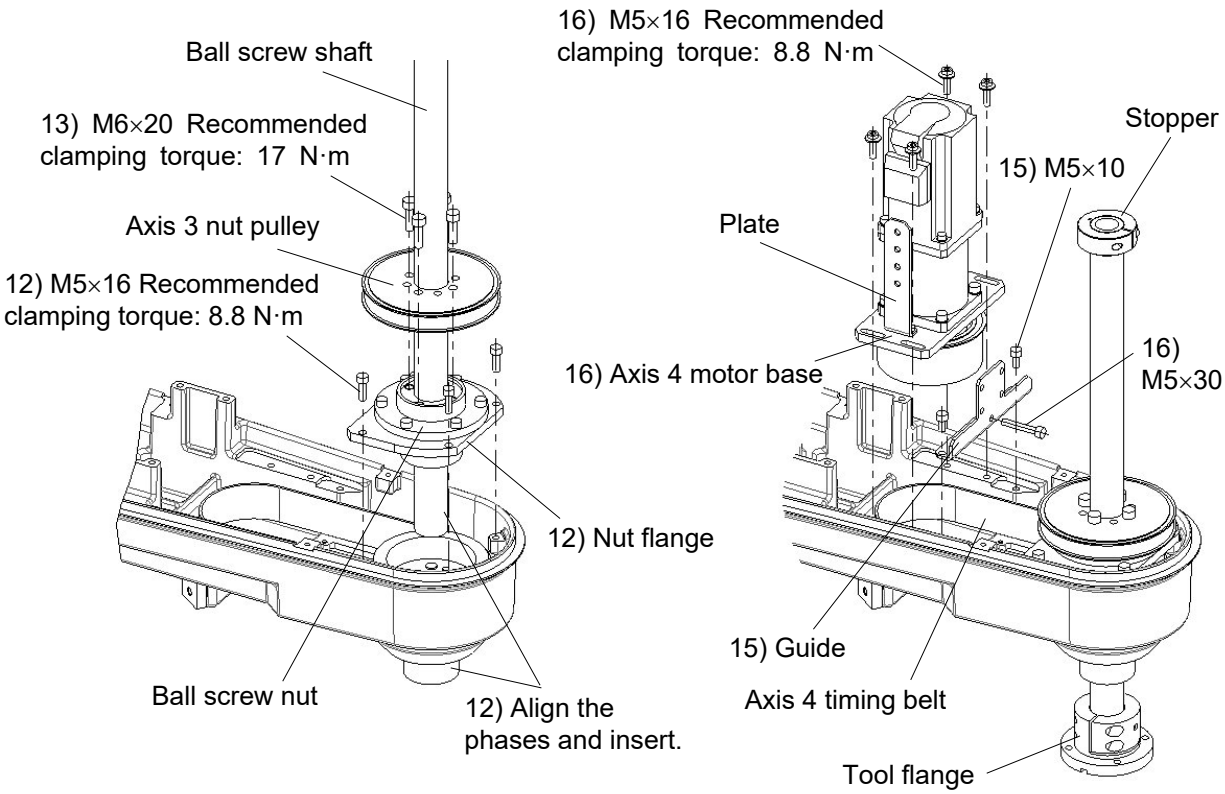
* A phase mark is provided on both the ball screw spline shaft and ball spline nut. For details, see Step 5) of Para. 2.7.5.

13) Attach the axis 3 nut pulley to the ball screw nut with the four (4) bolts (M6 × 20).

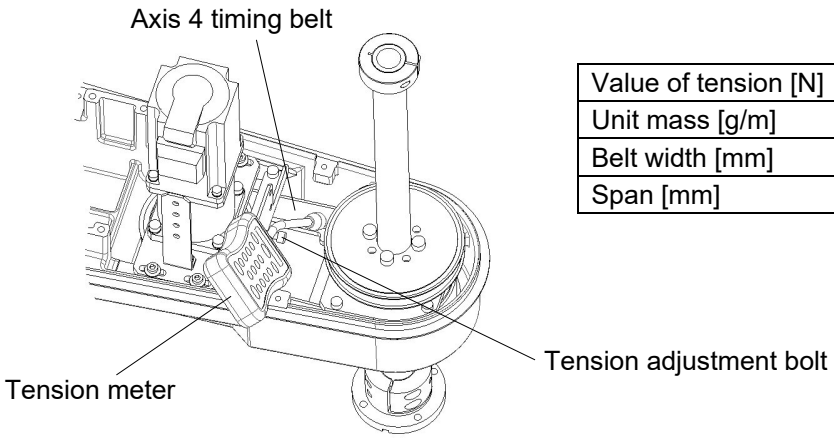
14) Attach the tool flange and stopper to the ball screw shaft.

15) Attach the guide to the arm 2 with the two (2) bolts (M5 × 10).

16) Temporarily set the motor base which was disconnected in Step 6) above to the arm 2 with the four (4) flange head bolts (M5 × 16) after setting the axis 4 timing belt to the axis 4 motor pulley. Also mount the axis 4 tension adjustment bolt (M5 × 30).



17) While pushing the axis 4 motor base by means of the axis 4 tension adjustment bolt, measure the tension on the tension meter. At a position where the tension has reached 320 to 380 N, tighten the flange head bolts.
The values used for the tension meter are tabled below.



Value of tension [N]	320 ~ 380
Unit mass [g/m]	4.1
Belt width [mm]	30
Span [mm]	160

18) Mount the axis 3 motor and timing belt. For the mounting procedures, see Para. 2.5.7 and Para. 2.6.3 above.

19) Arrange the connectors and cables as originally set.

20) Perform home setting for the axes 3 and 4.

21) Carry out a test operation of the axes 3 and 4 and make sure that the belt tension is appropriate.

22) Attach the arm 2 cover. Now replacement of the axis 4 timing belt completes.

2.6.5 Adjusting Timing Belt

Perform inspection and adjustment of each timing belt semi-annually (every six (6) months).

1) Remove the arm 2 cover. (See Para. 2.4.1.)

2) If the belt is loosened heavily, adjust its tension, referring to the following table. Note that the value of tension at replacement of timing belt (i.e., when mounting a new timing belt) differs from the value of tension at adjustment. (The values other than the tension remain the same.)

Axis	Value of tension [N]	Unit mass [g/m]	Belt width [mm]	Span [mm]
Axis 3	43 ~ 49	2.2	10	232
Axis 4	280 ~ 315	4.1	30	160

If the tension is appropriate, mount the arm 2 cover. Now the inspection completes.

3) When adjustment of the tension is required, loosen the flange head bolts (M5 × 16) securing the axis 3 and axis 4 motor bases, respectively. Measure the tension on the tension meter while adjusting the tension adjustment bolt (axis 3: M4 × 35, axis 4: M5 × 30).

4) When the tension has fallen under the appropriate range, tighten the flange head bolts.

5) Mount the arm 2 cover. Now the adjustment completes.

2.7 Filling Grease to Ball Screw Spline Unit and Replacement

The ball screw spline unit is to be replaced by our service engineer. If it is replaced by the customer, we will not guarantee any consequential trouble or accident.

! DANGER

- The ball screw spline unit should be replaced with a new one only after the controller power is turned off with the power supply plug removed. If the work is done while the power is turned on, you may get an electric shock or the robot may malfunction, which is very dangerous.

! CAUTION

- Because the timing belt, nut and pulley are disconnected, the mechanical home point shift will shift and proper control cannot be done. To avoid this, home return operation is necessary after replacement of the ball screw spline unit. For the home return procedures, see Section 4.

2.7.1 Type of Ball Screw Spline Unit

The ball screw spline unit used in this robot is shown below.
When you place an order for the new ball screw spline unit, specify the robot model (TH650A) and our drawing number.

Description	Stroke	Our drawing No.
Ball screw spline unit	200 mm	H846688
	400 mm	H846689

* Stroke 400 mm is optional.

2.7.2 Ball Screw Spline Unit Location

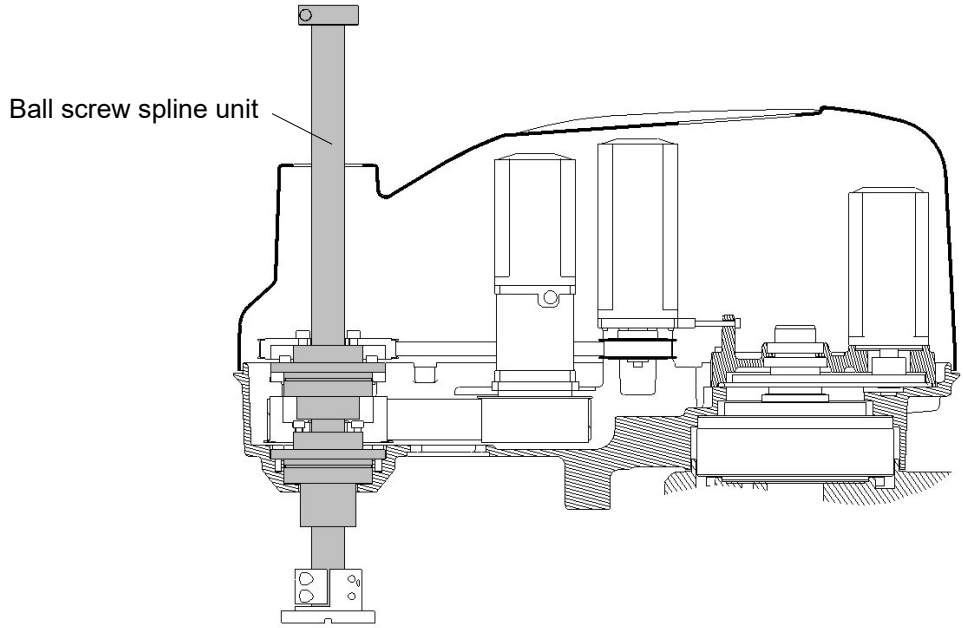


Fig. 2.12 Ball screw spline unit location

2.7.3 Filling Grease to Ball Screw Spline Unit

! CAUTION

- When vertically moving the ball screw by hand, take careful precautions not to have your hand or finger caught.
- As there is a fear that the grease drops, cover the peripheral equipment, etc.

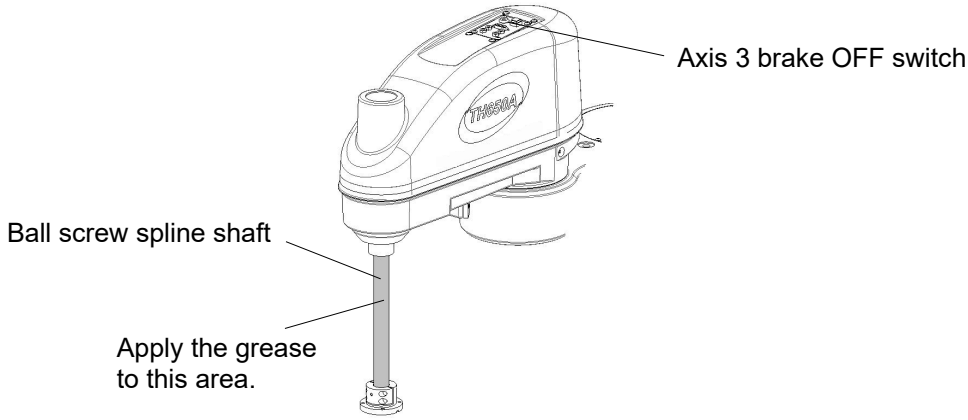
! CAUTION

- When the grease has run short, cut or scratch will be caused on the slide unit, etc., resulting in drop of the performance. To avoid this, take careful precautions to prevent shortage of the grease.
- Be sure to use the grease designated by SHIBAURA MACHINE.

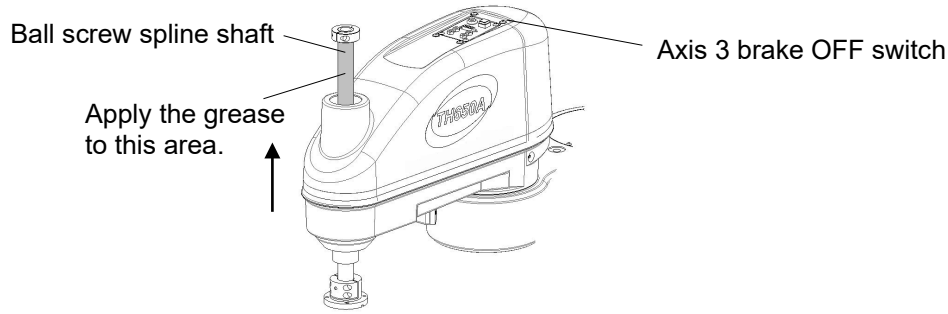
Fill the grease to the ball screw spline unit every three (3) months.
At daily inspection also, make sure that the ball screw spline unit is filled with a sufficient volume of grease.

Recommended grease	Maker
AFB grease	THK

1) Turn on the controller power and turn off the servo system.
2) Move the arm to a position where the axis 3 can be moved over the full stroke.
3) Push down the ball screw spline shaft to the lower limit while pressing the axis 3 brake OFF switch.
4) Directly apply the grease to the exposed shaft area by hand or finger.
   Apply the grease to such an extent that the shaft groove can be filled. Wipe out the surplus grease.



5) Push up the shaft up to the upper limit while pressing the axis 3 brake OFF switch.
6) Directly apply the grease by hand or finger to the shaft area sticking up from the cover. Apply the grease to such an extent that the shaft groove can be filled. Wipe out the surplus grease.



7) Move the shaft up and down repeatedly while pressing the axis 3 brake OFF switch to fill the grease uniformly. Wipe out the surplus grease. Now the work completes.

2.7.4 Dismounting Ball Screw Spline Unit

DANGER

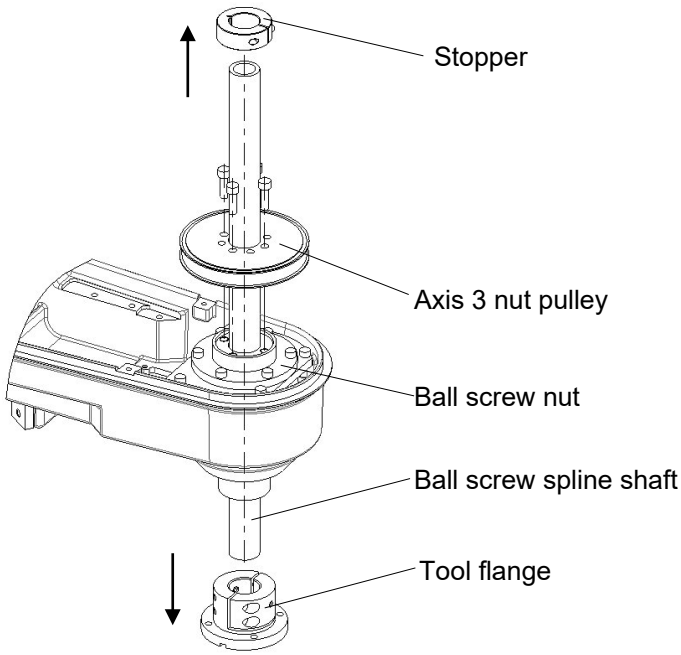
- The ball screw spline unit should be replaced with a new one only after the controller power is turned off with the power supply plug removed. If the work is done while the power is turned on, you may get an electric shock or the robot may malfunction, which is very dangerous.

CAUTION

- Handle the ball screw spline unit with extreme care. If the unit drops or an unusually large external force is exerted on it, it cannot function any further.
- Replacement of the ball screw spline unit involves mounting and dismounting of the axis 3 and 4 motors and timing belts. Also observe the cautions on each work.
- Because the motor, timing belt, nut and pulley are disconnected, the mechanical home point will shift and proper control cannot be done. To avoid this, home return operation is necessary after replacement of the ball screw spline unit. For the home return procedures, see Section 4.

In the descriptions on replacing the ball screw spline unit, the procedures for changing the axis 3 and 4 motors and timing belts are not included. For details, refer to the descriptions on replacement of each part.

1) Remove the arm 2 cover. (See Para. 2.4.1.)
2) Disconnect the hand, tool, etc. followed by the tool flange.
3) Disconnect the axis 3 and 4 motors and timing belts.
4) Remove the stopper attached to the top of the ball screw spline shaft.
   Once the stopper is removed, the shaft can slip out in the vertical direction.
   Take careful precautions not to drop it by its dead weight.
5) Disconnect the axis 3 nut pulley secured to the ball screw nut.

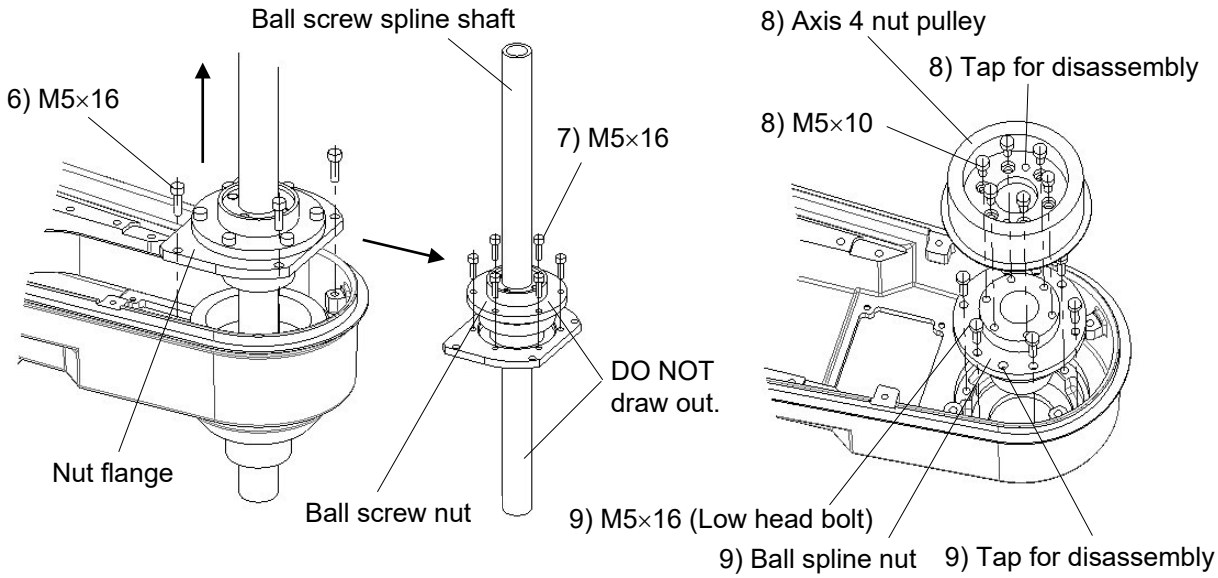


6) Remove the four (4) bolts (M5 × 16) securing the nut flange, then draw out the nut flange, ball screw nut integrated with the ball screw spline shaft toward the up direction.

7) Remove the six (6) bolts (M5 × 16) from the parts disconnected in Step 6) above, then remove the nut flange.

! CAUTION

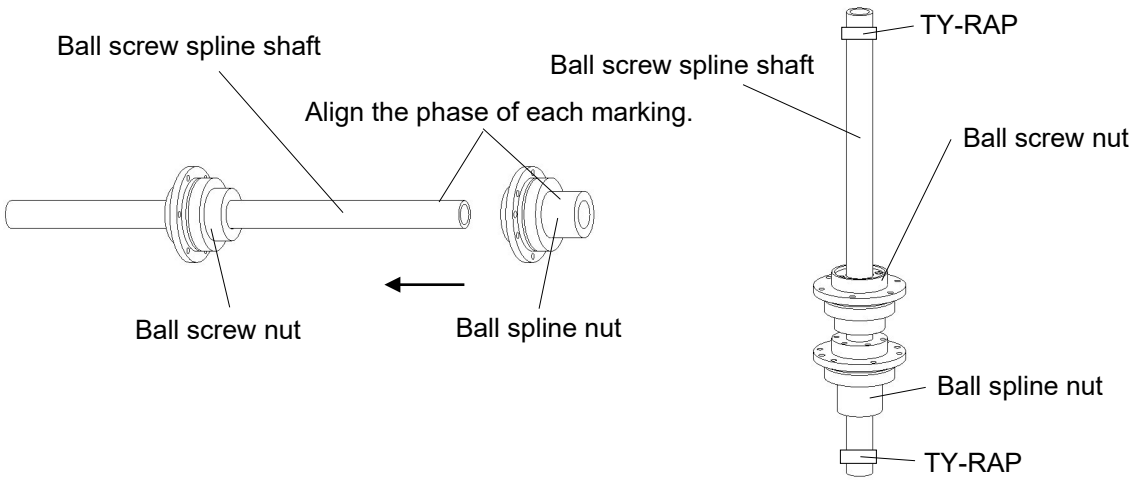
- When the ball screw nut integrated with the ball screw spline shaft is disconnected, take utmost care not to cause the ball screw spline shaft to come off. Otherwise, the ball in the ball screw nut will drop and the ball screw nut integrated with the ball screw spline shaft cannot function any further.



8) Disconnect the axis 4 nut pulley secured to the ball spline nut with six (6) bolts (M5 × 10). When it is hard to disconnect, screw the bolt into the tap for disassembly (M5) machined on the pulley and disconnect.

9) Remove the six (6) low head bolts (M5 × 16) securing the ball spline nut and draw out the ball spline nut upward. As the hole on each low head bolt, which is intended for using a wrench, is shallow, take care to handle the wrench. When the ball spline nut is hard to disconnect, screw the bolt into the tap for disassembly (M5) machined on the ball spline nut and disconnect.

10) Insert the disconnected ball spline nut into the shaft. Wind the TY-RAP around the vicinity of top and bottom of the shaft to prevent the nut from slipping off.



2.7.5 Mounting Ball Screw Spline Unit

! DANGER

- The ball screw spline unit should be replaced with a new one only after the controller power is turned off with the power supply plug removed. If the work is done while the power is turned on, you may get an electric shock or the robot may malfunction, which is very dangerous.

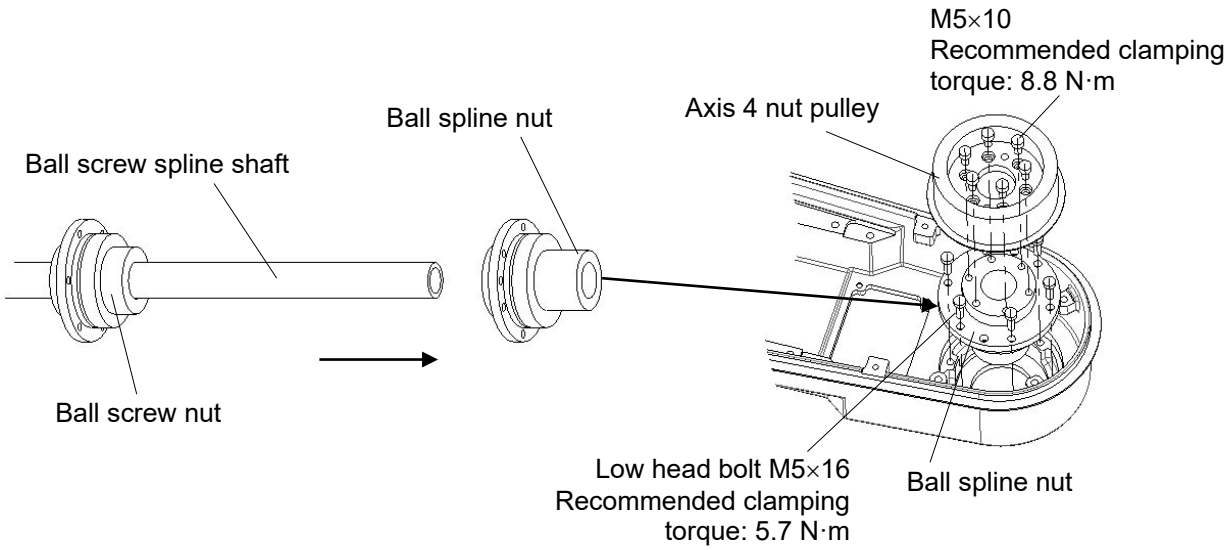
! CAUTION

- Handle the ball screw spline unit with extreme care. If the unit drops or an unusually large external force is exerted on it, it cannot function any further.
- Replacement of the ball screw spline unit involves mounting and dismounting of the axis 3 and 4 motors and timing belts. Also observe the cautions on each work.
- Because the motor, timing belt, nut and pulley are disconnected, the mechanical home point will shift and proper control cannot be done. To avoid this, home return operation is necessary after replacement of the ball screw spline unit. For the home return procedures, see Section 4.
- DO NOT pull out the ball screw shaft from the ball screw nut. Otherwise, the ball in the ball screw nut will drop and the ball screw nut integrated with the ball screw shaft cannot function any further.

1) Disconnect the ball spline nut from the new ball screw spline unit.

2) Secure the ball spline nut to the arm 2 with six (6) low head bolts (M5 × 16).
The clamping torque of each low head bolt is 5.7 N·m.
As the hole on each low head bolt, which is intended for using a wrench, is shallow, take care to handle the wrench.

3) Mount the axis 4 nut pulley on the ball spline nut with the six (6) bolts (M5 × 10).
At this time, the pulley will rotate and cannot be secured easily. Tighten the bolts only after the axis 4 motor is mounted.

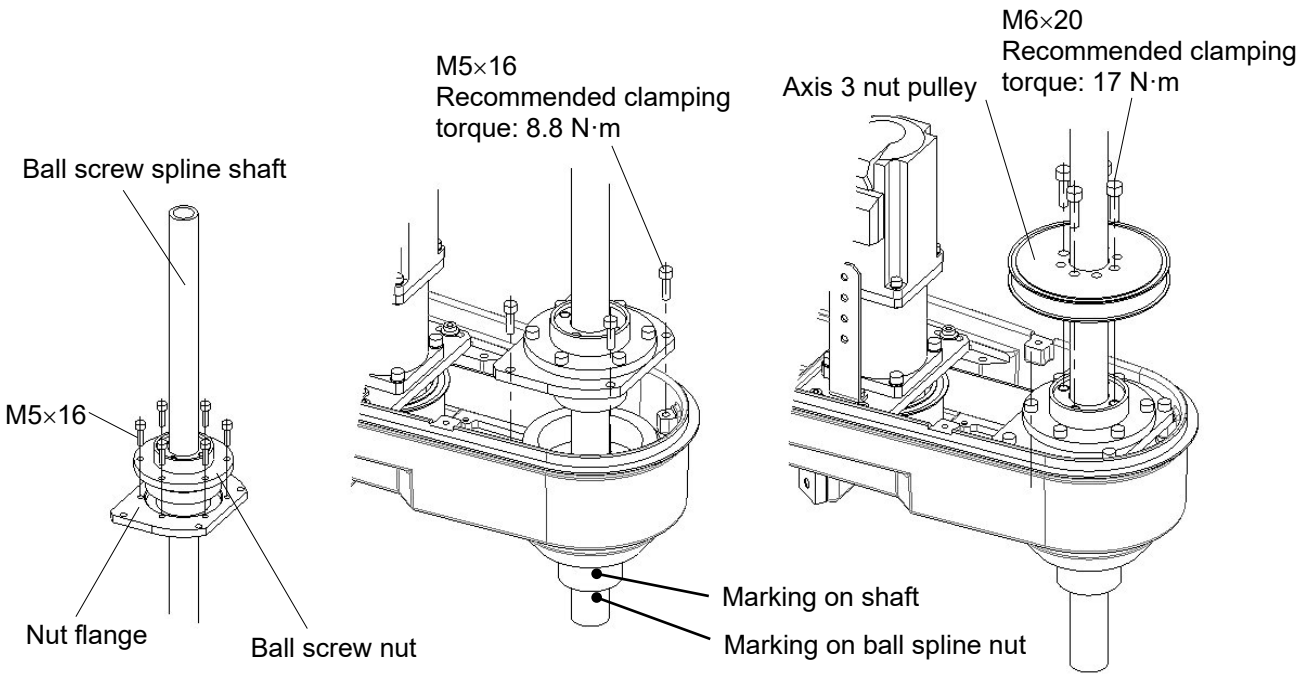


4) Mount the axis 4 timing belt and axis 4 motor, then adjust the belt tension.

5) Mount the ball screw nut integrated with the shaft onto the nut flange, then pass them through the ball spline nut. At this time, align the phases of both markings put on the tip of the ball spline nut and the shaft.

6) Secure the nut flange to the arm 2 with the four (4) bolts (M5 × 16).
At first, tighten the bolts temporarily. Move the shaft up and down by hand and make sure that it moves smoothly, then tighten the bolts completely.

7) Mount the axis 3 nut pulley with the four (4) bolts (M6 × 20).



8) Mount the tool flange and stopper on the ball screw shaft.

9) Mount the axis 3 timing belt and axis 3 motor, then adjust the belt tension.

10) Arrange the connectors and cables as originally set.

11) Perform home setting for the axes 3 and 4.

12) Carry out a test operation of the axes 3 and 4 and make sure that each part operates properly.

13) Mount the arm 2 cover. Now replacement of the ball screw spline unit completes.

2.8 Filling Grease to Reduction Gear and Replacement

The reduction gear is to be replaced by our service engineer. If it is replaced by the customer, we will not guarantee any consequential trouble or accident.

! DANGER

- The reduction gear should be replaced with a new one only after the controller power is turned off with the power supply plug removed. If the work is done while the power is turned on, you may get an electric shock or the robot may malfunction, which is very dangerous.

! CAUTION

- Because the reduction gear is disconnected, the mechanical home point will shift and proper control cannot be done. To avoid this, home return operation is necessary after replacement of the reduction gear. For the home return procedures, see Section 4.

2.8.1 Type of Reduction Gear

The reduction gears used in this robot are shown below.
When you place an order for the new reduction gear, specify the robot model (TH650A), axis name, type and our drawing number.

Description	Axis name	Type	Our drawing No.
Reduction gear	Axis 1	RV–27C–36–57	S816716
	Axis 2	RV–10C–27	S816717
	Axis 4	VRGF–15B60P–400	S829013

2.8.2 Reduction Gear Locations

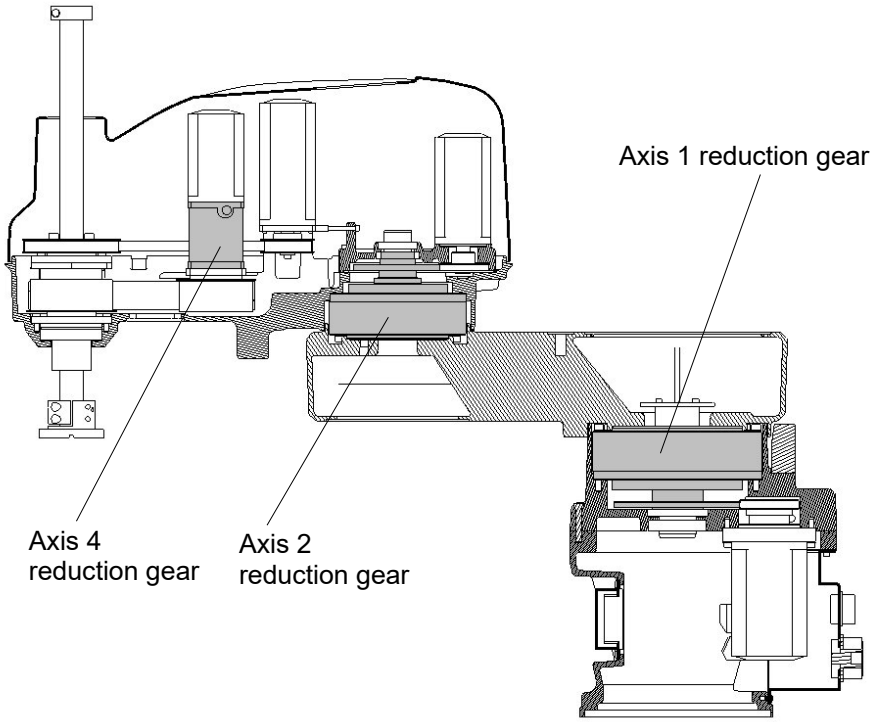


Fig. 2.13 Reduction gear locations

2.8.3 Filling Grease to Reduction Gear

When the reduction gears are used under the normal operating conditions, regular greasing of them is unnecessary. Replace the reduction gears at the time of overhauling. For the axes 1 and 2, however, if the reduction gears are used under the conditions of high operation duty and heavy load, fill the grease to them every 10,000 hours (i.e., approximately two (2) years). Greasing of the axis 4 reduction gear is unnecessary.

! CAUTION

- When the grease has run short, cut or scratch will be caused on the reduction gear interior or gear, resulting in drop of the performance. To avoid this, take careful precautions to prevent shortage of the grease.
- Be sure to use the grease designated by SHIBAURA MACHINE.
- Increase in internal pressure will adversely affect the starting torque and damage the internal seal. To avoid this, be sure to observe the grease filling volume.
- Be sure to restore the grease nipple and hexagon socket plug used for filling the grease as they were set originally. If they are left intact, internal pressure in the reduction gear will increase, resulting in malfunction at an early stage.

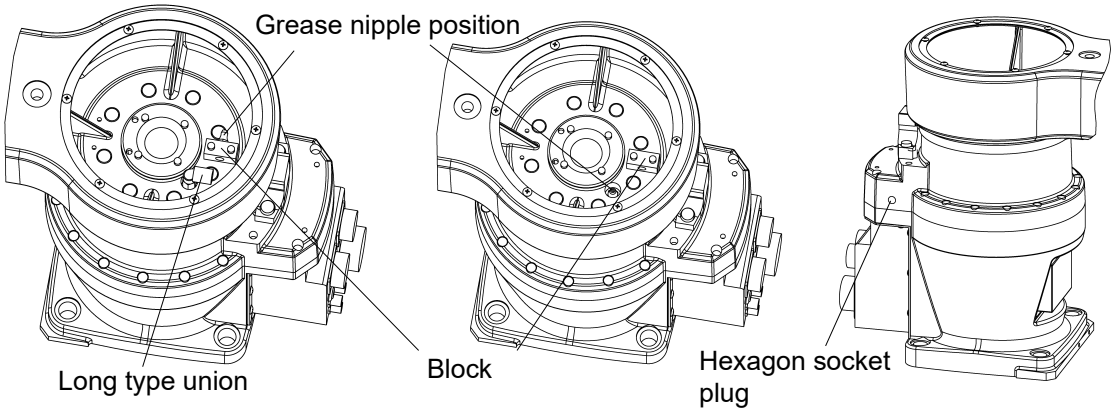
Grease filling volume

Grease filling volume for axis 1 reduction gear	Grease filling volume for axis 2 reduction gear
330 g	210 g

When you wish to place an order for the grease, contact us at the Service Department.

1) Filling grease to axis 1 reduction gear
   Remove the arm 1 top cover (arm cover 1) and disconnect the long type union in the arm 1. Remove the grease nipple attached to the block in the arm 1 and mount it on the place where the long type union was mounted before. Fill the grease from the grease nipple then. When this happens, remove the hexagon socket plug (Rc1/8) attached to the lateral side of the base and remove the old grease. Provide an oil pan, etc. to receive the old grease.

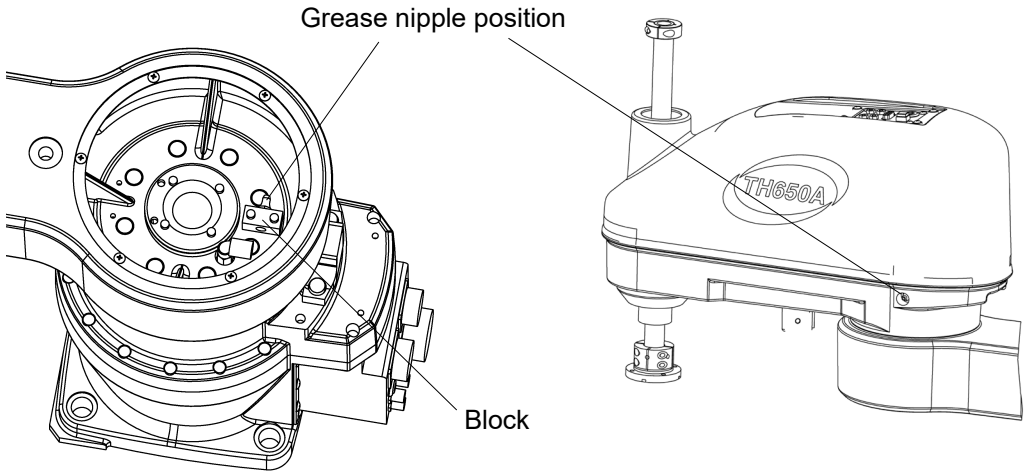
After the specified volume of grease is filled, attach the hexagon socket plug (Rc1/8) to the lateral side of the base and wipe out the grease left on the lateral side of the base and around the grease nipple. Then remove the grease nipple and mount the long type union again. Attach the removed grease nipple to the block in the arm 1 and cover the arm 1. Now greasing of the axis 1 reduction gear completes.
When attaching the hexagon socket plug, be sure to use a seal tape.
To circulate the grease throughout the reduction gear interior, it is recommended to perform running at a low speed for about five (5) minutes.



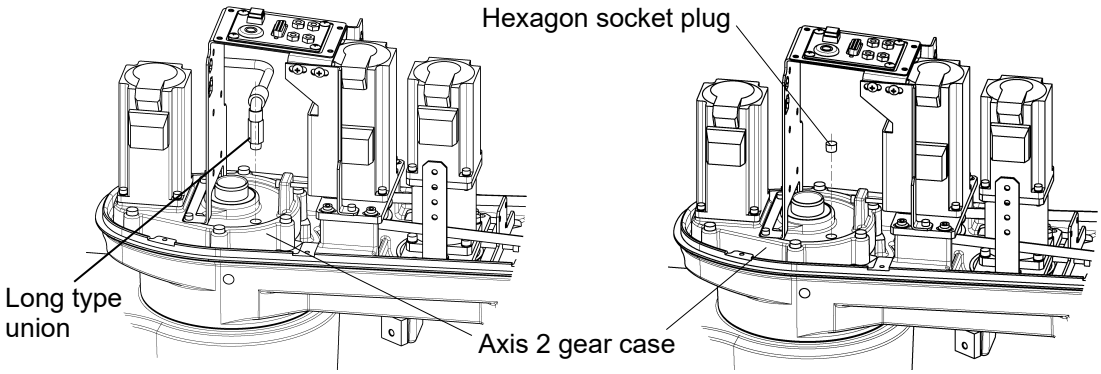
2) Filling grease to axis 2 reduction gear

- Remove the arm 1 bottom cover (arm cover 2), then the hexagon socket plug (Rc1/8) attached to the lateral side of the arm 2.

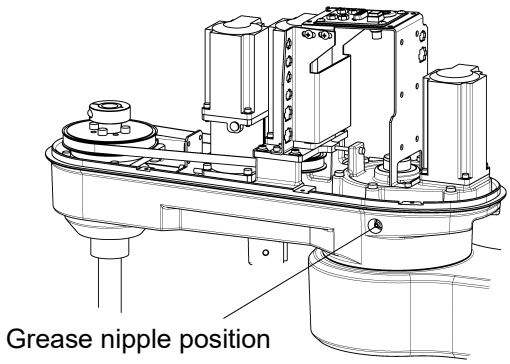
- Attach to the lateral side of the arm 2 the grease nipple which is attached to the block in the arm 1.

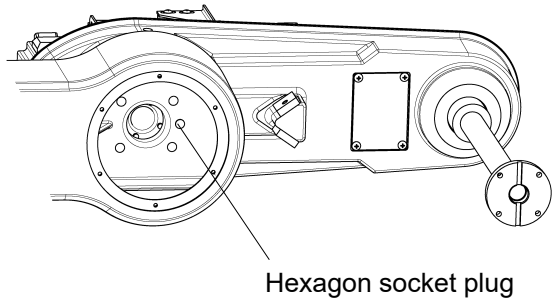


- Remove the arm 2 cover (see Para. 2.4.1), then the long type union secured to the axis 2 gear case. Attach the hexagon socket plug (Rc1/8) on the lateral side of the arm 2 to the place where the long type union was attached before.

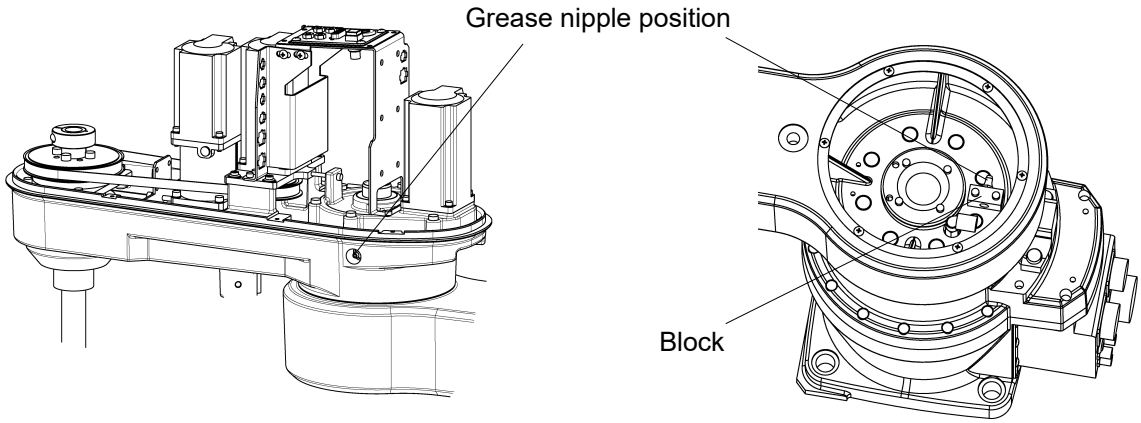


- Fill the grease from the grease nipple mounted on the lateral side of the arm 2. When this happens, remove the hexagon socket plug (Rc1/8) provided in the arm 1 and remove the old grease. Provide an oil pan, etc. to receive the old grease.
- After the specified volume of grease is filled, attach the hexagon socket plug in the arm 1 and wipe out the grease left in the arm 1 and around the grease nipple. Then set the arm 1 bottom cover.

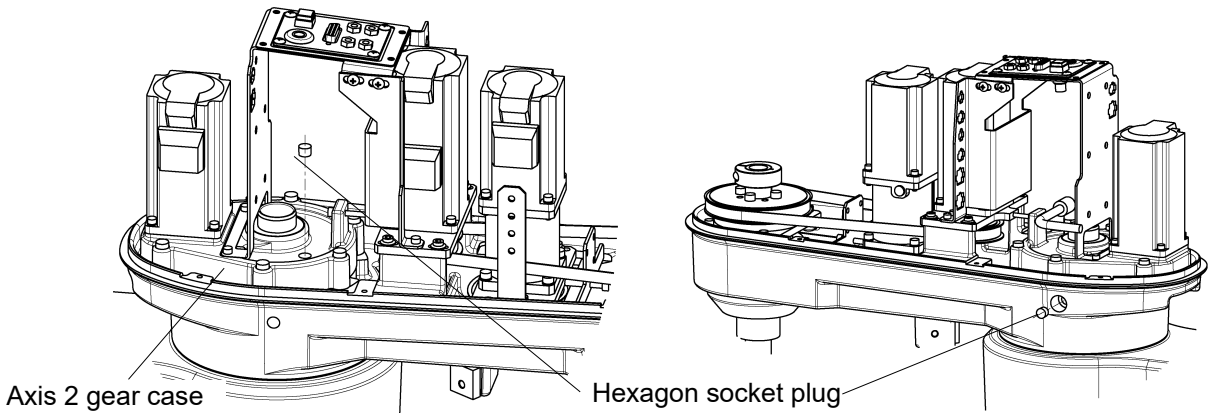




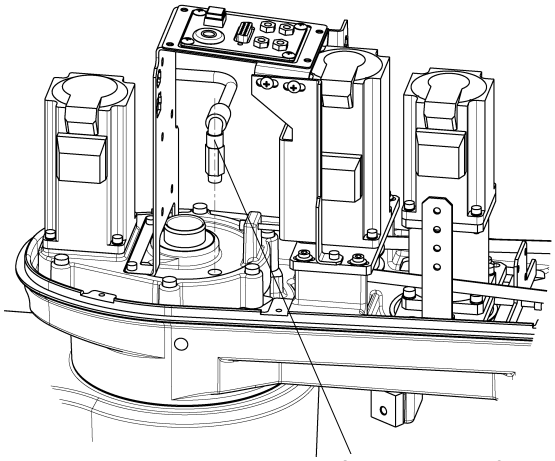
- Remove the grease nipple from the lateral side of the arm 2 and attach it to the block in the arm 1.

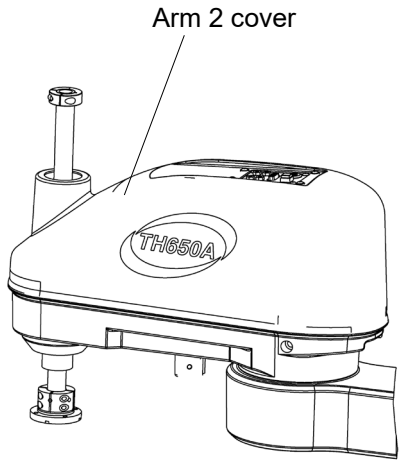


- Remove the hexagon socket plug (Rc1/8) attached to the axis 2 gear case and mount it on the lateral side of the arm 2.



- Attach the long type union to the axis 2 gear case again, and set the arm 2 cover to the arm 2. Now greasing of the axis 2 reduction gear completes. When attaching the hexagon socket plug, be sure to use a seal tape.





- To circulate the grease throughout the reduction gear interior, it is recommended to perform running at a low speed for about five (5) minutes.

2.8.4 Dismounting Axis 1 Reduction Gear

1) Remove the arm 1 top cover. (See Para. 2.4.2.)

2) Disconnect the cable connectors and air tubing connected in the arm 1.

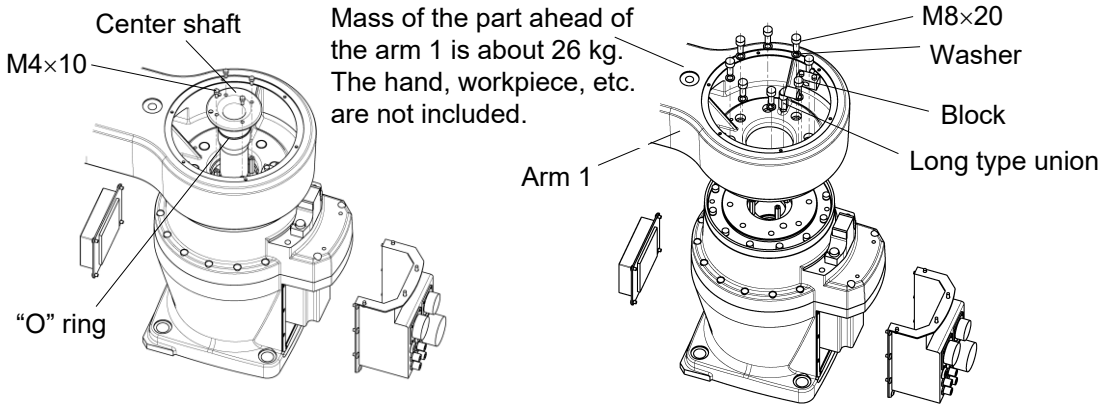
3) Remove the base cover and pull out the cables and air tubes up to the base interior. See Para. 2.4.3.

! CAUTION

- The arm should always be mounted and dismounted by two (2) or more persons. When removing the arm set bolts, take careful precautions because the arm will drop. Also, if an excessively large impact is imposed on the arm, the robot will be damaged.

4) Remove the four (4) bolts (M4 × 10) and disconnect the axis 1 center shaft. (The "O" ring is attached.)

5) While more than one (1) person is holding the arm, another person should remove the eight (8) bolts (M8 × 20) securing the arm 1, long type union and block where the grease nipple is secured, then disconnect the arm 1. Place the arm thus disconnected in such a manner that no impact is exerted on it.

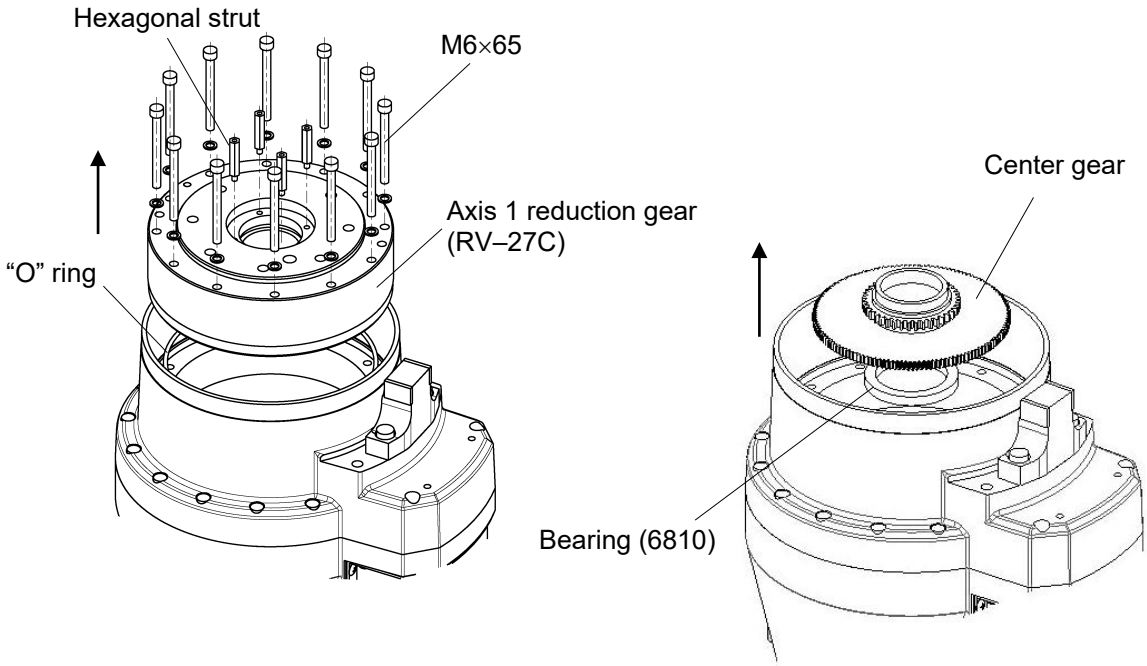


6) Remove the four (4) hexagonal struts attached to the reduction gear.

7) Remove the twelve (12) bolts (M6 × 65) securing the reduction gear, then draw out the reduction gear upward. If it is hard to disconnect, screw the bolt into the tap for disassembly (M6) which is machined on the reduction gear.
Take careful precautions not to miss the exclusive “O” ring attached to the bottom of the reduction gear.
When dismounting the reduction gear, it is recommended to prepare a waste cloth, etc. because the grease is filled to the vicinity of the center gear on the lower side.

8) After dismounting the reduction gear, remove the center gear. The center gear can be drawn upward as it is. The center gear and reduction gear form a pair. When replacing the reduction gear, be sure to replace the center gear also. The center gear is attached to the new reduction gear.

9) It is recommended to replace the bearing supporting the center gear, too.
Type of bearing used: 6810



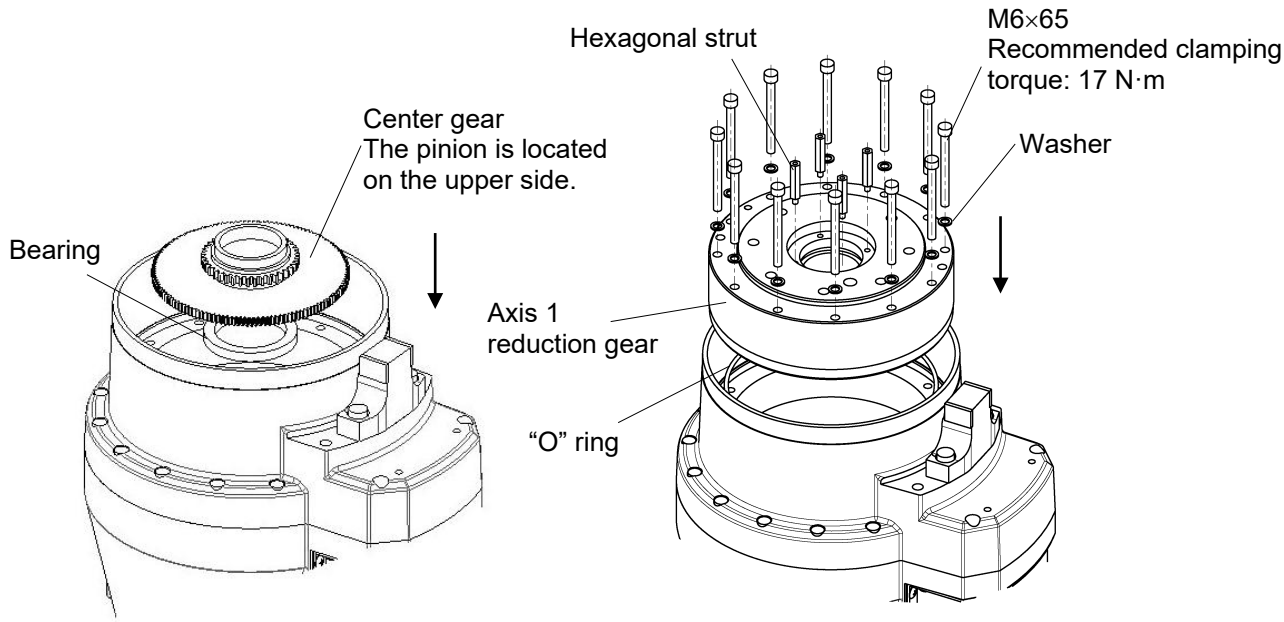
2.8.5 Mounting Axis 1 Reduction Gear

! CAUTION

- Handle the reduction gear with extreme care. If it drops or an unusually large external force is exerted on it, the reduction gear cannot function any further.
- Use the center gear which is attached to the new reduction gear.
  If the old center gear is used as it is, abnormal noise will be caused, the service life will shorten or positioning accuracy will deteriorate due to incompatibility with the new reduction gear.
- Also use the “O” ring attached to the new reduction gear. Be sure to set the “O” ring. Unless the “O” ring is set, grease will leak from the axis 1 reduction gear set surface. When mounting the reduction gear, take careful precautions not to break the “O” ring.

1) Mount the new bearing, and center gear attached to the new reduction gear.

2) Mount the reduction gear with the twelve (12) bolts (M6 × 65). Be sure to apply the Loctite adhesives to the threaded area of each bolt. Before mounting the reduction gear, set the “O” ring in the “O” ring groove on the bottom of the reduction gear. Attach the four (4) hexagonal struts. The “O” ring is attached to the new reduction gear.



3) Mount the arm 1 with the eight (8) bolts (M8 × 20). Be sure to apply the Loctite adhesives to the threaded area of each bolt. At this time, align the phases of the recess hole for the long type union of the arm 1 and reduction gear. Then mount the block where the long type union and grease nipple are secured.

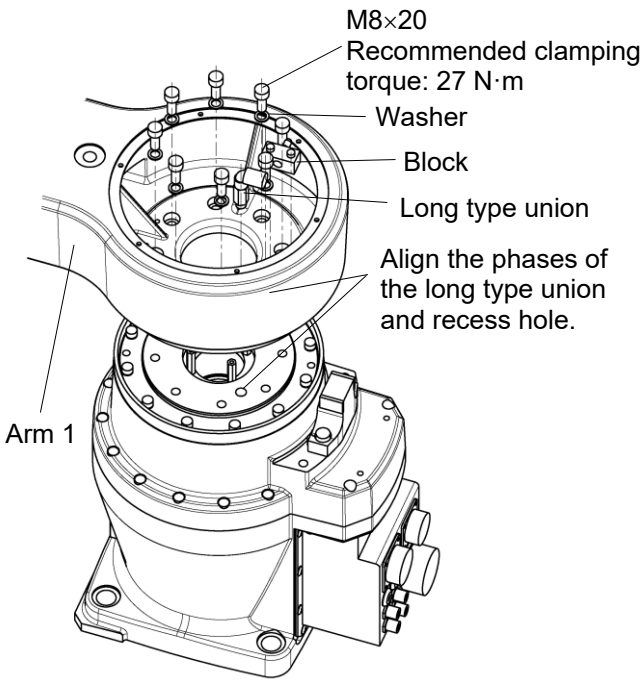
4) Mount the center shaft with the four (4) bolts (M4 × 10). After setting the “O” ring attached to the reduction gear to the center shaft, secure the center shaft to the reduction gear.

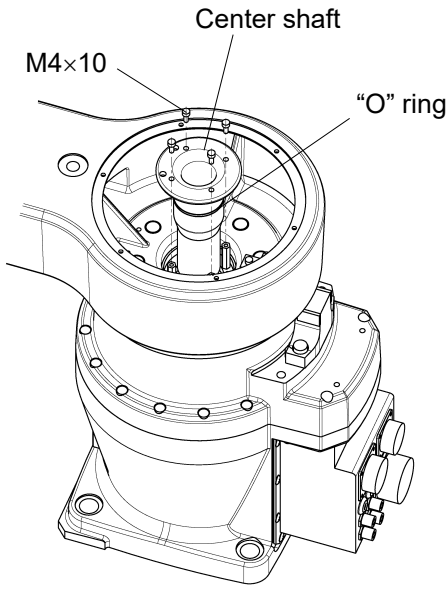
5) Pass through the center shaft the cables and air tubes which were drawn up to the base interior, then draw them out to the arm 1.

6) Connect the cable connectors and air tubes as originally set.

7) Mount the arm 1 cover and perform home setting for the axis 1 to axis 4. Now replacement of the axis 1 reduction gear completes.
As all cable connectors are disconnected, home setting of axes 2, 3 and 4 is necessary.

8) Carry out a test operation of the axis 1 and make sure that each part operates properly.



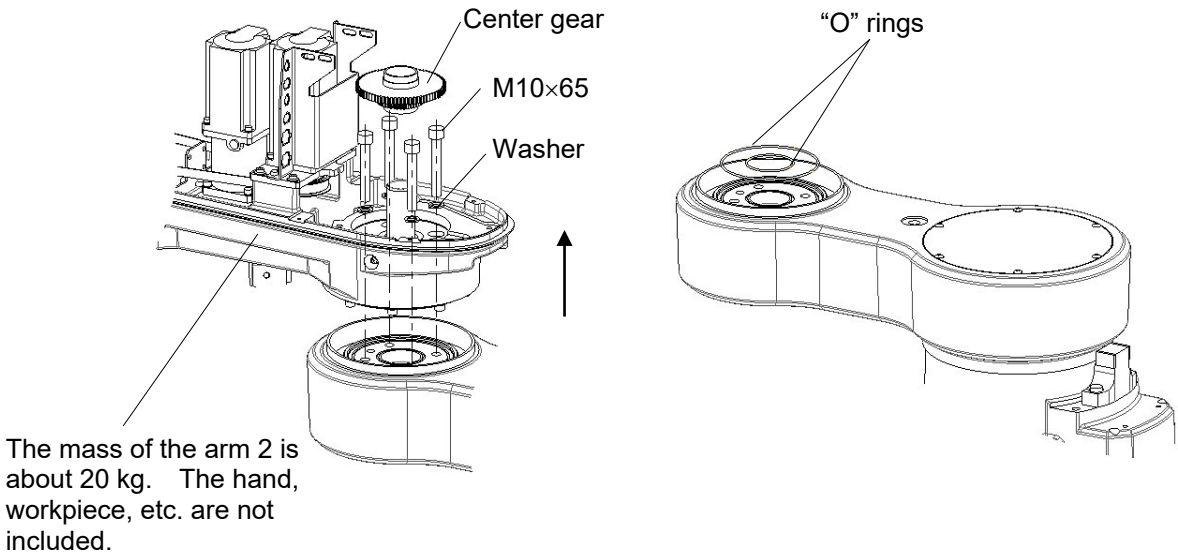


2.8.6 Dismounting Axis 2 Reduction Gear

1) Remove the arm 2 cover. (See Para. 2.4.1.)

2) Dismount the axis 2 motor, referring to Steps 1) through 9) of Para. 2.5.5. Remove the bearing from the disconnected gear case. It is recommended to replace this bearing also together with the reduction gear.
Type of bearing used: 6807

3) Draw out the center gear upward, remove the four (4) bolts (10 × 65) securing the reduction gear, then disconnect the arm 2. As the two (2) "O" rings are set on the axis 2 reduction gear set surface of the arm 1, be sure to remove them.

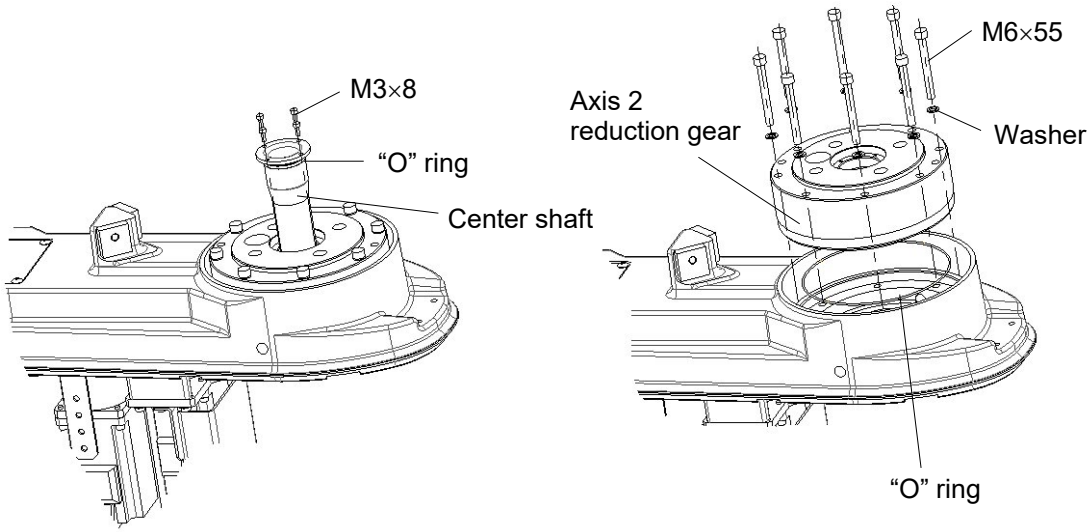


! CAUTION

- The arm should always be mounted and dismounted by two (2) or more persons. When removing the arm set bolts, take careful precautions because the arm will drop. Also, if an excessively large impact is imposed on the arm, the robot will be damaged.

4) Remove the center shaft secured to the axis 2 reduction gear with the four (4) bolts (M3 × 8).

5) Remove the eight (8) bolts (M6 × 55), then dismount the axis 2 reduction gear. Take careful precautions not to miss the exclusive "O" ring attached to the bottom of the reduction gear.



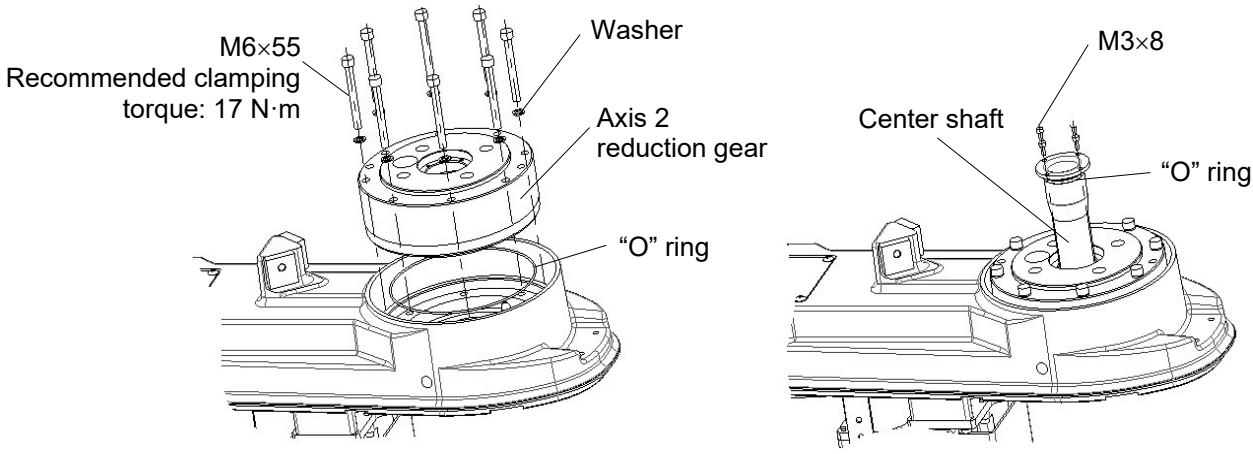
2.8.7 Mounting Axis 2 Reduction Gear

! CAUTION

- Handle the reduction gear with extreme care. If it drops or an unusually large external force is exerted on it, the reduction gear cannot function any further.
- Use the center gear which is attached to the new reduction gear.
  If the old center gear is used as it is, abnormal noise will be caused, the service life will shorten or positioning accuracy will deteriorate due to incompatibility with the new reduction gear.
- Also use the "O" ring attached to the new reduction gear. Be sure to set the "O" ring. Unless the "O" ring is set, grease will leak from the axis 2 reduction gear set surface. When mounting the reduction gear, take careful precautions not to break the "O" ring.

1) Mount the new reduction gear on the arm 2 with the eight (8) bolts (M6 × 55). Before mounting the reduction gear, set the "O" ring in the "O" ring groove on the bottom of the reduction gear.

2) Mount the center shaft on the axis 2 reduction gear with the four (4) bolts (M3 × 8). After setting the "O" ring attached to the reduction gear in the center shaft, secure the center shaft to the reduction gear.



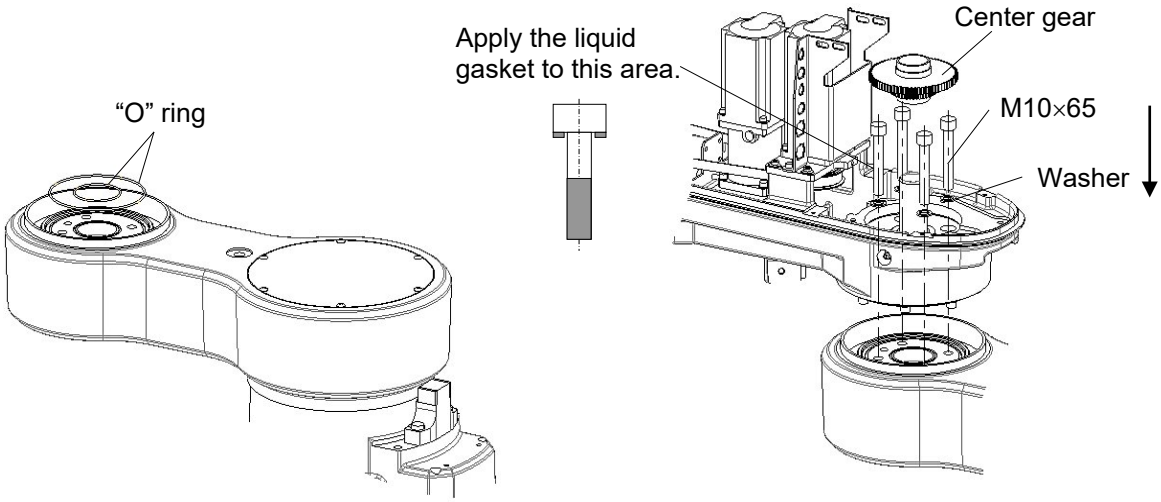
3) Set the two (2) "O" rings attached to the reduction gear in the "O" ring grooves machined on the arm 1.

4) Mount the arm 2 with the four (4) bolts (M10 × 65). When this happens, the clamping torque of each bolt should be 54 N·m. DO NOT tighten the bolts with the clamping torque exceeding 54 N·m. Be sure to apply the liquid gasket to the threaded area of each bolt.

5) Mount the new center gear on the axis 2 reduction gear.

! CAUTION

- Be sure to observe the clamping torque of the bolts (M10 × 65) securing the axis 2 reduction gear with the arm 1. If these bolts are tightened with clamping force exceeding 54 N·m, the axis 2 starting torque will increase.
- Be sure to apply the liquid gasket to the threaded area of each bolt (M10 × 65).



6) Mount the gear case attached with the new bearing onto the arm 2. For mounting the axis 2 motor, see Steps 5) through 9) of Para. 2.5.6 above.

7) Mount the arm 2 cover and perform home resetting for the axes 2, 3 and 4. Now replacement of the axis 2 reduction gear completes.
(Check the axis 1 for the coordinates and perform home setting as necessary.)

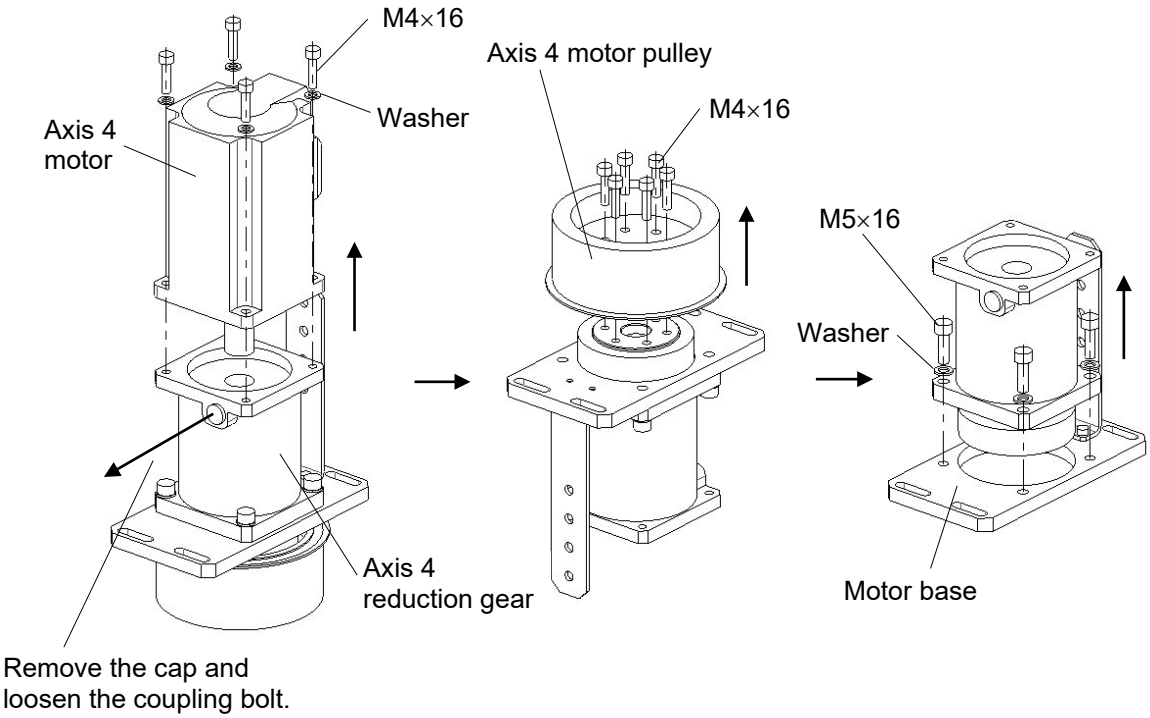
8) Carry out a test operation of the axis 2 and make sure that each part operates properly.

2.8.8 Replacing Axis 4 Reduction Gear

! CAUTION

- Handle the reduction gear with extreme care. If it drops or an unusually large external force is exerted on it, the reduction gear cannot function any further.
- Be sure to tighten the coupling connecting the reduction gear and motor shaft.

1) Remove the arm 2 cover. (See Para. 2.4.1.)
2) Disconnect the axis 4 motor and motor base, axis 3 motor and motor base, and axes 3 and 4 timing belts from the arm 2, referring to Steps 1) to 6) inclusive of Para. 2.6.4.
3) Remove the cap attached to the lateral side of the axis 4 reduction gear, then loosen the coupling bolt (M4 × 12) securing the axis 4 motor shaft and the input shaft of the axis 4 reduction gear.
4) Remove the four (4) bolts (M4 × 16) securing the motor, and draw out the motor upward.
5) Secure the pulley, using the hole for knock which is machined on the axis 4 motor pulley, remove the six (6) bolts (M4 × 16), then disconnect the pulley.
6) Remove the four (4) bolts (M5 × 16) securing the reduction gear to the motor base, then dismount the reduction gear.



7) Assemble the new reduction gear, motor and pulley in the reverse order of Steps 5) and 6) above.

8) Assemble the axis 4 motor and timing belt according to Steps 16) through 19) of Para. 2.6.4 above, then mount the arm 2 cover.

9) Perform home setting for the axes 3 and 4. Now replacement of the axis 4 reduction gear completes.

2.9 Replacing Position Detector (or Encoder) Batteries

! CAUTION

- The batteries should be disposed of according to the user's in-house regulations.
  NEVER drop the battery into fire. NEVER short-circuit, charge, disassemble or heat it. Otherwise, liquid leakage or rupture may be caused.

To keep the data of the position detector attached to the motor, they are backed up by batteries. Be sure to replace the batteries with new ones at the time of annual inspection. When the robot is not used for a long term (i.e., the batteries are left intact in the power OFF condition), replace the batteries at the time of robot startup. As a yardstick, the shutdown period is two (2) months.
(Three (3) size AA batteries)

! CAUTION

- When the battery voltage has dropped, a battery alarm will generate. If the batteries are replaced just after generation of the battery alarm, the battery voltage returns to normal with the battery alarm reset automatically. Unless the batteries are changed just after generation of the battery alarm, however, the battery voltage will drop further and a battery error will occur. Under this condition, position data detected by the encoder is not reliable. As a result, a position detection error occurs and the robot enters an emergency stop state so that the servo system cannot be turned on. If the power is turned off in this condition, the position data is lost.

  To avoid the above, be sure to replace the batteries with new ones at the time of annual inspection.

2.9.1 Battery Box Location

The battery box for the position detector is provided inside the battery box cover of the base unit.

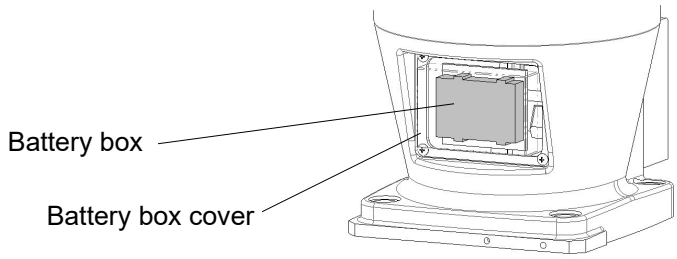


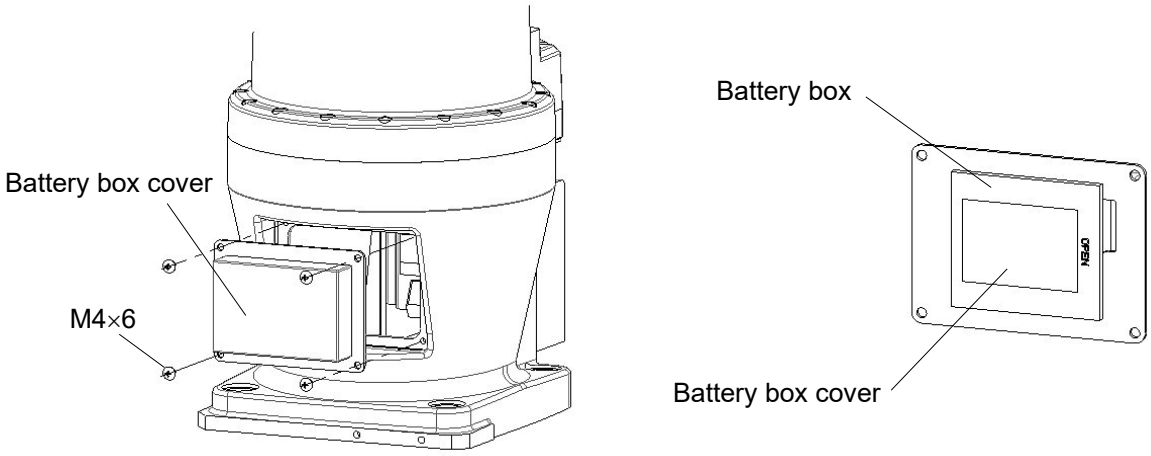
Fig. 2.14 Battery box location

2.9.2 Replacing Batteries

! DANGER

- Only after assuring the safe work, replace the batteries with new ones while the power is turned on and an emergency stop is effected on the robot.

1) Assure the safe work. Then keep the power ON and press the EMERGENCY pushbutton switch to effect an emergency stop on the robot. Unless the safe work can be assured, turn the power off.
2) Disconnect the battery box cover which is secured to the base unit with the four (4) truss head screws (M4 × 6). As the cables are connected with the cover, DO NOT pull the cover by force.
3) Remove the battery box cover followed by the batteries.
   Set the three (3) new batteries, taking care of the "+" and "–" polarities.
   To replace the batteries while the power is turned off, finish the replacement within five (5) minutes.



4) Attach the battery box cover to the base unit. Now replacement of the batteries completes.

2.9.3 Battery Error Code

When a position detector error including battery error has occurred, the following error code is shown on the error display.

8–065 Axis1 Encoder abnormal
8–097 Axis2 Encoder abnormal
8–129 Axis3 Encoder abnormal
8–161 Axis4 Encoder abnormal
8–193 Axis5 Encoder abnormal

When a battery alarm has occurred, the following error code is shown on the error display. When the following error code is shown in the error history at generation of a position detector error, the error can be identified as the battery error.

1–049 Axis1 Battery alarm
1–065 Axis2 Battery alarm
1–081 Axis3 Battery alarm
1–097 Axis4 Battery alarm
1–113 Axis5 Battery alarm

The battery error is one of the position detector errors (i.e., encoder errors), and there is other cause leading to the position detector error. The restoring method from the position detector error varies with the error contents generated. To properly execute the restoring method from the position detector error, therefore, read through and completely understand the descriptions carried in Section 4.

3. Maintenance of Controller

3.1 Cautions on Maintenance and Inspection

When performing maintenance and inspection of robot controller, follow the items given below so that the work can be carried out safely.

! CAUTION

- Before removing the controller cover for maintenance or inspection, be sure to turn off the main power switch of the controller.
  Also, turn off the power breaker when connecting or disconnecting the power cable.
- DO NOT touch the servo driver for at least two (2) minutes after the main power switch is turned off. The large-capacity capacitor in the servo driver may be charged with voltage, and touching the servo driver could lead to electric shocks.
- DO NOT disconnect the battery connector at other than battery replacement. Otherwise, files saved in the memory may be lost.

! DANGER

Be sure to turn off the main power switch. If the main power switch remains ON, the servo power printed board, servo printed board and switching power supply are charged, and touching them could lead to electric shocks.

3.2 Layout of Controller Parts

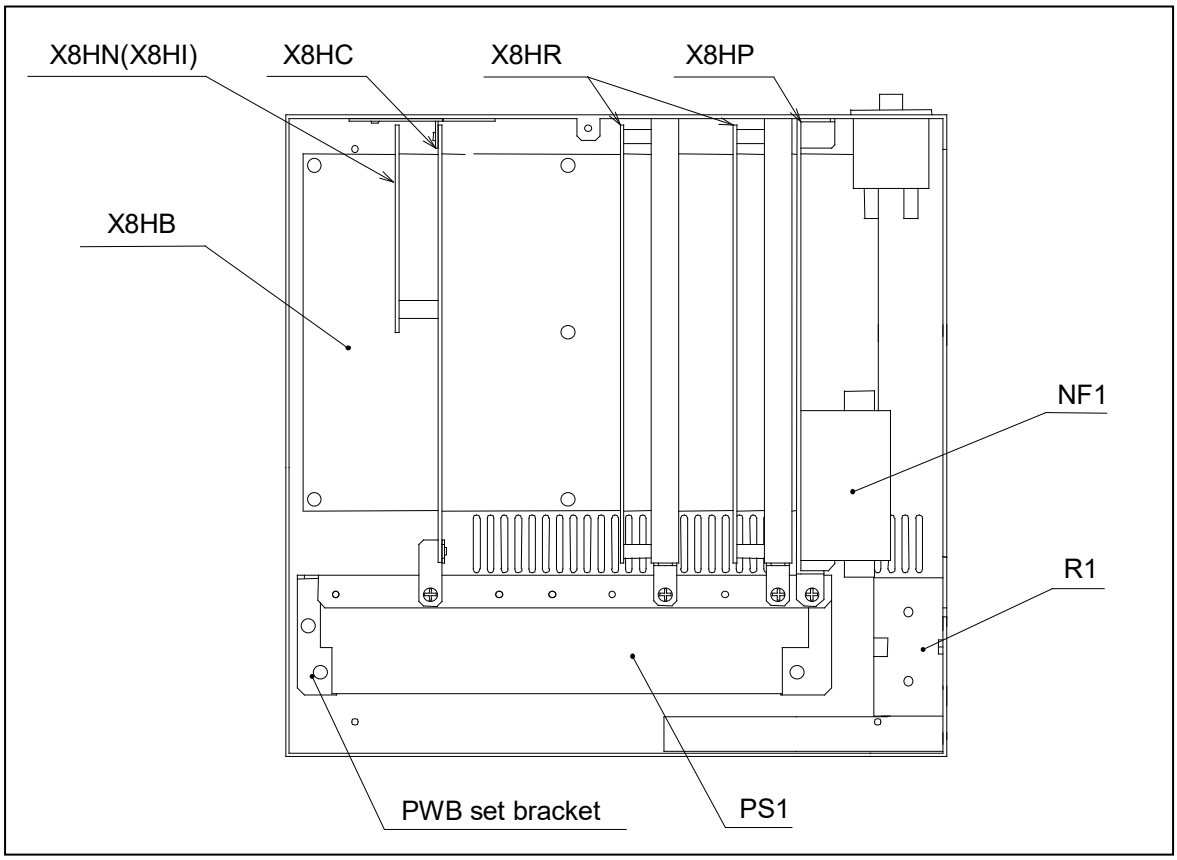


Fig. 3.1 Layout of controller parts

Part name	Descriptions
PS1	P5V, P24V switching power supply
PWB set bracket	Bracket for mounting PS1, etc.
X8HP	Servo power printed board
X8HC	Main control printed board
X8HN (X8HI)	I/O output printed board
X8HR	Servo printed board
X8HB	Mother board
R1	Regenerative resistor
NF1	Noise filter

3.3 Maintenance Procedures

3.3.1 Check of Controller for Air Vent Holes

If the air vent holes are blocked, the controller may overheat and malfunction. To avoid this, perform check on the air vent holes to make sure that air is flowing freely through them.

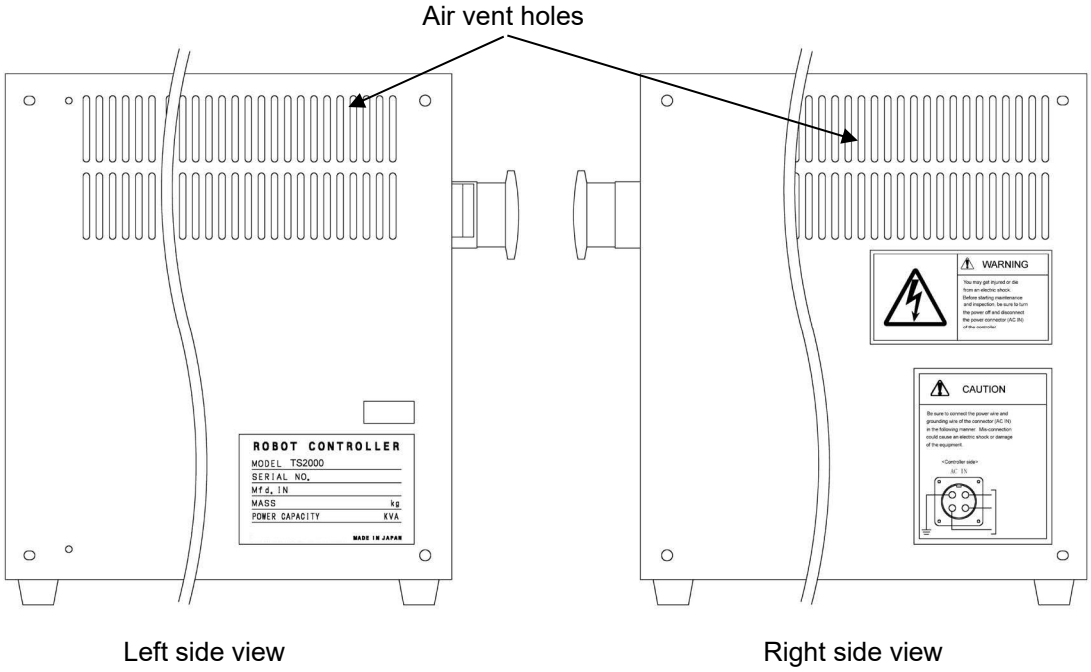


Fig. 3.2 Controller side views

a) Make sure that no obstacle stands in front of the air vent holes.
If there are any obstacles, move them away from the vent holes so that the air flow is not blocked.

b) Make sure that no contaminant is left on the air vent holes. If any contaminant is let on the vent hole, remove it.

c) If the controller is placed sideways, make sure that the air vent holes on the bottom surface of the controller are more than 11 mm away from the floor surface. Always keep the space of 11 mm or over. The length of rubber shoe is approximately 11 mm.

3.3.2 Check of Safety Devices for Function

Make sure that the emergency stop pushbutton switches equipped on the control panel and teach pendant work properly. Also make sure that respective safety devices connected with external operation input signals work correctly.

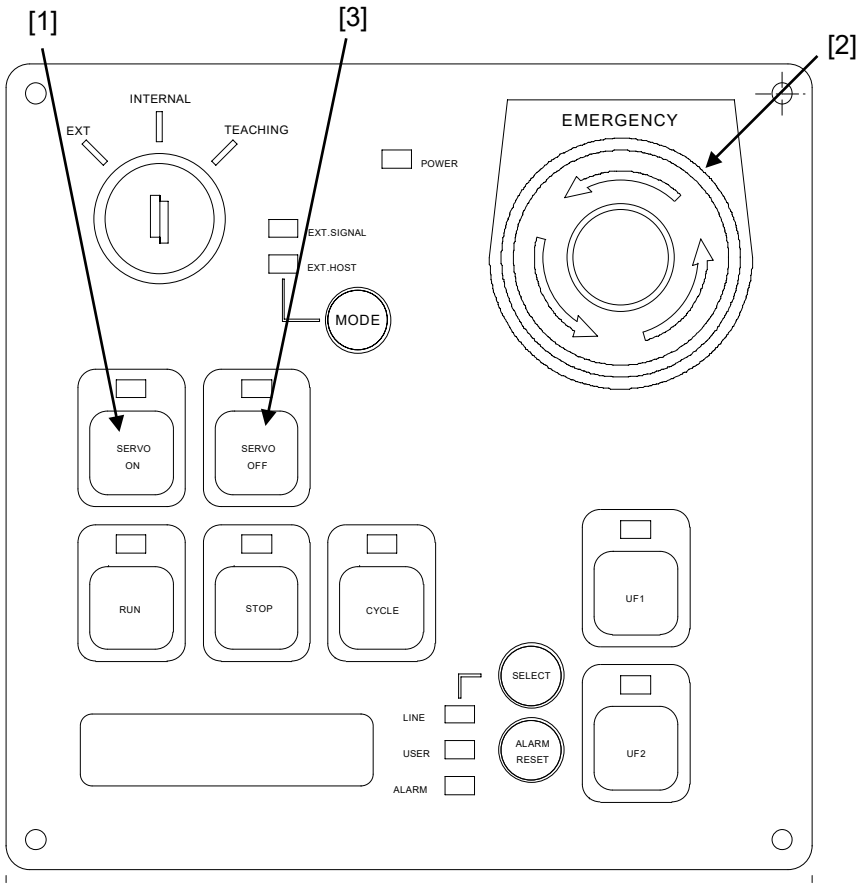


Fig. 3.3 Layout of control panel switches

a) Make sure that the EMERGENCY (emergency stop) switch [2] provided on the control panel functions properly.

1) Press the SERVO ON switch [1] provided on the control panel to turn the servos on and make sure that the SERVO ON switch [1] LED is illuminated.

2) Press the EMERGENCY switch [2] and make sure that the servos are turned off. The SERVO ON switch [1] LED goes out and the SERVO OFF switch [3] LED is illuminated. At this time, make sure that the EMERGENCY switch [2] remains depressed.

3) Turn the EMERGENCY switch [2] clockwise and make sure that the same switch turns off.

b) Make sure that the EMERGENCY STOP switch [6] provided on the teach pendant functions properly.

1) Press the SERVO ON switch [4] on the teach pendant to turn on the servo power, and make sure that the SERVO ON switch LED [5] is illuminated.

Note: The servo power cannot be turned on in the TEST RUN mode.

2) Press the EMERGENCY STOP switch [6] on the teach pendant and make sure that the servos are turned off. The SERVO ON switch LED [5] goes out.

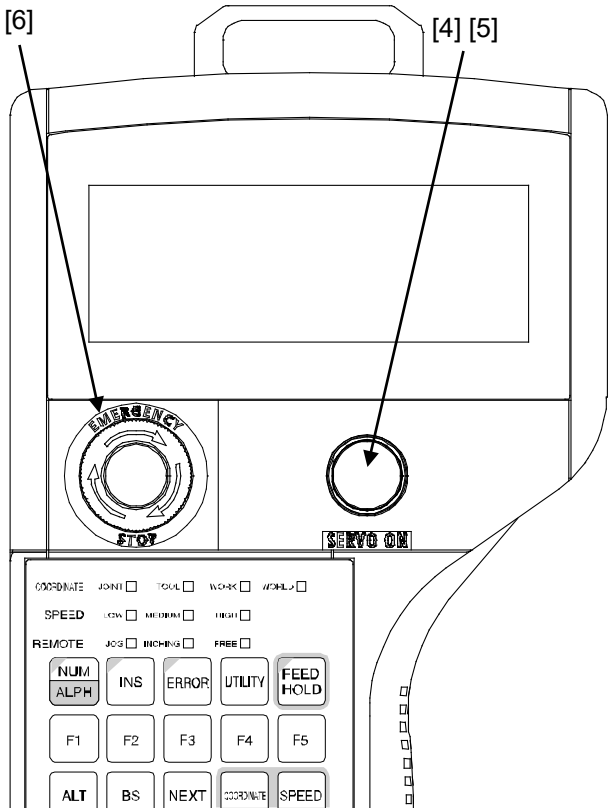


Fig. 3.4 EMERGENCY STOP pushbutton switch equipped on teach pendant

At this time, make sure that the EMERGENCY STOP switch [6] remains depressed.

3) Turn the EMERGENCY STOP switch [6] clockwise and make sure that the same switch turns off.

c) Make sure that each safety device connected with the external operation signal works properly.

1) Press the SERVO ON switch [1] equipped on the control panel to turn on the servo power. Then turn on (or trip) the safety devices connected with external operation input signal "emergency stop" (such as external emergency stop switch, photoelectric type sensing safety device and foot switch) to make sure that the servo power is turned off.

2) Set ON the safety devices connected with external operation input signal "low speed command", and make sure that each signal functions properly.

3.3.3 Battery Replacement

The memory equipped on the X8HC printed board of the TS2000 robot controller is backed up by a lithium battery to save the data. Replace the battery every five (5) years. The lithium battery will turn its life when used for a predetermined time. If it is used, neglecting the life, the battery voltage will drop to below the voltage required for keeping the memory data, resulting in data being lost and faults caused by leakage of the battery liquid.
If the following alarm has occurred, replace the battery with a new one.

- 1–145 MAIN Battery alarm

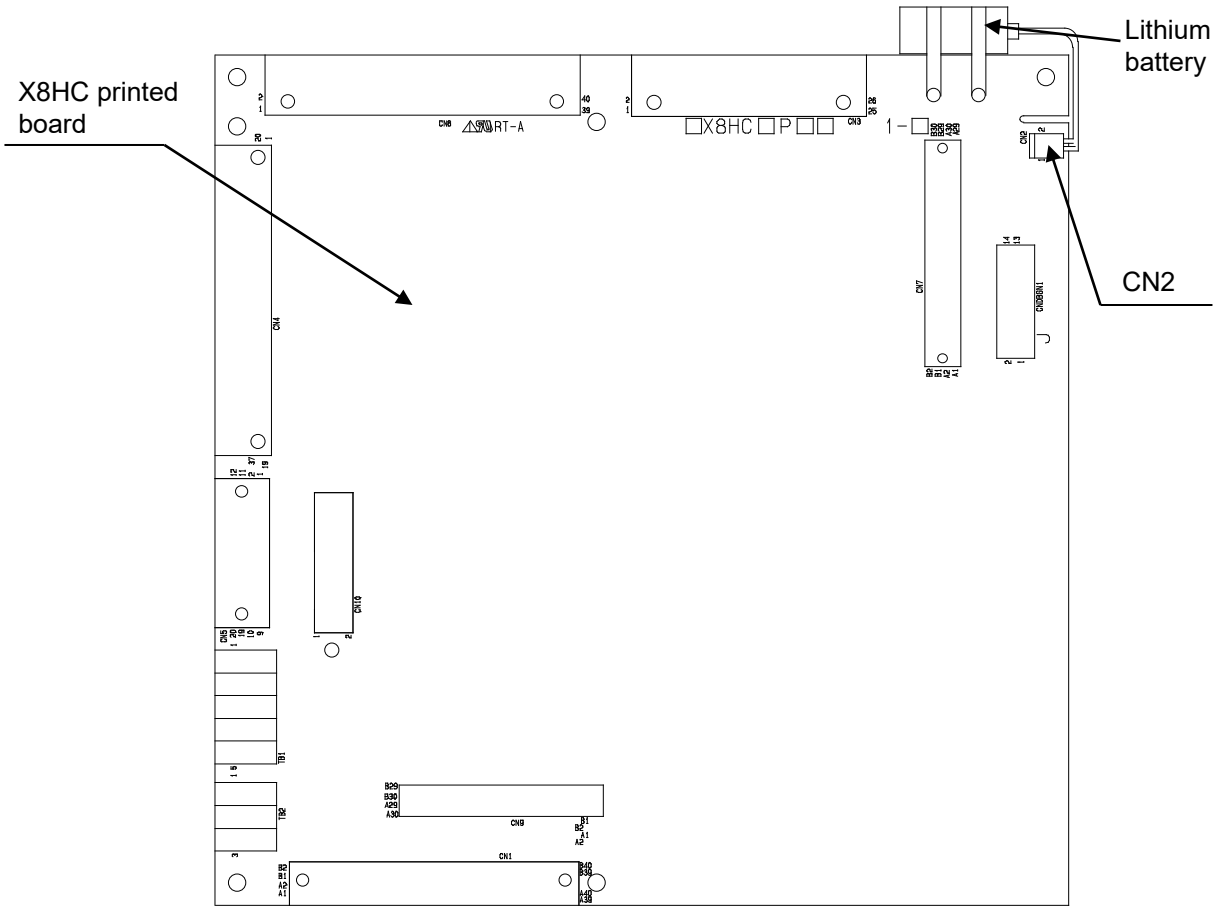


Fig. 3.5 Location of lithium battery

Type of battery	Manufacturer	Battery service life
ER3V–C	Toshiba Battery	10 years (at room temperature)

Note: The battery life is ten (10) years at the room temperature. As it differs with the external environment such as operating temperature and humidity, we recommend to replace the battery every five (5) years.
Only the battery shown above should be used. As this is an exclusive battery, contact SHIBAURA MACHINE at order entry.

[Battery replacement procedures]

1) If the battery is to be kept removed for more than five (5) minutes, copy all programs and parameter data stored in the memory to the personal computer. In the worst case, all memory data are lost.

2) Set OFF the main power switch.

3) Prepare a new battery, then remove the cover from the controller.

4) Remove the battery connector from CN2 of the X8HC printed board, then disconnect the battery from the TY-RAP securing the battery.
(For the battery location, see Fig. 3.5.)

5) Set the new battery immediately. (If the battery is kept removed for more than one (1) minute, all memory data are lost.)
Even if not used, the lithium battery capacity decreases by natural discharge. Use a lithium battery which is within three (3) years (kept at the room temperature) from purchasing.

6) Mount the cover on the body.

7) If the battery is kept removed for more than one (1) minute, turn on the main power switch and load the programs and parameter data from the floppy disk unit or personal computer in which the TSPC (option) is already installed.
If the controller has detected a battery voltage drop alarm even before the five (5)-year replacement period, replace the battery immediately.

! CAUTION

- Waste battery should be disposed of according to the user's in-house regulations.
- NEVER drop the battery into fire. NEVER short-circuit, charge, disassemble or heat it. Otherwise, liquid leakage or rupture may be caused.

3.3.4 Replacement of Switching Power Supply Unit

The life of the switching power supply unit (DC5V, DC24V) used in the robot controller differs with the operating conditions. This unit uses an aluminum electrolytic capacitor. If the load current is large, the running time is long and the ambient temperature is high, the life will reduce.
Though the life varies largely with the user's operating conditions (5 ~ 10 years), replace the power supply unit every five (5) years for preventive maintenance.

[Replacement procedures]

1) Turn off the main power switch.

2) Turn off the power breaker.

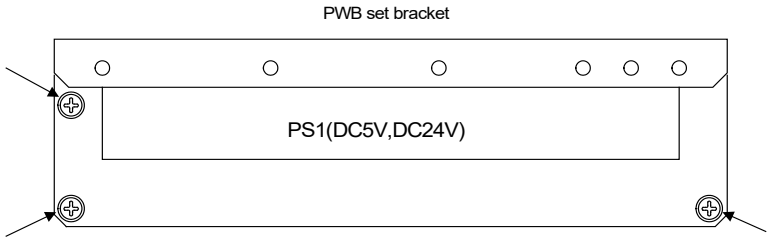
3) Remove the cover from the controller.

4) Remove all connectors in the control panel, which are connected to the X8HC, X8HRA (2 pcs.) and X8HP printed boards. (For the location of each printed board, see Fig. 3.1.)

5) Remove the X8HC, X8HN (X8HI), X8HRA (2 pcs.) and X8HP printed boards from the controller.

6) Disconnect the cables connected to the parts on the PWB set bracket.

7) Remove the screws (3 places) which secure the controller and PWB set bracket.(For the location of the PWB set bracket, see Fig. 3.1.)



Upper side view

Fig. 3.6 PWB set bracket (upper side view)

8) Disconnect the PWB set bracket from the controller.

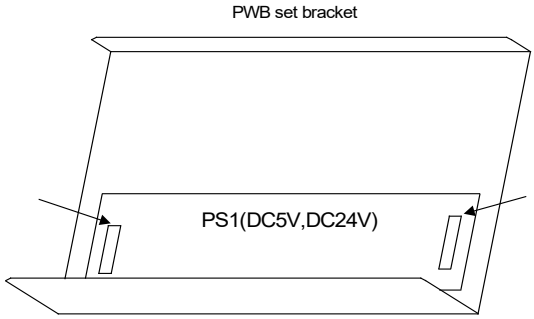


Fig. 3.7 PWB set bracket

9) Remove each power supply unit from the PWB set bracket, and set a new power supply unit.

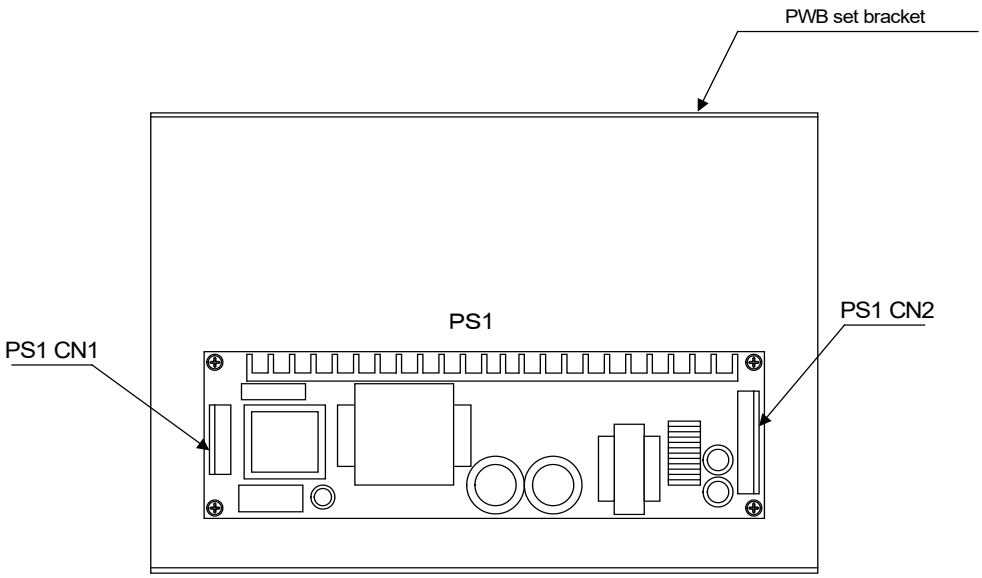


Fig. 3.8 PWB set bracket (front view)

10) Restore the PWB set bracket to the controller, where the new power supply unit is set. Then connect the connector.

11) Secure the PWB set bracket to the controller, using the three (3) screws. (See Fig. 3.6.)

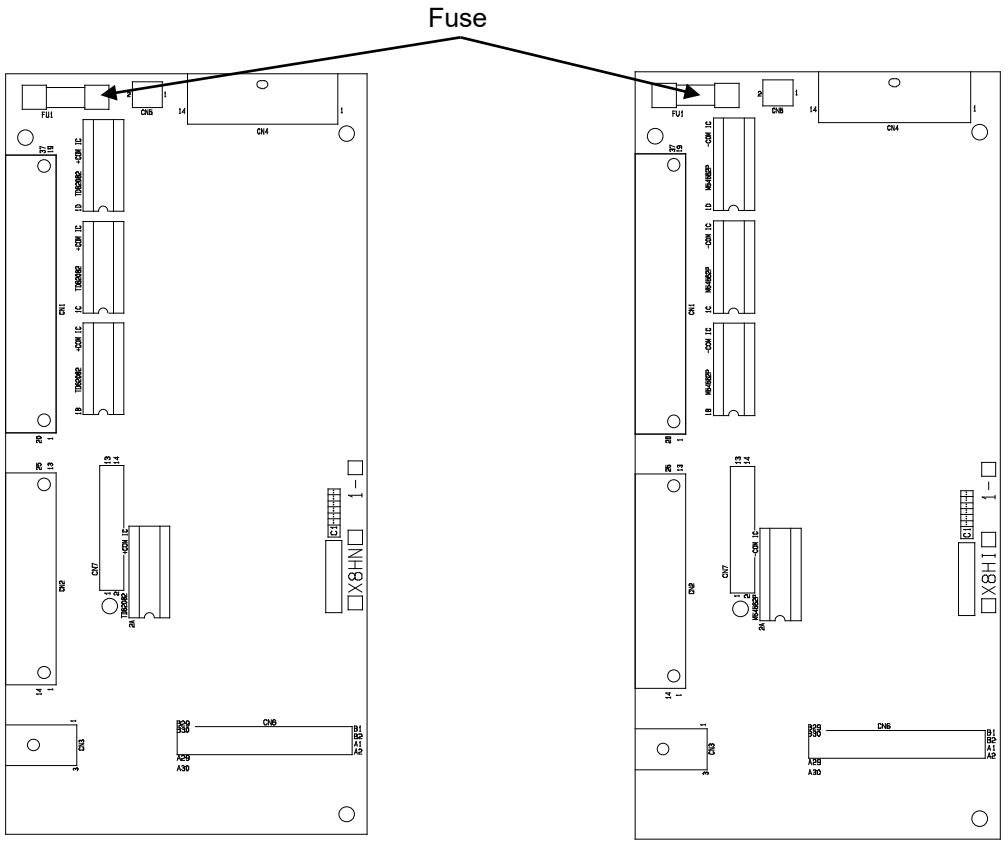
12) Mount the disconnected printed boards on the controller.

13) Connect the disconnected connectors with respective printed boards.

14) Attach the cover to the controller.

3.3.5 Replacement of Fuse (X8HN, H8HI Printed Boards)

If the current exceeding the specified current has run through the I/O unit, the fuse of the X8HN (X8HI) printed board will blow out. If the alarm saying "I/O FUSE Broken (8–273)" is displayed, replace the fuse with a new one.
If the fuse of the output module has blown out, examine and identify an fault circuit, remedy the cause, then replace the fuse.



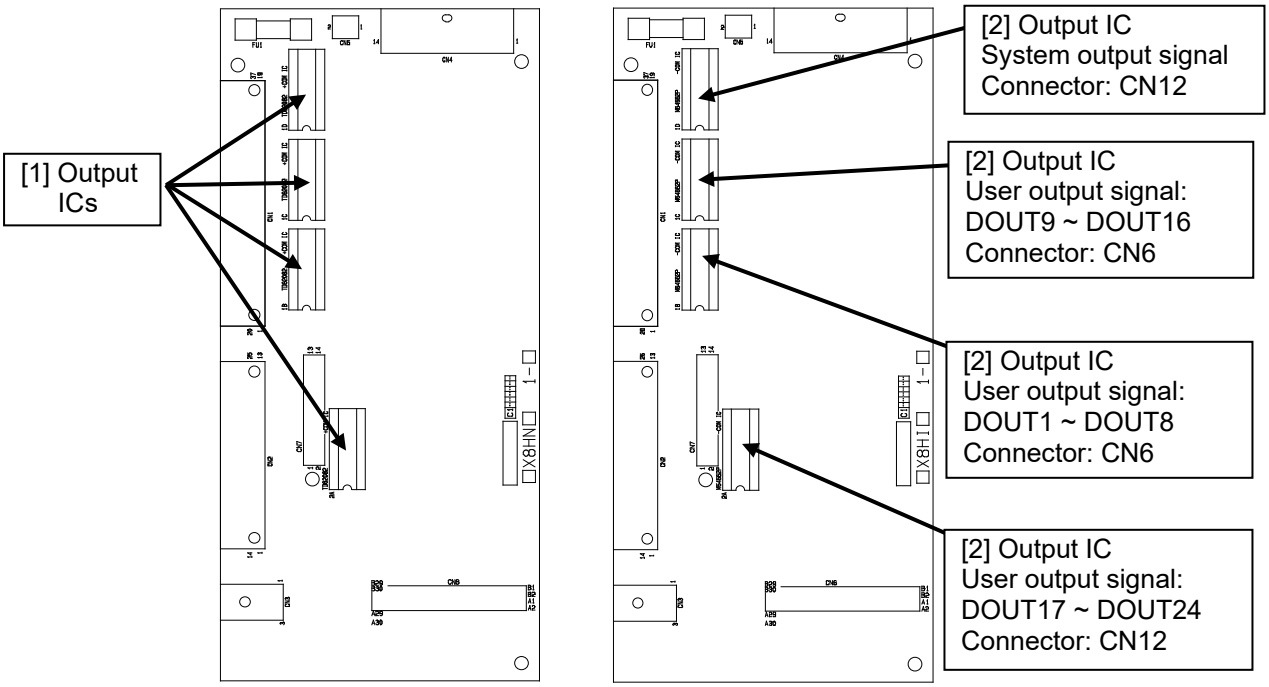
Type of fuse	Manufacturer
51NM030H	PICO

[Replacement procedures]

1) Turn off the controller main power.
2) Remove the cover attached to the fuse unit as shown above.
3) Remove the fuse and set a new fuse.
4) Mount the removed cover.
5) Turn on the controller main power and make sure that no error will occur.

3.3.6 Replacement of Output ICs (X8HN, X8HI Printed Boards)

If the current exceeding the specified current has run through the output unit, the ICs on the X8HN (X8HI) printed board will be damaged.
When this happens, replace the ICs.
Before the replacement, examine and identify an fault circuit, remedy the cause, then replace the ICs.
The relationship between the output ICs [2] and output signals is shown below.



	Name of board	Type of IC	Manufacturer
[1]	X8HN	TD62082AP	Toshiba Corp.
[2]	X8HI	M54562P	Mitsubishi Electric

[Replacement procedures]

1) Turn off the controller main power.
2) Remove the cover from the controller.
3) Disconnect all connectors connected to the X8HC printed board in the controller.
4) Disconnect the X8HC, X8HN (X8HI) printed boards from the controller.
5) Remove the ICs shown above from respective sockets.
6) Attach the new ICs to the sockets.
7) Mount the X8HC, X8HN (X8HI) printed boards on the controller.
8) Connect all disconnected connectors to the X8HC printed board, then attach the controller cover.
9) Turn on the controller main power.

4. Robot Home Point and Position Detector Error

4.1 Setting Robot Home Point

Before delivery from our plant, home point setting is performed for the robot after its arm is secured with clamp for home point setting.
At the time of home point setting, position data of the motor position detector (i.e., encoder) is backed up by batteries, and coordinates of the robot need not be set each time the power is turned on.
The position data of the motor position detector comes in the two (2) types; home position data within one (1) full turn of motor and motor multi-turn data.

Home position data within one (1) full turn of motor:
At home point setting operation of the robot (i.e., ZEROP operation and REORG operation), the data is written into the parameter file. As the data is backed up by batteries, the parameters should be loaded again from the attached system disk at replacement of the main control printed board.

Motor multi-turn data:
This data is kept in the encoder by battery backup, which differs from the backup battery of the main control printed board. When the battery voltage drops, correct multi-turn data cannot be guaranteed and the encoder itself outputs an alarm. This data is set to zero (0) at home point setting operation of the robot (i.e., ZEROP operation and REORG operation). Likewise, it is set to zero (0) also by multi-turn data reset operation.

4.2 Position Detector Error

If a position detector error (i.e., encoder error) occurs, either of the following errors is shown on the error display of the robot controller.

Error code	Error content
8–065	Axis1 Encoder abnormal
8–097	Axis2 Encoder abnormal
8–129	Axis3 Encoder abnormal
8–161	Axis4 Encoder abnormal
8–193	Axis5 Encoder abnormal

The position detector error is an error detected by the encoder itself. Even if the data in the parameter file is lost (or changed) to change the home point coordinates of the robot, for instance, a position detector error will not occur. If the machine coordinates have changed without causing a position detector error, the mechanical connecting position of the servo motor and mechanical unit shifted or the data in the parameter file described above was changed. (Ex.: Tooth skip of timing belt, etc.)

Errors detected by the encoder itself include the battery voltage drop, error caused by temperature rise in the encoder, counter overflow, internal counter data inconsistency, etc. Among these errors, the error which occurs most frequently is the battery voltage drop which is caused by the absence of maintenance at specified change intervals (once a year) due to a long-term shutdown or cable breakage.
To prevent the position detector error, therefore, replace the batteries on a regular basis. For the battery replacement, see Para. 2.9.
Also, a heavy vibration should not be exerted on the robot or the robot should not be moved suddenly by hand during power OFF. Especially, when transporting and storing the robot, secure the robot in a posture for shipment, using the attached clamps.

Restoration from position detector error:
The restoring operation from the position detector error comes in the following four (4) manners.

[1] Encoder error reset operation: See Para. 4.3.2.
[2] HOME operation: See Para. 4.4.
[3] ZEROP operation: See Para. 4.5.
[4] Multi-turn data reset operation: See Para. 4.6.

(1) When the mechanical connecting position with the motor has changed after motor or belt replacement:

A position detector error may occur after replacement and adjustment of the servo motor or timing belt. When this happens, execute the following restoring operation.

1) Execute “[2] HOME operation.”
2) Execute “[3] ZEROP operation.” At this time, the contents in the parameter file (i.e., home position data within one (1) full turn of motor) are updated. Rewriting of the data into the system disk is necessary.
3) Make sure that the error has been reset, referring to the error display.
4) In the “[2] HOME operation” and “[3] ZEROP operation”, some error may be caused according to the adjusting method. Re-teaching of the teach points may be required under some circumstances.

After resetting the error, effect an emergency stop, manually move each axis to the motion limit and make sure that a soft stroke limit error generates. If the mechanical interference is caused before generation of the soft stroke limit error, repeat the restoring operation in the manner described above.

(2) When a position detector error has occurred during normal operation with the mechanical connecting position unchanged:

A position detector error may occur after battery voltage drop, or battery or cable replacement. When this happens, execute the following restoring operation.

1) Execute “[1] Encoder error reset operation.”
2) Make sure that the error has been reset, referring to the error display.

After resetting the error, effect an emergency stop, manually move each axis to the motion limit and make sure that a soft stroke limit error generates.

! DANGER

- When moving the robot by hand while the power is turned on, be sure to assure the safe work and effect an emergency stop beforehand.
- In the above situation, if the work is to be done while the axis 3 motor brake OFF switch is pressed, be sure to perform the work by two (2) persons.
  One person should carry out the work while the other is monitoring the work outside the hazardous zone. The latter person should watch the work and be ready to turn off the controller switch immediately if any abnormality occurs.
  Once the controller power is turned off, the motor brake actuates even if the axis 3 motor brake OFF switch is kept pressed.
- If the axis 3 motor brake OFF switch is pressed while the robot is carrying a heavy workpiece, the axis 3 may drop suddenly. To avoid this, the customer should take all necessary measures by themselves.

2.1) **When the error has been reset:**

Turn on the controller power only (DO NOT turn on the servo power), and make sure by moving the arm, etc. by hand that the position detector error will not occur again. Then turn off the controller power and turn on again to make sure that the same error will not recur.

2.1.1) **When the error occurs again:**

The position detector has possibly malfunctioned.

2.1.2) **When the error will not occur again:**

Turn on the servo power. Select the TEACHING mode and gradually move each axis at low speed to make sure of its soft stroke. If the soft stroke or machine coordinate is abnormal, perform “[4] Multi-turn data reset operation.”

2.2) **When the error is not reset:**

Replace the batteries, then perform “[1] Encoder error reset operation” again. Make sure again that the error has been reset, referring to the error display.

2.2.1) **When the error has been reset:**

Execute the same operation as in Para. 2.1) above.

2.2.2) **When the error cannot be reset:**

The position detector has possibly malfunctioned, or the encoder cable has broken.

4.3 Restoring Operation from Position Detector Error

Make sure on the error display that a position detector error is generated. Identify the type of the position detector error and position data on the encoder status display, then perform each restoring operation.

4.3.1 Encoder Status Display

Call the encoder status screen on the teach pendant display, and make sure of the position data and the type of position detector error there. The type of the position detector error (i.e., error status) is expressed as shown below. The motor multi-turn data, home position data within one (1) full turn of motor and error status are displayed under the columns of [MULTI], [SINGLE] and [Err-df], respectively.
To call the encoder status screen, observe the following procedures.

Table 4.1 Error status table

Error status	Description
0000	Signifies the normal condition.
0100	This error occurs when a heavy vibration has been exerted on the robot or when the robot has been moved fast by hand while the power drive cable was disconnected during power OFF. The multi-turn data has possibly shifted.
0200	Because axis feed was not executed after power ON, the encoder resolution has deteriorated. During axis feed, the encoder resolution returns to normal. No alarm will generate.
8000	The battery voltage has dropped. Replace the batteries.
4000	The battery voltage has dropped further than the above. An error has possibly occurred in the multi-turn data.
4200	Errors 4000 and 0200 have occurred at the same time.
C200	Errors 8000, 4000 and 0200 have occurred at the same time.

[1] Press the “UTILITY” key provided on the teach pendant. Press the [NEXT] key until [ENC] is displayed on the function menu.

[2] Press the [ENC] (F5) key, and the encoder status screen appears on the display.



If the error status shown in the error status table above is displayed, you can restore from the position detector error.

4.3.2 Encoder Error Reset Operation

When the mechanical connecting position of the servo motor and mechanical unit remains unchanged, that is, after battery replacement due to battery voltage drop or cable replacement due to cable breakage, execute this operation.

[1] Call the encoder status screen on the teach pendant display, referring to Para. 4.3.1 above.

[2] Just after the encoder status screen is called, the cursor points out the [Err-df] column on the [E1] line. Move the cursor to a line for the system you wish to execute the encoder error reset operation by means of the cursor keys (i.e., "↑" and "↓" keys).

	MULTI	SINGLE	Err-df	ID
E1	0	00001234	0200	10
E2	0	00001234	0200	10
E3	0	00001234	**4200**	10
E4	0	00001234	0200	10
E5	0	00000000	0000	00

ERROR **RESET**

[3] To reset the axis 3 encoder error, for instance, move the cursor to the [Err-df] column on the [E3] line and press the [RESET] (F3) key. If everything is OK, press the "EXE" key. Now the axis 3 encoder error has been reset.

[4] Call the normal error screen, press the [RESET] (F3) key to reset the error of "8–129 Axis3 Encoder Abnormal."

4.4 Restoring Home Position Data by HOME Function

Batteries for backing up the position information of the position detector are housed in the base unit of the robot. Unless the robot has been used for a long term, the battery voltage drops and the home position data will be lost. Also, when the battery voltage for backing up the internal memory of the X8HC printed board has dropped, user's set programs, parameters and absolute coordinates will be lost. To avoid this, replace the batteries on a regular basis.

4.4.1 Outline of HOME Function

This function memorizes a specific coordinate value for each axis and restores the original home position data by using the HOME function when the machine home position data has been destroyed.

The set points are provided for HOME1 to HOME4, and the axis 1 to axis 3 plus (+) side mechanical stopper positions are set in HOME3, and the axis 1 to axis 3 minus (–) mechanical stopper positions in HOME4. The axis 4 is set at the 0° position according to the home point match-mark.
For HOME1 and HOME2, you can specify any values. It is recommended that when building up a system, you specify the position for securing the robot end.

(Page 1)

USER	HOME1	HOME2	
(J1)	0	0	[deg]
(J2)	0	0	[deg]
(J3)	0	0	[mm]
(J4)	0	0	[deg]
(J5)	0	0	[mm]

SET TEACH

(Page 2)

SYSTEM	HOME3	HOME4	
(J1)	**1183468**	– 1180474	[deg]
(J2)	1450716	– 1450720	[deg]
(J3)	1239092	– 36711	[mm]
(J4)	0	0	[deg]
(J5)	0	0	[mm]

SET TEACH

Caution: This function is used to restore the machine home point of each axis. NEVER use the function at other than the setting for restoration.

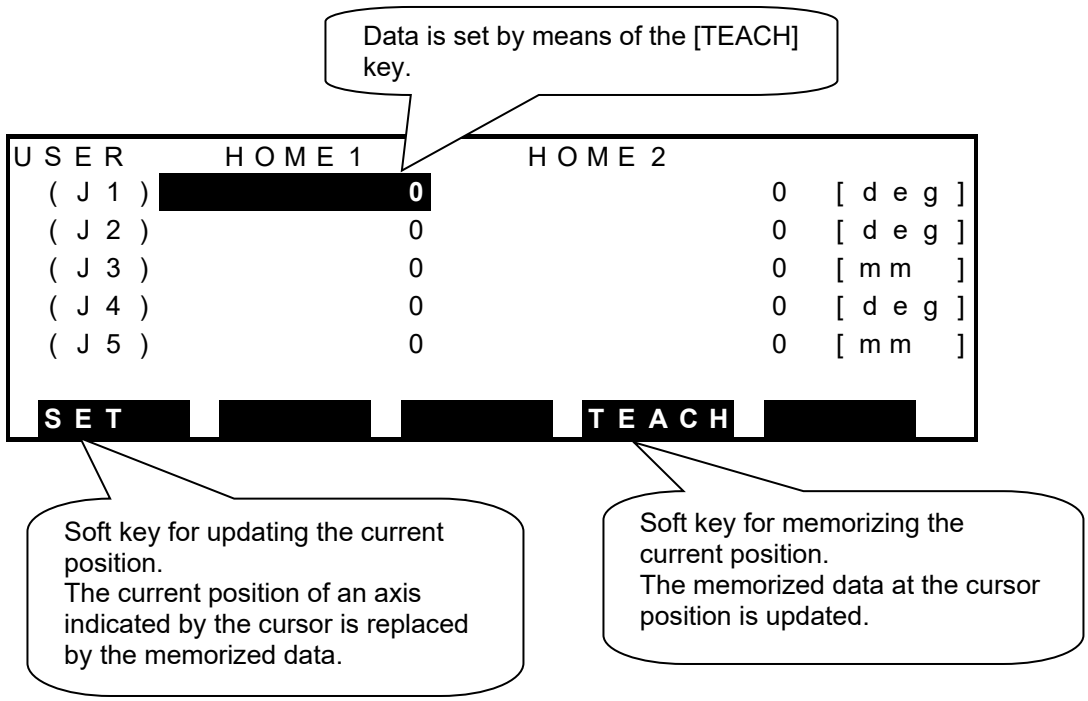
4.4.2 Setting HOME1 and HOME2

1) Guide the robot to secure the tool shaft at a desired position.

2) Turn off the servo power.

3) Press the “UTILITY” key equipped on the teach pendant.

4) Press the [NEXT] key until the [REORG] menu appears, then press the [REORG] key.



5) When the above screen appears, memorize the position where the robot was secured for each axis by moving the cursor and using the [TEACH] mode.

4.4.3 How to Restore Data by HOME1 or HOME2

1) Guide the robot and secure it at the position set in HOME1 or HOME2 above.
2) Turn off the servo power.
3) Press the “UTILITY” key equipped on the teach pendant.
4) Press the [NEXT] key until the [REORG] menu appears, then press the [REORG] key.
5) Move the cursor to each axis under HOME1 of the REORG screen, then press the [SET] key to update the coordinate value set in HOME1 to the position where the robot was secured. Thus, the machine home position can be restored.

! CAUTION

- The axis 3 and axis 4 of the robot are interlocked with each other. Unless the specified procedures are followed, therefore, the current position of the robot may shift.
  When restoring the home point of the axis 3 (or axis 4) by using the [REORG] function, be sure to restore the home point of the axis 4 (or axis 3) also.

  Operating procedures

  1) Secure the axes 3 and 4 to each mechanical stopper or specified position.
  2) Move the cursor to (J3) on the screen and press the [SET] key to restore the axis 3 home point.
  3) Likewise, move the cursor to (J4) on the screen and press the [SET] key to restore the axis 4 home point.

4.4.4 Locations of Robot Home Point Match-Marks

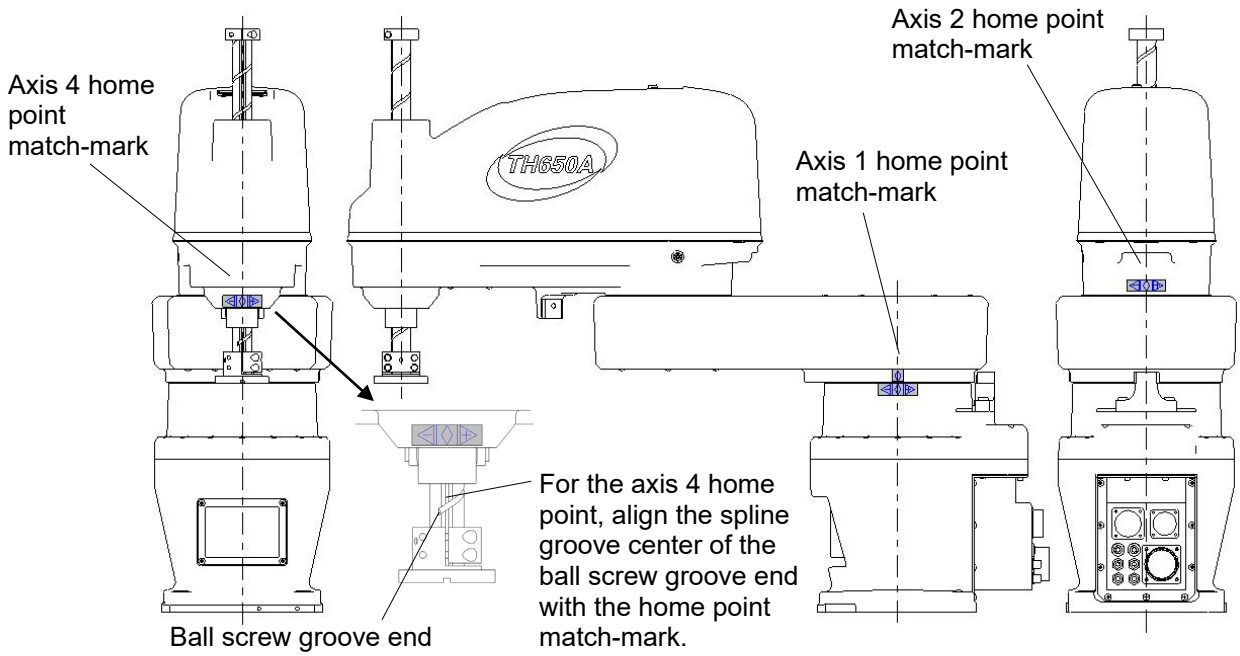
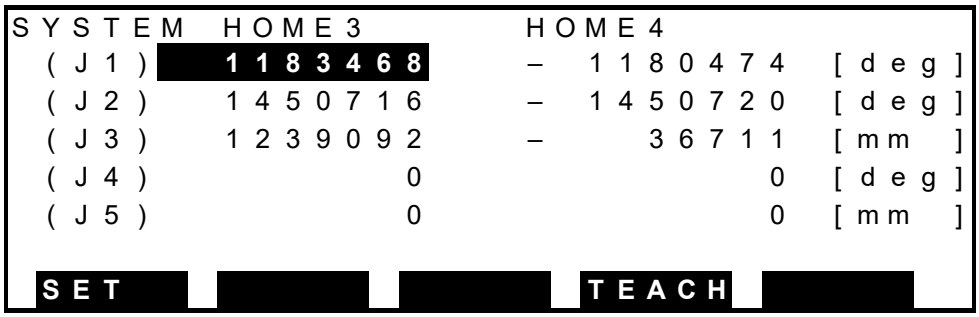


Fig. 4.1 Locations of home point match-marks

4.4.5 How to Restore Data by HOME3 or HOME4

The plus (+) side mechanical stopper positions and minus (–) side mechanical stopper positions of the axes 1 to 3 are factory-set in HOME3 and HOME4, respectively. The axis 4 is set to the home point match-mark.
The machine home point can be restored, using the above.
This paragraph deals with the restoration procedures, using HOME3.

1) Turn off the servo power.
2) Press the “UTILITY” key provided on the teach pendant.
3) Press the [NEXT] key until the [REORG] menu appears, then press the [REORG] key.
4) Press the [NEXT] key to call Page 2.



5) Contact the axis 1 to the plus (+) side mechanical stopper by hand.

6) Move the cursor to (J1) of HOME3, then press the [SET] key. Now, the machine home point of the axis 1 is restored.

7) Likewise, contact the axis 2 to the plus (+) side mechanical stopper by hand.

8) Move the cursor to (J2) of HOME3, then press the [SET] key. Now, the machine home point of the axis 2 is restored.

9) Press the [ESC] key to escape from the REORG screen once.

10) Turn on the servo power, and set the axes 3 and 4 in the servo-free state.

11) Set the axis 4 to the home point match-mark.

12) Contact the axis 3 to the plus (+) side mechanical stopper by hand.

13) Turn off the servo power.

14) Repeat Steps [2] to [4] inclusive to call the HOME screen again.

15) Move the cursor to (J4) of HOME3, then press the [SET] key. Now, the machine home point of the axis 4 is restored.

16) Move the cursor to (J3) of HOME3, then press the [SET] key. Now, the machine home point of the axis 3 is restored.

17) Press the [ESC] key to escape from the HOME screen. Now, the operation has finished.

4.5 Restoring Home Position Data by ZEROP Function

Batteries for backing up the position information of the position detector are housed in the base unit of the robot. Unless the robot has been used for a long term, the battery voltage drops and the home position data will be lost. Also, when the battery voltage for backing up the internal memory of the X8HC printed board has dropped, user's set programs, parameters and absolute coordinates will be lost. To avoid this, replace the batteries on a regular basis.

When the robot arm is moved fast during power OFF, the absolute coordinate counter cannot follow the robot motion, resulting in destruction of the home point data.
This paragraph deals with the restoring procedures when the home position data was destroyed by mistake.
This restoring operation may cause some error according to the adjusting method.

4.5.1 Preparation before Restoring Operation

1) Turn on the controller power and align the home point match-marks of each axis in the manual mode.
2) If this is not possible, mark the reference position of each axis and jot down its coordinate beforehand.
3) Move the axis 3 down to near the lower limit and set it to the dimension shown in the figure below (i.e., distance between the boss end surface of tool shaft and the tool flange end surface).

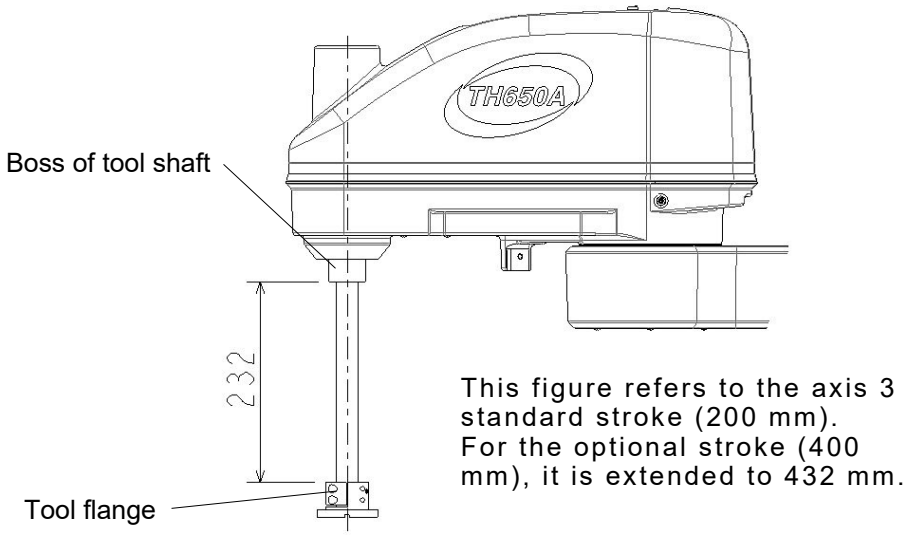
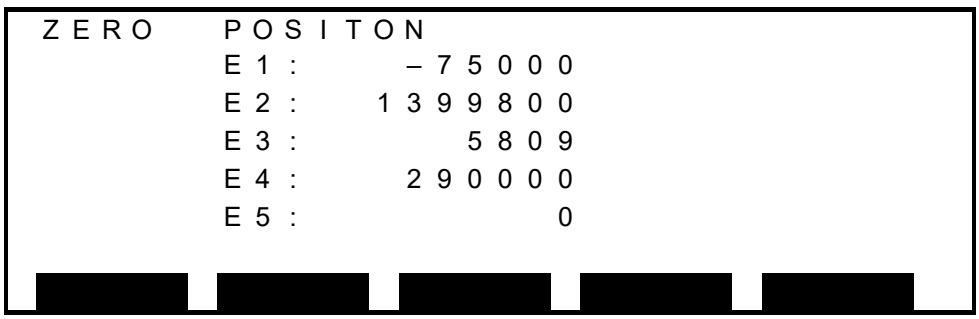


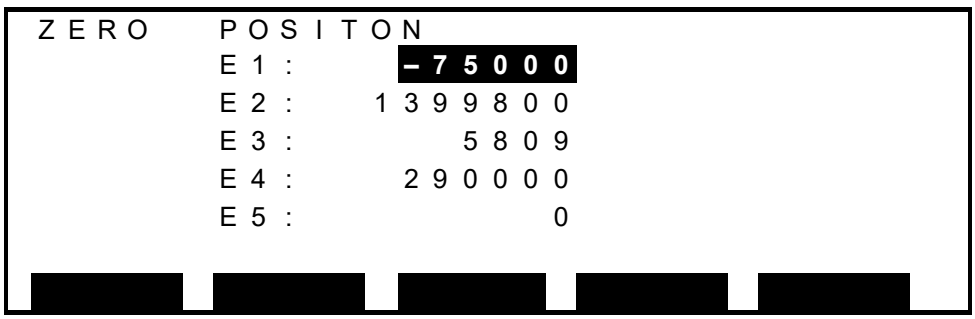
Fig. 4.2 Dimension of axis 3 home point

4.5.2 Setting New Home Position Data

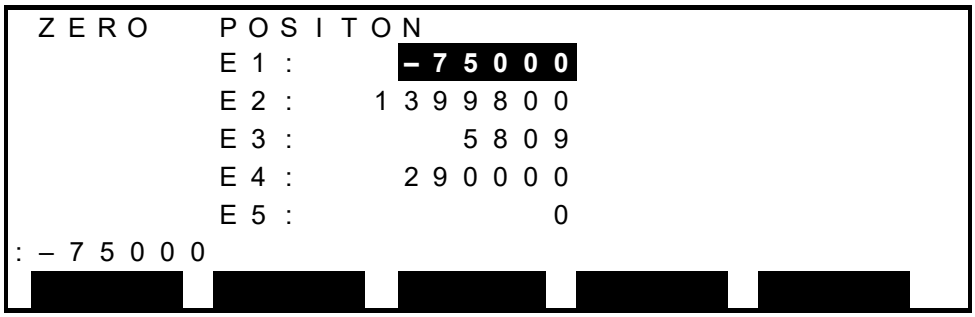
1) Select the TEACHING mode by means of the master mode switch.
2) Press the “UTILITY” key equipped on the teach pendant.
3) Press the [NEXT] (F6) key on the teach pendant twice.
4) Turn off the servo power.
5) Press the [F3] key on the teach pendant to select the ZEROP mode.
6) The following home point setting screen is called.



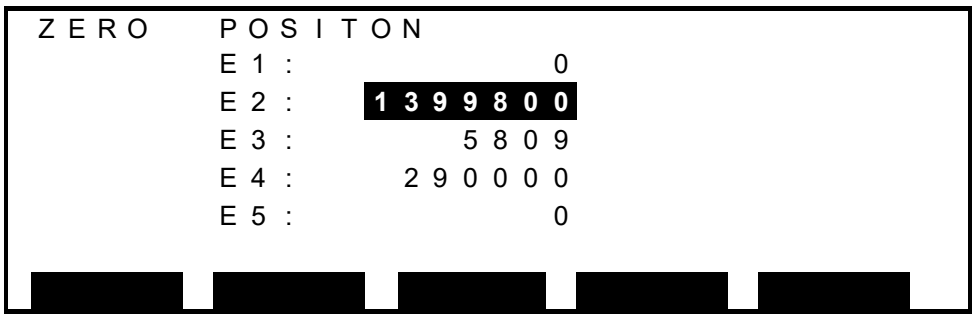
7) Press the “ALT” and “0” keys on the teach pendant at the same time.
8) The following home point editing screen appears.
9) The numerical value of E1 (axis 1) is inversely displayed.



10) Press the “EXE” key on the teach pendant, and the value of “E1” is displayed at the lower left side. Set zero (0) in lieu of this value and press the “EXE” key. Then the E1 value becomes zero (0).
11) Unless the axis could be set to the home point match-mark, enter the coordinate value registered beforehand.



12) When the "↓" key on the teach pendant is pressed, the following E2 value is displayed by reversed image. Perform setting in the same manner as above.



! CAUTION

- The axis 3 and axis 4 of the robot are interlocked with each other. Be sure to strictly observe the order of the home point setting.

  E1 (axis 1) → E2 (axis 2) → E4 (axis 4) → E3 (axis 3)

13) Turn off the controller power, then turn it on again.

4.5.3 Confirming Home Position

1) Make sure of the robot current position according to the POS data of the UTILITY mode. Guide the robot to the approximate home position by manual operation and confirm the position. Unless it could be located at the home point match-mark, make sure that the approximate position is the reference position marked in advance.

2) Execute the following program and make sure of the robot machine home position. Unless the robot could be located at the home point match-mark, DO NOT use the following program.
Create a program in which the robot moves to the position recorded beforehand, referring to the following program. Then make sure that the robot moves to the reference position mark according to the program thus created.

```
PROGRAM ZERO
 SPEED=30
 MOVEA 3,100
 MOVEA 1,0
 MOVEA 2,0
 MOVEA 3,0
 MOVEA 4,0
END
```

4.5.4 Saving Data

Save the system parameters in the system disk.
Connect the personal computer and controller using TSPC (option), then save the “MACHINE.PAR” file from the robot controller to the hard disk, etc. of the personal computer. Save the home position data in the system parameter file.
The home position data is the data characteristic of each robot, which differs with the robot you use. This data is necessary when you effect a cold start and restore the home position data again. Be sure to save the data.

Now, the operation completes.

4.6 Restoring Home Position Data by Multi-Turn Data Clear

1) Home point match-marks are provided for the axes 1, 2 and 4. When only the multi-turn data of these axes has been cleared, move the robot to each home point match-mark position. For the axis 3, move it to the position shown in Fig. 4.2 above.
2) Turn off the servo power, select the TEACHING mode by means of the master mode switch, then press the “UTILITY” key equipped on the teach pendant.
3) When the [UTILITY] screen is called, press the [NEXT] key once, then press the [F5] function key corresponding to the [ENC] menu.
4) When the encoder information screen appears, press the “ALT” and “0” keys at the same time to cancel the editing lock mode.
5) Set the cursor to an axis whose multi-turn data is to be reset to zero (0), and press the [F3] function key corresponding to the [RESET] menu.
6) Move the robot to the teach points, etc. and make sure that the robot can be located at them exactly. (Thus, the robot can be restored completely to the original condition.)

5. Replacement Parts for Maintenance

5.1 List of Replacement Parts – Main Robot

	Part name	Type	Our Dwg. No.	Unit code	Maker	Q'ty	Remarks
1	AC servo motor	MSMD092S2U	S746326	Y610A3400	SHIBAURA MACHINE	1	Axis 1
2		MSMD062S2U	S746327	Y610A3410		1	Axis 2
3		MSMD042S2V	S746336	Y610A3420		1	Axis 3
4		MSMD042S1C	S816701	Y610A3950		1	Axis 4
5	Reduction gear	RV–27C–36.57	S816716	Y610A3960	SHIBAURA MACHINE	1	Axis 1
6		RV–10C–27	S816717	Y610A3970		1	Axis 2
7		VRGF–15B60P –400	S829013	Y610A3JD0		1	Axis 4
8	Input gear		S816839	Y610A3990	SHIBAURA MACHINE	1	Axis 1
9			S816840			1	Axis 2
10	Timing pulley		M198788 (ball screw side)	Y610A39A0	SHIBAURA MACHINE	1	Axis 3
11			M198789 (motor side)			1	Axis 3
12			M198785 (ball screw side)			1	Axis 4
13			M198786 (motor side)				Axis 4
14	Timing belt	BG–705–UP–3M–10–HC	S816773	Y610A39A0	SHIBAURA MACHINE	1	Axis 3
15		BG–575–UP–5M–30–HC	S816774			1	Axis 4
16	Ball screw spline unit		H846688 (Z200)	Y610A3JE0	SHIBAURA MACHINE	1	Axis 3
17			H846689 (Z400)	Y610A3JF0		1	Axis 3
18	Grease	VIGO grease	For reduction gear		Nabtesco		Axes 1 , 2
19		AFB grease (Lithium-base grease)	For ball screw		THK		Axis 3
20	Encoder backup battery	Size AA alkali battery				3	All axes

! CAUTION

- With the exception of the encoder backup batteries and grease, all parts listed above are custom-made to SHIBAURA MACHINE specifications. Contact SHIBAURA MACHINE at order entry.
- Replacement of the parts is performed by SHIBAURA MACHINE service engineer. We will not guarantee any trouble or accident caused after the customer replaced any part or grease by themselves.

5.2 List of Replacement Parts – Controller

No.	Part name	Type	Unit code	Maker	Q'ty	Remarks
1	PS1 (Switching power supply)	LEB100F–0524		Cosel	1	
2	Main power switch	IR–11–AB–10A–1B		Nippon Thermo	1	
3	Lithium battery	ER3V–C		SHIBAURA MACHINE	1	
4	Fuse	51NM030H		PICO	1	For X8HN, X8HI printed boards
5	Printed board	X8HC (Main control)	Y610A2100	SHIBAURA MACHINE	1	
		X8HJ (Connector)	Y610A2320		1	
		X8HN (I/O board)	Y610A2120		1	Minus (–) common
		X8HI (I/O board)	Y610A2130		1	Plus (+) common
		X8HRA (Servo)	See Para. 5.3.		2	Two (2)-axis control per board
		X8HP (Servo power supply)	Y610A2190		1	
		X8HBA (Mother board)	Y610A2140		1	Standard: Four (4)-axis control
		X8HBB (Mother board)	Y610A2110		1	Special five (5)-axis control is possible.
6	Operation panel unit	X8HO	Y610A2310		1	
7	Teach pendant	TP1000	Y610A2600		1	With 5 m-long cable
8	System disk	TS2000SYS			1	

5.3 List of Servo Amplifier Module

Type of servo amplifier module		
Type	Unit code	Remarks
X8HRA 10A/30A	Y610A28L0	Axes 1 and 4
X8HRA 10A/15A	Y610A28E0	Axis 2 and 3